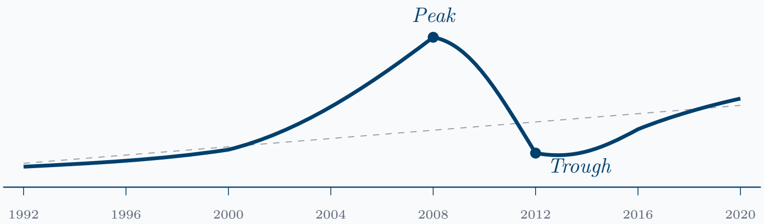
Master Thesis

Housing bubble dynamics in Denmark

From regional price exuberance to household loss sales in the Danish housing market



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Declaration of Authorship

We hereby declare that this thesis is our own original work, written independently and in collaboration as specified. All sources and references used have been properly cited. No part of this thesis has been submitted for any other academic degree or professional qualification.

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Abstract

This thesis investigates the formation and geographic propagation of housing price bubbles across Denmark over 1992–2023. We construct a repeat-sales price index from the universe of property transactions, validated against the Statistics Denmark benchmark, and apply a date-stamping procedure to identify explosive episodes at the regional level. We then develop a contagion framework based on time-varying explosiveness measures to trace how bubbles spread from core to peripheral markets. Finally, we link bubble exposure to household-level outcomes during the 2008 crisis through a Cox proportional hazards model, quantifying the role of illiquidity and negative equity.

Contents

1 Introduction

The 2007–2009 housing crisis posed a host of questions for economists that, even fifteen years later, remain only partially answered. The framework seeks to address some questions regarding this topic, such as: what constitutes a housing bubble? How can it be detected? When does it begin and when does it end? Does it spread geographically, and if so, in which direction? When the cycle eventually turns, which households end up selling at a loss the homes that they had just bought?

These questions are usually asked separately, by different literatures, using different data. In this thesis, these questions are asked together, on the same country, with the same data, in order to understand Danish housing bubbles in a holistic way.

Denmark is an unconventional setting for examining speculative dynamics, as its small and peripheral nature leaves it largely absent from standard bubble narratives, which predominantly focus on the United States. It is, however, almost ideal for the empirical question that is answered here. Statistics Denmark's administrative registers cover the entire population at the individual level and can be linked across decades through identifiers. Transactions, property characteristics, and household balance sheets can be combined to form a highly distinctive and comprehensive dataset. Very few countries offer this combination, and fewer still let it be used at the granularity required here.

At the market level, it was initially expected that bubble-like episodes would be found in the Danish housing sector, perhaps more strongly in urban areas. It was also expected that explosive dynamics originated in the primary urban centers and propagates outward. At the household level, the prior expectation was that financial variables observed shortly after purchase, particularly loan-to-value and liquidity position, would carry predictive content for the probability of a subsequent open-market loss sale.

The analysis confirms some, but not all, of the initial expectations. The following conclusions were reached:

The mid-2000s bubble episode was widespread across Denmark: the mid-2000s boom shows up, depending on how it is measured, in all or almost all of the areas of the country. At the same time, exuberance in the 2020s was shorter-lived and more localized, especially around the Capital Region.

It is also found that contagion almost vanishes once bootstrap inference is applied. Copenhagen, the area with the longest and richest bubble history, generates no statistically significant transmission paths to any other area. Aarhus produces a small number of significant pairs, but their geographic pattern does not support a coherent propagation mechanism. Whatever common pattern the bubble chronologies share looks more like a national co-movement than a sequential diffusion.

At the household level, the financial controls carry predictive content, but the sign of the leverage effect runs against the initial prior. Realised loss sales are over-represented among initially liquid borrowers, while the high-leverage / low-liquidity cell is the most under-represented in the failure events. The natural reading is that the most constrained households did not sell at a loss; they did not sell at all. Equity lock-in absorbs them into the right-censored portion of the sample, and the voluntary-sale segment is left with the households that could still afford to transact.

The rest of the thesis follows the following empirical pipeline: Section 2 reviews the four literatures the work draws on. Section 3 describes the registers, the cleaning, and the variables. Section 4 develops the methodology in four layers, price indices, bubble detection, contagion, and survival analysis, in the order in which their outputs feed one another. Section 5 presents the empirical results, Section 6 summarizes the findings and Section 7 discusses what further research and improvements could be done in the future.

This thesis provides an integrated empirical analysis of Danish housing bubble dynamics, combining price-index construction, bubble detection, contagion analysis, and household-level loss-sale behaviour within a common data framework.

2 Literature Review

2.1 House Price Index Methodologies

Accurately measuring price dynamics in the housing market represents an empirical challenge. The primary difficulty comes from the extreme heterogeneity of real estate assets and the continuous shift in the composition of transacted properties over time. Relying on simple aggregate metrics, such as calculating the mean or median sale price in a given time period, would introduce severe composition bias, distorting the true appreciation or depreciation of the overall market.[1]

To overcome these measurement issues, economists rely on specialized methodologies designed to estimate the true, underlying evolution of prices. Specifically, two widely established and complementary approaches dominate the literature: the repeat-sales method and the hedonic pricing method. Both techniques aim to solve the exact same problem, that is comparing equivalent assets in a market where every house is unique, but they achieve this through different statistical mechanisms.

2.1.1 Repeat-Sales & Hedonic Pricing Methodology

To construct real estate price indices, Bailey et al. (1963) introduced a linear regression approach based on the logged price differential between two sales of the exact same house. The core assumption is that this logarithmic difference captures the true shift in the market index, accompanied by a homoscedastic error term. Rather than comparing different properties over time, they track the exact same houses sold in two distinct time periods. By calculating the price change between these transactions, the repeat-sales method holds all observable and unobservable[2] physical characteristics of the property constant. Aggregating the price variations of thousands of these transaction pairs yields a macroeconomic index stripped of quality shifts, reflecting the pure underlying market trend.

Although their idea turned out to be effective for constructing a house price index, it suffers from a major drawback. It relies on the assumption of homoscedastic error terms. This is, however, a very strong and unrealistic assumption. Houses remain on the market for varying lengths of time; some sell more quickly than others. As a result, the error term becomes heteroscedastic. Case and Shiller (1987, 1989), revolutionized the repeat-sales methodology by introducing weights into the regression to account for heteroscedasticity.

An alternative to this first approach is to estimate a hedonic model. The empirical application of this framework to the real estate market was introduced by Kain and Quigley (1970). Using an extensive survey of housing characteristics, the authors applied factor analysis to identify the most significant qualitative attributes, successfully estimating their implicit prices for both owners and renters. Unlike the repeat-sales method, which discards all single-transaction properties from the sample (with the consequence of reducing sample size and risking selection bias) the hedonic method uses all available

[1] Composition bias is particularly severe across business cycles. During periods of market expansion, the volume of traded properties often increases, while during contractions, the transaction sample might be skewed by distressed sales or foreclosures, artificially altering aggregate means.

[2] While highly effective in controlling for time-invariant characteristics, the repeat-sales method assumes that the physical quality of the house remains constant between sales, potentially ignoring unobserved renovations or natural depreciation.

transactions in the dataset. This method decomposes a property's final sale price into the implicit marginal value of its observable attributes. By statistically controlling for these physical and spatial traits through regression analysis, the model isolates the pure temporal component of the price.

2.1.2 Other Methodologies

Using both the methodologies previously described, a hybrid framework was introduced by Case and Quigley (1991). Their argument is that both approaches systematically discard usable information: the repeat-sales method drops every property sold only once, while the hedonic method ignores the structure that emerges when the same house transacts more than once. Case and Quigley combine both types of observation in a single estimator.

Another approach is the Sale Price Appraisal Ratio (SPAR) method used by Bourassa et al. (2006), which builds the index from the ratio between transaction prices and administrative assessed values. The main pitfall of this approach is that SPAR is dependent on the timing and revision rules of the underlying valuation system, which makes it less suited to a monthly repeat-sales framework like the one used in this thesis.

Pace et al. (1998) developed spatiotemporal autoregressive models that capture the interdependence between neighbouring property prices and their evolution over time, with Dubin et al. (1999) providing a synthesis of the wider class of spatial autoregressive methods used in real estate research. The slow-moving spatial heterogeneity that a spatial weight matrix is designed to capture is largely absorbed by area-by-time fixed effects in standard repeated-sales specifications, which limits the marginal gain of these techniques in aggregate index construction.

More recently, machine learning methods have been increasingly applied to real estate valuation, with Baldominos et al. (2018) providing a representative application to asset valuation and market identification. As Mullainathan and Spiess (2017) clarify these techniques are designed for prediction rather than parameter estimation.

Time-varying hedonic models have also proven valuable for addressing temporal instability in price-quality relationships, although allowing coefficients to drift over time addresses only one of the difficulties commonly associated with hedonic, and these models may still be sensitive to issues such as omitted variable bias.

Smoothing and variance reduction methods have been applied to repeat-sales indices to reduce noise, however, their natural application is at the testing stage rather than at the index-construction stage.

In general, repeated sales and hedonic models can differ in their specifications, and any combination of spatial, learning-based, or time-varying refinements must still seek to deliver a transparent price series.

2.1.3 The Guren Framework: Micro-foundations of Price Momentum

These indices allowed researchers to test economic theories. In the previously mentioned Case and Shiller (1989) paper, the authors utilized their repeat-sales dataset to test whether the housing market is actually efficient. They demonstrated that, unlike liquid stock markets, real estate prices violate weak-form efficiency[3]. Aggregate house price dynamics show a degree of inertia, which means that a city-wide increase in housing prices predicts continued increases in the following year. While individual house prices contain significant idiosyncratic noise, this predictable macroeconomic momentum shows that aggregate housing markets are inefficient.

It is exactly upon this foundational discovery of price momentum that the critical contribution of Guren (2018) builds. He leverages both repeat-sales and hedonic methodologies to demonstrate that momentum is not a statistical artifact, but instead the result of structural and behavioral market frictions. Instead of using these indices to construct aggregate macroeconomic trends, he applies them at the microeconomic level to estimate the expected market value of individual properties at the exact moment they are listed for sale.

Guren demonstrates that sellers face a concave demand curve with respect to relative price as listing a property above the estimated quality-adjusted market average causes the probability of sale to drop precipitously.[4] Underpricing the property below market value results in only negligible benefits in terms of sale speed or transaction probability.

This asymmetry generates strategic complementarity. Sellers are incentivized to price their homes in strict conformity with the market average and to anchor their expectations to recent neighborhood transactions. When the housing market is hit by an exogenous shock, sellers adjust their list prices very slowly and partially to avoid deviating from the rest of the active listings. At the aggregate level, this microeconomic pricing inertia causes the overall price index to absorb economic shocks gradually, ultimately generating the exact long-lasting momentum historically observed by the Case-Shiller indices.

2.1.4 Structural Limitations of Price Indices

Despite the robustness of these methodologies, they are characterized by structural limitations that can introduce significant measurement errors. While Case and Shiller initially highlighted the vulnerability of the repeat-sales model to unobserved quality changes in their papers, the deeper statistical constraints of the methodology are extensively analyzed by Clapham et al. (2006). Specifically, they illustrate how the repeat-sales model is affected by left and right censoring. This occurs because the index only considers properties that sell at least twice within the observed sample period. For this reason, homes that were sold for the first time before the start of the data collection (left-censored) or those that have not yet undergone a second transaction by the end of the sample (right-censored) are excluded.[5]

This censoring could, potentially, lead to selection bias, as the index effectively monitors only the most

[3] In financial economics, weak-form efficiency implies that all historical price data is already reflected in the current asset price. If a market is weak-form efficient, prices should follow a random walk.

[4] In the real estate literature, this phenomenon is closely tied to "Time-on-Market". Properties that are initially overpriced tend to stay active on the market longer, sending a negative signal to prospective buyers and eventually forcing severe price cuts.

[5] In many empirical applications, the repeat-sales requirement forces researchers to discard a vast majority of the available transaction data, drastically reducing the statistical predictive power of the index and, potentially, producing biased estimates.

frequently traded segment of the market. The model remains very vulnerable to physical alterations of the housing stock; improvements or renovations between sales can be mistaken for market appreciation, while physical depreciation may lead to an underestimation of the real price trend.

The hedonic approach, while avoiding the censoring issues by utilizing all transactions, is primarily constrained by omitted variable bias and multicollinearity. Even with extensive datasets, it is nearly impossible to quantify all attributes that influence a buyer's decision, such as interior finishes or local micro-environmental factors.[6] Hedonic indices are structurally vulnerable to broader econometric specification problems, requiring massive amounts of granular data.

2.1.5 Temporal Instability and Index Revision

Building upon these structural constraints, another critical limitation of housing price methodologies is their temporal instability. In Clapham et al. (2006) the authors evaluate how historical index values change ex-post as new transaction data becomes available, a property that is absolutely crucial for the settlement of real estate financial derivatives.

Their findings expose an important weakness of the repeat-sales methodology since it is prone to substantial and systematic historical revisions. Because the model relies entirely on paired transactions, a house sold today that was sold previously during the sample introduces new information for all periods. The econometric estimation must recursively recalculate the index for all those past quarters. They demonstrate that these revisions are not random noise, but are systematically biased downward[7].

In contrast, the authors find that hedonic price indices exhibit significantly greater stability over time. Because hedonic models estimate the implicit prices of housing attributes using the cross-section of transactions within a specific period, the arrival of new sales data primarily updates the current period's index value without recursively rewriting history[8].

[6] If unobserved characteristics are highly correlated with observed attributes, the estimated coefficients will be biased, distorting the accuracy of the quality adjustment. This issue is compounded by severe *multicollinearity* (the inherent correlation between the included variables themselves), which inflates standard errors.

[7] This systematic downward revision is largely driven by unobserved physical depreciation. As time passes, older homes that have gradually deteriorated are eventually sold, yielding lower-than-expected prices. When the repeat-sales model incorporates these new data pairs, it interprets the lower prices as a historical overestimation of the broader market, thereby retroactively downgrading the past index values.

[8] While pooled hedonic models might experience slight revisions if the coefficients of housing characteristics are constrained to be constant over the entire sample period, time-varying or chained hedonic models effectively lock in historical estimates, immunizing them against the arrival of future data.

2.2 Bubble definitions and empirical detection

2.2.1 Definitions and mechanisms of bubbles

The term bubble is used in several related but distinct ways in economics. A common starting point is that an asset price contains a bubble when it exceeds the value justified by expected future payoffs. That definition is simple in principle, but difficult in empirical work because fundamental value is not observed. It depends on assumptions about expected payoffs, discount rates, risk, taxes, market structure and information. For that reason, the literature does not provide a single empirical measure that can be taken directly to the data. Some of the papers define bubbles through no-arbitrage asset-pricing restrictions, while others emphasize expectations, coordination problems, credit conditions or deviations from trend.

A useful way to read these definitions is to separate three questions. The first is whether the price is above a benchmark fundamental value. The second is why the deviation can persist. The third is how the deviation can be detected in data. Different authors try to answer different parts of this sequence. Some are concerned with rational expectations and no-arbitrage restrictions, while others emphasize limits to arbitrage, coordination or credit.

Blanchard (1979) is an early reference for rational speculative bubbles. The paper shows that bubbles and crashes can be compatible with rational expectations, including cases in which a bubble ends and is followed by a crash. This separates the concept of a bubble from irrationality. A price path may be rational for investors if it is supported by expectations about future prices, even if it is risky and eventually collapses. Blanchard also notes that bubbles may take many forms, which helps explain why they are hard to reject or confirm empirically.

Stiglitz (1990) provides a broader and more intuitive definition. A bubble exists when the reason the price of an asset is high today is that market participants believe it can be sold at a high price tomorrow, while fundamental factors do not appear to justify its price. This definition links bubbles to expectations and resale value. It also points to fragility: if the expectation of future appreciation changes, the support for the current price can weaken quickly. Stiglitz's discussion does not reduce bubbles to a single stochastic process. It covers rationality, multiple equilibria, arbitrage limits and transversality conditions, all of which affect whether a bubble can exist and persist.

Froot and Obstfeld (1991) broaden the rational-bubble framework with the concept of intrinsic bubbles. In their model, the bubble is not an extrinsic process separated from fundamentals. Instead, its variability is linked to fundamentals, like dividends, through a nonlinear relationship. A departure from a linear present-value model may reflect a bubble, but it may also reflect fads, time-varying discount factors, changing dividend growth expectations, or regime changes. The paper therefore supports a cautious interpretation of empirical evidence: unusual price dynamics can be consistent with bubbles but that is not the only possible explanation.

Other theoretical papers explain why bubble-like prices might persist. Abreu and Brunnermeier (2003) study rational arbitrageurs who recognize mispricing but cannot coordinate their exit. Because selling too early can be costly when prices continue rising, arbitrageurs might ride the bubble until enough investors attempt to exit. The paper does not provide a detection method, but it explains why mispricing may survive even when sophisticated agents are aware of it. Allen and Gale (2000) give a different mechanism based on credit and financial intermediation. In their model, agency problems under debt finance lead investors to bid up risky asset prices because they do not bear the full downside risk. The result can be a cycle of credit expansion, asset-price increases, collapse, default and financial distress.

2.2.2 Empirical tests and identification problems

A direct empirical approach to rational bubbles emerged with tests based on integration, cointegration and explosiveness. Diba and Grossman (1988) define a rational bubble as a self-confirming component of price that is not part of fundamentals, or that depends on relevant variables in a way not captured by the fundamental valuation model. Under a no-arbitrage restriction, such a bubble component has explosive properties. Their testing logic is that if dividends and unobserved fundamentals are not themselves explosive, stock prices should not be less stationary than fundamentals in the absence of rational bubbles. They test this using unit-root and cointegration arguments and interpret their evidence as being against a class of explosive rational stock-price bubbles, linking rational bubbles to observable time-series properties. Their results show evidence against explosive rational bubbles in U.S. stock prices. However, the inference is conditional on assumptions about fundamentals.

Evans (1991) criticizes this methodology. He shows that standard unit-root and cointegration methods have low power against periodically collapsing rational bubbles. In his model, a bubble grows explosively during expansion phases but occasionally collapses to a smaller value. Because the explosive phases are interrupted, the full sample may not look explosive. A test designed to detect one persistent explosive component can therefore have low power against a process that is explosive only over subperiods.

Evans's critique shifted attention away from full-sample explosiveness and helped motivate methods that search for local explosive behaviour. If explosive behavior is temporary and followed by collapses, a full-sample test might average over normal, explosive, and collapsing phases and fail to detect the bubble. Therefore, this kind of test can fail even if the series contains intervals that would clearly look explosive when examined locally. The important takeaway here is that the question is no longer only whether the whole sample rejects a no-bubble null or not, but whether the data contains localized periods where the process changes into an explosive regime even if it later returns to less explosive dynamics. This is the setting in which recursive procedures become useful.

Gürkaynak (2008) surveys the prior literature on bubble testing. His main conclusion is that econometric tests cannot establish the existence of asset-price bubbles with complete certainty. For many findings interpreted as bubbles, an alternative model with time-varying or regime-switching fundamentals can generate similar data. This does not make bubble tests useless, but it narrows what they can claim. A test can reject a particular no-bubble model, or identify dynamics associated with exuberance, without proving that prices exceeded true fundamental value. This distinction is important for the language used in empirical work.

The survey also clarifies why the empirical literature moved toward more modest claims. If a test depends on an assumed model of dividends, rents or discount rates, rejection can always be challenged by specifying a different model of fundamentals. Conversely, a method that avoids modelling fundamentals directly cannot claim to have measured mispricing. Later recursive right-tailed unit-root tests would occupy this second position. They do not estimate a full fundamental value; instead, they detect periods of explosive or mildly explosive behavior. For this reason, their results are often described as evidence of exuberance or explosive dynamics. The bubble interpretation is an economic reading of that statistical pattern, not a theorem that rules out all fundamental explanations.

2.2.3 Recursive tests for explosive behaviour

Building on earlier right-tailed unit-root approaches to rational bubbles, Phillips, Wu and Yu (2011) developed the recursive forward-regression Supremum Augmented Dickey-Fuller (SADF) test designed to overcome the low power of full-sample tests against periodically collapsing bubbles found by Evans (1991). Unlike in left-tailed tests (Dickey and Fuller, 1981), the relevant alternative hypothesis in bubble detection is not stationarity, but explosive dynamics (a root above unity over part of the sample). Phillips, Wu and Yu (2011) introduced a practical recursive approach for detecting mildly explosive behaviour. Their main detector uses an expanding sample with a fixed initial start point and a moving endpoint. The sequence of right-tailed ADF statistics is then used to test whether explosiveness appears at some point in the sample. Because the statistic is computed recursively, it can also be used for date-stamping the origination and collapse of exuberance, rather than only for a full-sample rejection.

The empirical application in Phillips, Wu and Yu (2011) is the 1990s Nasdaq exuberance episode. They examined logarithmic real Nasdaq prices and dividends and found evidence of explosiveness in prices but not in dividends. Their date-stamping procedure places the emergence of exuberance in the mid-1990s, before Alan Greenspan's famous 1996 remarks on "irrational exuberance" (Greenspan, 1996). The paper established the recursive explosive-root approach and showed how particular episodes of exuberance can be date stamped instead of just limiting itself to detecting if the whole sample is or is not exuberant.

Phillips and Yu (2011) applied this same logic to the subprime crisis. They date episodes of financial exuberance in several markets, including a rent-adjusted U.S. house-price index, crude oil prices and bond spreads, finding statistically significant bubble characteristics in all three, with the real-estate bubble emerging in February 2002.

Phillips, Shi and Yu (2015a, 2015b) address the main limitation of the original PWY recursive approach: long samples might contain more than one episode of exuberance and collapse. If the recursion always begins at the first observation, earlier collapses can weaken the evidence for later bubbles, which might not show as significant in PWY. The PSY procedure generalizes the earlier test by allowing both the start and end points of the regression window to vary. This flexible-window design searches over a wider set of subsamples, making it better suited to recurring episodes. The Generalized Supremum Augmented Dickey-Fuller (GSADF) test is used to test whether the series as a whole contains some explosive behaviour, and is itself a supremum of all the different Backward Supremum Augmented Dickey-Fuller (BSADF) tests (one for each time period except for a minimum window period at the beginning of the sample), whose sequence is used to date origination and termination points.

The distinction between the earlier PWY logic and the later PSY logic is best understood through the treatment of the sample window. The original PWY detector is still effective when the main empirical problem is just a single episode of exuberance. In a longer time series, however, the statistic calculated at a later date can be affected by observations from earlier booms and collapses. PSY allows the starting point of the regression window to move as well as the endpoint. This means that a later episode can be evaluated using subsamples that are not forced to include the whole earlier history.

Date-stamping is handled through the time path of the backward statistic. At each endpoint, the method looks backward over admissible starting points and keeps the strongest evidence of explosiveness. Bubble origination is dated when this statistic first exceeds the relevant right-tail critical value (derived from Monte Carlo simulations based on the null hypothesis of a random-walk, instead of the alternative hypothesis of exuberance), and termination is dated when it falls back below the critical value after imposing a minimum-duration condition for bubbles, in order to avoid flagging short, likely spurious, results as true bubbles.

The main paper, Phillips, Shi and Yu (2015a), applies the method to monthly S&P 500 data from 1871 to 2010, using as their main series the price-dividend ratio. The flexible-window PSY method identifies several well-known episodes of exuberance and collapse, while the original PWY method is more conservative in the same long sample. The comparison illustrates the practical reason for the PSY extension: a method designed around one dominant bubble may fail to recover a richer sequence of episodes.

The companion paper, Phillips, Shi and Yu (2015b), provides the limit theory for real-time detectors. It models mildly explosive episodes embedded in longer stochastic-trend periods and derives consistency results for dating origination and collapse under multiple-bubble alternatives. It also compares the moving-window detector with the original PWY strategy, sequential PWY and other procedures. These results support the use of the flexible-window detector when several exuberance episodes are suspected. The theoretical results are conditional on explicit assumptions, and implementation still requires choices about window length, lag specification, critical values and minimum duration.

Several implementation choices follow from this literature. Minimum window length affects the types of episodes that can be detected. Lag specification affects finite-sample size and power, with the authors suggesting a low, fixed lag (in their S&P 500 application, PSY use $k = 0$, and simulations show lag overspecification can create size distortions, especially for GSADF). Critical values are non-standard because the statistics are recursive suprema and must therefore be simulated. Minimum-duration rules are used to avoid classifying very short statistical spikes as bubbles. These different modelling choices might affect the outcome of the tests.

The PSY framework should be interpreted as just a detector of explosive dynamics. It does not estimate a structural bubble component. Under the maintained assumption that fundamentals are not themselves explosive, explosive price behaviour is difficult to reconcile with a standard no-bubble benchmark. If fundamentals are explosive, or if omitted fundamentals generate explosive movements in prices, a price-only test cannot separate those cases from non-fundamental exuberance.

The use of logarithmic prices is also supported within this literature. Phillips, Wu and Yu (2011) test logarithmic real Nasdaq prices and dividends. Phillips, Shi and Yu (2015a) state that the pricing-equation argument can apply to logarithmic asset prices and dividends under a log-linear approximation, and they report a robustness check using the logarithm of the real S&P 500 price index instead of the price-dividend ratio. These examples do not imply that log price indices prove overvaluation. They show that recursive explosive-root tests can be applied to logged prices when the interpretation is limited to explosive price dynamics.

2.2.4 Trend-based boom indicators and the role of an additional filter

A separate literature identifies asset-price booms using deviations from trend. This literature does not test for explosive roots or estimate bubble components. It instead classifies booms or financial imbalances, often for macro-financial monitoring.

Borio and Lowe (2002) examine asset prices, credit and financial stability. Their use of gaps from trends is part of an early-warning framework, not a proof that trend gaps identify bubbles. Detken and Smets (2004) and Adalid and Detken (2007) develop related boom-bust classifications using deviations from recursively estimated trends. These papers motivate the use of trend gaps as indicators of accumulated imbalances while leaving causality and fundamental mispricing as separate questions.

Goodhart and Hofmann (2008) apply a similar approach to house prices. They study the interaction between house prices, money, credit and macroeconomic activity across industrialized countries. In their boom analysis, house-price booms are defined as sustained positive deviations of real house prices from a one-sided Hodrick-Prescott trend (Hodrick and Prescott, 1997). The classification identifies elevated price regimes, not bubbles in the PSY sense. It is therefore best used as a boom filter rather than as an alternative bubble test.

2.3 Bubble contagion

The previous section reviewed the definition and detection of asset bubbles. The next question is different. Once those episodes have been identified, how should their geographic propagation be studied? The literature for bubble contagion is smaller and less settled than the literature on bubble detection itself. While there are some studies that try to isolate explosive or non-fundamental dynamics and then trace how those dynamics move from one market to another, the bulk of the literature limits itself to examining how housing price movements spread across regions in general. Finally, the last part of this section is concerned with a review of relevant bootstrapping methods, which is required in order to do inference for the statistics in the chosen contagion model.

2.3.1 Housing price spillovers

A natural starting point is the regional housing price diffusion literature. Well before recursive right-tailed tests for bubble detection became standard, it had already been documented that price movements often begin in major metropolitan areas and then spread outward with a delay. MacDonald and Taylor (1993) provide early time series evidence for a British regional housing price ripple effect. Using quarterly regional housing price data for 1969-1987, they test for long-run interregional relationships and short-run dynamics, finding numerous cointegrating relationships across regional prices. Their impulse-response analysis of a shock to Greater London prices shows a “ripple down” pattern: prices in other regions initially rise following the London shock and then fall back over time. Therefore, the results could be read as pointing to regional price spillovers being present in the data.

Meen (1999) builds on this empirical literature by asking what might explain that ripple effect rather than simply documenting it. He reviews migration, equity transfer, spatial arbitrage and regional economic growth as possible explanations, but in the end argues that the effect can arise even without direct spatial spillovers between regions. He shows that common national shocks can produce different regional price responses when regional housing markets have different structures, captured by spatially patterned coefficient heterogeneity. In this interpretation, the ripple effect is caused by the fact that different regional housing markets respond differently to the same national economic conditions, and not a direct spillover caused by migration or arbitrage.

Holly et al. (2011) examine how housing price shocks spread across UK regions using quarterly real house-price data for London and 11 other regions from 1974 to 2008. The paper develops a spatio-temporal model that separates London’s role as the dominant region from local neighbourhood spillovers. It finds that shocks to London house prices spread immediately to other regions and then diffuse further over time through regional interactions. London has persistent long-run effects on most regional markets, while shocks from other regions have only temporary effects on London. The study also shows that New York house prices affect London directly, suggesting that international shocks enter the UK housing market through London. Overall, the paper supports a UK “ripple effect” in which London leads national house-price diffusion.

Finally, Miao et al. (2011) examine return and volatility spillovers across 16 U.S. metropolitan housing markets using Case-Shiller indices from 1989 to 2006, applying VAR models for return transmission and a multivariate GARCH-BEKK model for volatility spillovers. They find that housing markets are spatially connected, particularly within geographic regions, with some evidence that markets such as New York, San Francisco, and Miami play more influential roles. Their results also suggest that volatility linkages are present across several regional markets, while Central and Mountain markets are relatively less affected by external shocks. Overall, the study indicates that housing-market linkages became stronger during the 1999–2006 boom period than in the earlier, calmer period.

Taken together, this literature suggests that housing price movements often diffuse across regions from central markets outward. However, such ripple patterns may reflect either true spillovers or common shocks with heterogeneous regional responses. Furthermore, the literature examined so far does not answer the more specific, and more relevant to the analysis, question of whether speculative episodes, and not just price dynamics in general, spread from one region to another. For that, the relevant literature is admittedly narrower.

2.3.2 Bubble contagion

One way of analyzing the transmission of exuberance is to separate the non-fundamental pressures. Costello et al. (2011) address this using a present-value model, in which fundamental house prices are derived from expected future income and a time-varying discount rate. They then examine spillovers in the gap between actual and fundamental prices, rather than in prices themselves. Their results show persistent deviations across Australian capital cities, strongest in New South Wales from around 2000, and uneven spillovers, with New South Wales more exposed and Western Australia and Canberra relatively insulated. The study does not use the PSY procedure, instead it shows another defensible way of identifying exuberant periods and modelling their spillovers. However, their methodology depends heavily on the assumptions made on the present-value model.

The analysis by Greenaway-McGrevy and Phillips (2016) is more appropriate for the chosen bubble detection model. It is the most notable example of a study that begins with recursive bubble dating and then turns explicitly to geographic transmission. Their first stage identifies regional housing bubbles in New Zealand following the PSY procedure. They found a broad bubble that began in 2003 and collapsed over mid-2007 to early 2008, with Auckland, Wellington, Christchurch and Hamilton all showing episodes of exuberance. The second stage then asks whether a, sometimes lagged, measure of local explosiveness in Auckland helps explain the corresponding measure in other metropolitan centres. Instead of imposing a constant coefficient, the study allows the strength of transmission to vary over time, introducing the idea that the strength and direction of the spillovers could be different in bubble and non-bubble periods.

From their first stage, Auckland emerges as the main candidate core market, and the contagion regressions in the second stage suggest that bubble conditions in Auckland were followed by successive episodes in Christchurch, Hamilton and neighbouring territorial authorities within the wider Auckland metropolitan area. At the same time, the responses are not stable across the sample. In the upswing of the mid-2000s, Auckland's influence is stronger and more clearly positive. In later periods, exuberance becomes more Auckland-specific, and the responses elsewhere become weaker. That result cautions against treating geographic contagion as a fixed parameter. It suggests instead that transmission is episodic, varying with the character of the cycle and with the degree to which exuberance is local or national.

For the present study, however, the most important limitation of Greenaway-McGrevy and Phillips is the lack of inference in their process. The time-varying contagion profiles are presented as descriptive objects, and the study does not provide any formal statistical inference for those paths. Their purely descriptive profile is still informative, but it leaves open the question of if or when the estimated transmission coefficient is statistically significant from zero. That gap is what makes the bootstrap literature relevant here, as a way to extend that methodology.

2.3.3 Bootstrap inference

The discussion now proceeds to compare different bootstrapping methods found in the literature in order to determine which one is most appropriate for the needs of the analysis.

The first candidates come from the general literature on dependent data. Künsch (1989) proposed the moving block bootstrap for stationary observations, resampling overlapping blocks of consecutive data so that local dependence is retained in the bootstrap sample. Politis and Romano (1994) modified that idea with the stationary bootstrap, where block lengths are random and geometrically distributed. Their aim was to obtain a resampled series that is stationary conditional on the data, something the moving block bootstrap does not guarantee. For time series like the monthly housing data used here, both procedures are appealing because they preserve short-run dependence much better than i.i.d. resampling. However, block bootstrapping requires the original data to be stationary, which is far from guaranteed in this setting.

A related branch of the literature replaces block resampling with autoregressive approximation. Bühlmann (1997) studies the sieve bootstrap, where a linear process is approximated by an autoregression whose order grows with the sample size, and the bootstrap sample is generated by resampling from the fitted residuals. In his simulations and theoretical results, the sieve bootstrap performs especially well for linear processes with weak dependence that dies out sufficiently fast, and his later review (Bühlmann, 2002) presents block, sieve and local bootstrap methods as the main nonparametric options for stationary time series. These procedures could also have been considered here. Even so, they were developed for stationary time-series settings in which the main difficulty is serial dependence. The problem addressed here is narrower and less standard, because the object of inference is built from recursive right-tailed bubble statistics and the volatility of the underlying innovations is not necessarily constant over time.

That volatility issue appears clearly in the unit-root literature. Cavaliere and Taylor (2007) show that permanent changes in volatility make the null distribution of standard unit-root tests depend on the variance profile of the shocks. Their solution in that paper is not a bootstrap, but a simulation-based procedure that uses an estimated variance profile to obtain valid critical values without imposing a parametric model for volatility. For present purposes, the paper matters because it states the problem in the right way: once volatility is non-stationary, textbook asymptotics are no longer enough. Cavaliere and Taylor (2008) then move to a bootstrap solution. They propose wild-bootstrap unit-root tests for time series with non-stationary volatility and show that the bootstrap statistics share the same limiting null distribution as the original statistics under that class of volatility processes. This provides a direct bridge from generic time-series resampling to volatility-robust bootstrap inference in unit-root environments.

The wild-bootstrap literature itself comes from regression analysis. Wu (1986) is an early reference on resampling methods in regression under heteroskedasticity, and Mammen (1993) studies wild-bootstrap procedures for least-squares estimators and linear contrasts in linear models where the conditional variance can depend on the regressors. In that setting, the point of the wild bootstrap is to preserve heteroskedasticity rather than to treat the disturbances as identically distributed.

The jump to bootstraps of bubble tests is taken by Harvey et al. (2016). They study the PWY procedure in the presence of non-stationary volatility and show that a standard procedure can be severely oversized and can spuriously indicate explosive behaviour. Their solution is a wild bootstrap applied to the first differences of the data, designed to reproduce the non-stationary volatility pattern in the innovations. They find that the bootstrap version restores correct asymptotic size under a broad class of volatility processes.

Phillips and Shi (2020) extend this type of bootstrap to the PSY procedure. Their composite bootstrap is built for a recursive testing environment, where the problem is not only heteroskedasticity in the innovations but also the accumulation of false positives across the whole sequence of evolving tests. The procedure starts by estimating the ADF regression under the null on the full sample and collecting the corresponding residuals. It then generates bootstrap data recursively from the fitted null model, but replaces the original disturbances with a wild-bootstrap innovation constructed from two ingredients: a random weight and a residual drawn with replacement from the estimated residuals. In this way the resampled series preserves the null of no explosiveness while allowing the volatility pattern in the shocks to be carried into the bootstrap exercise. For each bootstrap sample, the full PSY sequence is recomputed over the monitoring window and the maximum statistic over that sequence is retained. After many repetitions, the critical values are taken from the empirical distribution of those maxima. This last step is what addresses multiplicity: inference is based on the distribution of the most extreme statistic generated by the entire recursive sequence, not on the distribution of a single regression. The result is a procedure tailored to the same recursive structure used for date-stamping bubbles, while remaining robust to the heteroskedasticity problem stressed by Harvey et al. (2016). In that regard, the attractiveness of this method is that it is not a generic bootstrap, but that it is specially constructed for recursively generated BSADF statistics. That is the reason why the inference procedure used here is mainly based on this study.

2.4 Open-market loss transactions during the 2008 crisis

2.4.1 Loss Aversion and the Identification of Loss Sales

During an economic downturn, as prices decline, most realized losses reflect general market conditions rather than individual financial constraints. This section asks when a completed sale made at a loss can be informative about the interaction between realised housing losses and the seller's balance-sheet position. In other words, the goal is to identify the behavioural, structural, and institutional mechanisms that shape the decision of selling at a loss in an economic downturn.

The behavioral starting point is Genesove and Mayer (2001). They show that homeowners anchor their reservation prices to the nominal purchase price. Sellers facing a nominal loss tend to set asking prices above the prevailing market value, accepting longer times-on-market. Nominal anchoring acts as a first-order driver of pricing and listing duration in housing markets. To estimate this effect they use a Cox proportional-hazards model on listing data between 1990 and 1997. The Cox specification identifies how a prospective loss, defined as the gap between the original nominal purchase price and the expected market value, shifts without imposing a parametric form on the underlying time-on-market distribution. The estimated effect of the loss variable on the hazard is large and statistically significant across specifications. Listings with larger prospective losses sell substantially more slowly, and the result survives controls for property characteristics, neighborhood, and time.

Using comprehensive Danish administrative data, Andersen et al. (2022) estimate a structural model of the listing decision that incorporates reference dependence, loss aversion, and financial constraints. The recovered loss-aversion parameter implies that nominal losses reduce household utility roughly 2.5 times as much as equivalent gains increase it, a magnitude large enough that completed sales below the purchase price are unlikely to reflect frictionless optimisation. They also document a striking empirical regularity as reference dependence is clearly weaker among households facing tighter financial constraints. The behavioral friction is what makes the loss informative, and its weakening among constrained households makes that informativeness one-sided.

A general housing crash can produce nominal losses for sellers who simply follow the market and have no specific household-level pressure. It also results in an increase in the amount of forced sales, in the form of foreclosures.

Campbell et al. (2011) show that forced sales (foreclosures and death-related liquidations) trade at a discount relative to comparable unforced transactions in comparable properties in the exact same area. They also show that this discount propagates to nearby houses, however this effect is very short lasting and extremely localized.

2.4.2 The Equity Lock-In Mechanism

The behavioural channel is one source of resistance to realising a loss. A second source is mechanical and structural rather than psychological. The theoretical anchor for this channel is Stein (1995), who develops a model of the housing market with binding down-payment constraints. In Stein's framework, owners with low equity who cannot raise the down-payment for a replacement property are unable to move even when continuing to occupy the current home is suboptimal. The reason is that selling an underwater property forces the seller to bring cash to the closing table to retire the residual mortgage, and households without that liquid buffer are locked in. The model predicts that declines in equity should reduce trading volume, and that the effect should be concentrated among households with low liquid assets relative to mortgage exposure.

In the conventional default literature, liquid assets act as a buffer against income shocks and reduce the failure rate. In Stein's setting, however, liquidity operates as the vehicle by which a constrained owner can execute a loss sale. A highly leveraged household with sufficient liquid assets retains the option to bring cash to closing, while a highly leveraged household without that buffer may remain locked in. The expected association between liquidity and the hazard of an open-market loss sale is conditional on the leverage position. The interaction between leverage and liquidity seems to be a structural object of interest in this context.

Ferreira et al. (2010) provide direct empirical confirmation of the Stein mechanism. They show that owners with negative equity have lower mobility rates than otherwise comparable owners with positive equity. The phenomenon they describe arises from the constraints imposed by leverage. For these reasons it supports the use of leverage measures as predictors of sale outcomes and it predicts that the observed hazard of loss-realising sales should take into consideration the fact that the most exposed households are prevented from completing a sale. A direct implication for hazard models of loss-realising sales is that the most constrained households are differentially absent from the set of completed transactions. Their selection out of the sale margin biases downward the estimated hazard ratio on loss positions, since the loss-sales that do occur are drawn disproportionately from households with the liquid resources to execute them.

The systemic relevance of these balance-sheet conditions is established by Mian and Sufi (2010). Their evidence shows that the depth of the U.S. housing downturn was concentrated in geographic areas with the highest pre-crisis household leverage, and that the recession was disproportionately driven by household-level deleveraging. Their result motivates the use of a leverage-based stratification in empirical models that study this period. If pre-crisis leverage is the macroeconomic catalyst of the downturn, then categorising households by their entry leverage position is the first step in identifying the cross-section of vulnerability.

2.4.3 The Double-Trigger Hypothesis

The equity lock-in mechanism explains why households resist realising a loss. It does not explain when a loss sale actually occurs. In the specific case of foreclosures, the household-finance literature addresses this question through the Double-Trigger hypothesis, formalised by Foote et al. (2008). They show empirically that the majority of underwater U.S. borrowers continued to service their debt and retain their homes during the 2007–2009 crisis indicating that negative equity is a necessary but not sufficient condition for default. The default decision is typically triggered by a concurrent shock to the household's ability to pay, with income and employment shocks being the strongest predictors. The "double trigger" refers to the joint occurrence of negative equity and a binding cash-flow shock.

Elul et al. (2010) and Gerardi, et al. (2018) refine and extend this evidence. Their identification strategies differ in the treatment of the second trigger with Elul et al. (2010) using credit-card utilisation as a proxy and treating it as a real-time signal of liquidity stress, while Gerardi et al. (2018) rely on direct survey matches to track individual household balance sheets and income shocks as the second trigger. Both papers confirm that negative equity and cash-flow stress interact non-linearly, meaning that their joint occurrence raises the failure rate by substantially more than the sum of the two separate effects, which is the empirical version of the joint-trigger prediction.

The Double-Trigger logic was developed primarily in the U.S. setting, where several states permit non-recourse mortgage lending. Under non-recourse rules, the borrower's liability is bounded by the value of the collateral, and a sufficiently underwater borrower retains the option to walk away from the loan without further claim. This option introduces strategic default into the empirical landscape.

The Danish institutional setting differs in a way that simplifies identification. Danish mortgages are full-recourse meaning that residual debt after a forced sale or foreclosure follows the borrower personally and is not extinguished by the loss of the asset. As discussed by Andersen et al. (2020) using Danish administrative data, this regime structurally weakens the financial incentive for strategic default. Within this regime, the decision to execute a discounted market sale is more likely to reflect constraints than opportunistic optimisations.

2.4.4 Econometric Foundations in Survival Analysis

Translating these behavioral and financial theories into empirical models often requires survival analysis techniques. The methodological foundation of this approach was introduced by Cox (1972). This paper presents the proportional hazards model and the partial likelihood estimator, which together form the standard framework for time-to-event analysis[9]. This framework allows researchers to estimate how controls shift the instantaneous risk of an event without imposing parametric assumptions on the baseline hazard, making it useful in settings where the timing of the failure event is itself an object of interest.

Three diagnostic concerns dominate the application of the Cox model to this kind of data. The first is the proportional-hazards assumption: the hazard ratio between two households with different controls must be approximately constant over the observation window. Grambsch and Therneau (1994) introduced a test to check whether the proportional hazard assumption holds by analysing the residuals behaviour; if the assumption holds the Schoenfeld residuals should exhibit no systematic drift over time. Therneau and Grambsch (2000) show how this framework can be extended in two relevant ways. Stratification can be used as a remedy for non-proportional baseline hazards across groups, while time-splitting and interval censoring can remedy non-proportional covariate effects driven by information decay.

[9] Historically, this model has primarily been used for medical analysis; for this reason most of the existing literature comes from that field.

The second concern is functional-form adequacy. The specification link test used here is based on an idea of Tukey (1949) and was further developed for generalized linear models by Pregibon (1980). This functional form test screens the chosen specification for omitted nonlinearities, missing interactions, or other structural sources of misspecification. Its adaptation to the Cox setting retains the original intuition: if the linear index already captures the systematic component of the relationship, an additional regressor constructed as the squared linear index should not enter significantly.

The third concern is multicollinearity among the financial controls. These controls describe pieces of the same household balance sheet, and the conceptual overlap usually shows up as statistical correlation. The standard diagnostic is the variance inflation factor, which depends only on the regressor matrix and not on the dependent variable (Belsley et al., 1980). This is convenient here, because the Cox partial likelihood does not produce a residual variance from which the VIF could be read directly: the workaround is to compute it from an auxiliary OLS regression on the same right-hand-side variables.

Taken together, the proportional-hazards test, the link test, and the VIF diagnostic provide evidence that a survival model is correctly specified, that its core assumptions hold and that the estimated coefficients are not affected by multicollinearity.

3 Dataset Construction and Cleaning

This section describes the administrative data used in the empirical estimations. The primary source is the registry data provided by Statistics Denmark (*DST*), which offers detailed, population-wide micro-level information spanning several decades. Through anonymized personal identification numbers and property codes, real estate transactions and physical housing characteristics are linked to the socioeconomic and demographic profiles of owners, making it possible both to construct the price indices and to carry out household-level analyses of open-market loss transactions.

The dataset brings together multiple registry modules that are organized into five main categories. The first contains identifier keys, linking individuals to residential units. The second covers household characteristics, with demographics and family structure taken from `BEF`, income, assets, and debt from `IND`, tax deductions for home improvements from `INDSLUT`, and education from `UDDA`. The third category describes property ownership history through `EJER`. The fourth category contains `nbpf2013`, which is used to harmonize pre- and post-2007 municipality codes. It also contains the physical housing attributes contained in `BOL`, and the official property tax valuations recorded in `EJVK`. Finally, the fifth module contains the administrative real estate transaction register `EJSA2023`, which provides all property sales records up to 2023. Descriptions of all data variables and filters are provided in the appendix section “Data Description”.

3.1 Housing Characteristics Dataset

First, the "Housing Characteristics" dataset is constructed as a granular panel from raw administrative registers covering physical attributes and property tax valuations.

To maintain a consistent sample representative of the standard residential market, the dataset is restricted to regular owner-occupied single-family houses, apartments, and summerhouses privately owned by households. Commercial buildings, agricultural estates, non-owner-occupied units, and properties owned by corporate entities, housing associations, or the state are excluded to prevent structural biases in price estimation. Special attention is given to the 2007 Danish municipality reform, which introduces structural breaks in the administrative panel: the procedure harmonizes municipality codes across the timeframe and removes duplicate addresses or overlapping property records.

The filtering procedure operates through a categorical tagging variable that flags observations failing one of eight validity criteria. These include wrong property type (commercial or agricultural use), non-private ownership, duplicates flagged directly by Statistics Denmark, suspiciously large or small living areas, missing or inconsistent room counts, repeated address entries, and the two patterns of address merging and splitting induced by the 2007 municipality reform (see Table *Filtering rules for the housing characteristics dataset* in the Appendix). The physical characteristics of the surviving properties are then harmonized across all years, standardizing continuous variables such as living area, number of rooms, number of bathrooms, and building age.[10]

[10]Where appropriate, missing values for secondary physical attributes (such as bathrooms or showers) are conservatively set to zero to prevent excessive observation loss during the regression analysis.

3.2 Price Index Dataset

The final transaction dataset is built from the administrative real estate transaction register `EJSA2023`. The objective is to isolate genuine market transactions from anomalous transfers and to merge these sales records with the harmonized physical property characteristics described above.

Administrative transaction registers contain noise, multiple preliminary records for a single trade, non-market transfers (e.g. intra-family), and idiosyncratic price outliers. To ensure that the pricing model accurately reflects standard market dynamics, the sample is restricted to regular market trades between private households; institutional trades, forced auctions, and sales involving non-monetary obligations are excluded. Once purified, the transaction sample is merged with the physical characteristics panel. Transaction prices are then converted into real terms and adjusted for statistical outliers before compilation into the final analytical dataset.

The procedure is divided into a pre-merge cleaning phase, a merging phase, and a post-merge filtering phase. Pre-merge, observations are dropped with a buying price of zero. When multiple records exist for the same property within the same municipality on the same reference date, a prioritization algorithm selects the most reliable information, retaining non-extreme prices and prioritizing the oldest date, thereby collapsing the data into one row per trade. The sample is further restricted to a single observation per property per year.[11]

Invalid transactions are filtered through a two-stage tagging variable. The pre-merge stage identifies non-household trades, administratively flagged extreme prices, special circumstances involving non-monetary obligations, foreclosures, and intra-family transfers. The post-merge stage catches issues that emerge only once physical and price information are jointly available, such as properties that fail to match physical characteristics, transactions with implausibly low or high real prices, and inconsistent living-area records (see Table 15 in the Appendix). After isolating the appropriate sales types (family houses, apartments, and summerhouses), the transaction records are merged with the housing dataset based on municipality, property ID, and year of sale[12].

Before finalizing the dataset, nominal sales prices are deflated using the Consumer Price Index to generate real prices in constant 2015 Danish Kroner. All flagged observations are removed, and after a uniqueness check by property ID and purchase year, the finalized panel dataset is ready for index estimation.

3.3 Open-Market Loss Analysis Dataset

The panel dataset constructed above is optimized for estimating aggregate housing price indices, where the methodological objective is to isolate the unobserved market value of the properties. The micro-level analysis at the end of this thesis, however, shifts the analytical focus from properties to individuals: to

[11] While this restriction introduces a theoretical selection bias, the incidence of properties bought and sold multiple times within a single calendar year is negligible relative to total transaction volume, and this restriction is necessary to facilitate a clean merge with the yearly housing characteristics panel.

[12] If characteristics are missing in the purchase year, the algorithm searches one year ahead, then sequentially up to five years backward.

identify the household-level dynamics underlying open-market loss transactions, a distinct secondary dataset has to be constructed through a specific sequence of merges.

This sequence is required to overcome a fundamental linkage limitation: the ownership registers cannot be merged directly with the transaction records because they do not share a unique identifier. The physical property dataset acts as the necessary bridge. The housing characteristics dataset is first merged with the ownership registers, exploiting the property and municipality identifiers shared between the two. This step maps each physical house to the household owning it. Because subsequent analysis tracks personal outcomes at the time of sale, some checks are applied at this stage to resolve ambiguous ownership and deal with duplicate address registrations and mismatched property-to-person identification links. Properties that cannot be unambiguously assigned to a distinct household entity are dropped.

Second, since the housing characteristics dataset shares unique identifiers with the transaction register, the combined owner-property panel is merged with the finalized `hedonicdata` dataset. By retaining only the exact intersection of these datasets, the sample is restricted to transactions with a perfect match across all three dimensions: physical characteristics, validated market prices, and unambiguous owner identity.

Following the integration of the transaction, property, and ownership registers, auxiliary administrative variables (such as detailed property tax assessment fields) that are not pertinent to the research question of this thesis are dropped. Because the analysis of open-market loss transactions requires complete socioeconomic profiles, observations missing crucial demographic information, specifically household income or owner education, are dropped.

3.4 Original Variables Descriptions

To keep the code tractable, after observation filtering only the variables relevant to the analysis are retained. These fall into two groups corresponding to the two analytical applications. The first group contains the core variables used in the price index estimation: the repeated-sales identifier and counter, geographic codes for the property and for the owner's residence, the exact purchase date, nominal and real transaction prices (both in levels and logs), and the property attributes that enter the hedonic specification (log living area, rooms, bathrooms, and building age). The second group contains the additional household-level variables required for the open-market loss dataset: demographic information on the household members (number of adults, age, gender, education in years and by degree category), income, net wealth, and the full breakdown of financial assets and liabilities used to construct the leverage, liquidity, and wealth controls, together with administrative dummies tracking household dissolution and total properties owned.

The foundational code used to construct the majority of these financial variables was originally developed by researchers at Copenhagen Business School (CBS) for a separate study.[13].

[13]The construction code for the financial variables was adapted from Andersen et al. (2022), and the variables used in the models were re-audited and validated. All financial variables in the Open-Market Loss Dataset are measured on 31 December

3.5 Derived Variables for the Empirical Analysis

Building on the merged dataset described above, which links repeated property transactions to year-end household balance-sheet information around the purchase year via geographic and temporal identifiers, a final set of variables is derived for the empirical analysis.

For the price index estimation, the sixteen geographic areas obtained by merging municipalities, the time elapsed between repeated sales (used both as a filtering criterion and as the basis for the GLS weights), the raw and area-specific growth rates entering the outlier-detection step, the property-specific and time fixed effects estimated within the repeat-sales regression, and the resulting area-level and national price indices normalized to a 2015 base are constructed.

For the open-market loss analysis, the derived variables are organized into four blocks. The first defines the loss-sale indicator, combining underperformance relative to the local benchmark, a nominal loss, and a resale before the trough after a bubble-window purchase. The second contains financial explanatory variables: a continuous LTV measure and its four-category discretization, together with the decile ranks for household liquidity, net financial wealth, and income that operationalize the household's structural position at the time of purchase. The third block collects the geographic and temporal fixed effects (area, municipality, and year of purchase) used to absorb spatial heterogeneity and cohort-level entry conditions. The fourth block contains the temporal variables: the static markers for the start and end of the detected bubble (April 2005 and January 2007), the peak (August 2007) and trough (January 2012) in the samples, and the two survival-time measures used across the six model specifications. The first of these, `k_months` together with its annualized version `k_years`, measures the holding period from the exact purchase date to either the date of sale or right-censoring at the trough, and underlies the baseline models. The second, `duration_crisis`, is a synchronized exposure metric calibrated to the distance between the sale date (or trough) and the market peak, used in the peak-to-trough specifications to measure the time each household spent under stress within the acute deflationary phase.

A complete description of all registry modules, filtering rules, original variables, and derived variables described in this section is provided in the Appendix, under the section *Dataset description.*

To analyze property ownership dynamics and market exits, the dataset is structured into "transaction pairs." Each observation represents a unique ownership cycle consisting of two distinct transactions: the first corresponds to the initial acquisition, while the second corresponds to the subsequent sale of the same property. This pairwise structure is essential for the identification strategy described in the methodology, since it allows the exact calculation of holding periods and the determination of the nominal capital gain or loss realized upon exit, which together define the loss-sale event used in the Cox proportional hazards model.

In Summary:

Step	Datasets joined	Output
Raw registers	`BOL`, `BEF`, `IND`, `INDSLUT`, `UDDA`, `EJER`	⟶ Ownership panel
	`BOL`, `EJVK`, `nbpf2013`	⟶ Housing characteristics
	`EJSA2023`	⟶ Transactions
Merge A	Housing characteristics + Transactions	⟶ `hedonicdata`
	Joined on property ID × year. Used for price index estimation.	
Merge B	Ownership panel + Housing characteristics	⟶ Owner–property panel
	Joined via shared `BOL` identifier (bridge). Direct merge with `EJSA2023` not possible.	
Merge C	Owner–property panel + `hedonicdata`	⟶ Open-market loss dataset
	Retains transactions with complete property and household information.	

Figure 1: Registry modules and merge sequence. Two final datasets are produced: hedonicdata, used for price index estimation, and the open-market loss dataset, used for the survival analysis. The sequential merge structure is necessary because the ownership registers do not share a direct identifier with the transaction register EJSA2023 and require the housing characteristics dataset as a bridge.

4 Methodology

4.1 Declaration of AI Usage

Generative Artificial Intelligence (GenAI) tools were used in an auxiliary role during the preparation of this thesis. Their use was limited to three specific tasks: refining the wording of the manuscript (proofreading for typographical and grammatical errors and suggesting stylistic improvements), assisting with routine coding tasks where applicable (syntax queries in `Stata` and debugging, and the formatting of figures and tables), and helping to summarise results once they had been produced. All outputs from GenAI were manually reviewed.

4.2 Code Availability

To facilitate replication, all programming code underlying this thesis is publicly archived and citable at:

`https://doi.org/10.5281/zenodo.20142133`

The repository contains the code scripts used to construct the housing transactions panel, the price-index estimators, the PSY date-stamping and bootstrap contagion routines, and the survival analysis pipeline. Because the underlying microdata are subject to Statistics Denmark's confidentiality rules, the repository contains no individual-level records; access to the original registers must be obtained independently. The archive contains exclusively code written by the authors for the purposes of this thesis. The original codebase of Andersen et al. (2022), on which parts of the empirical strategy of this thesis build conceptually, is *not* redistributed here; what is published is an independent implementation for this thesis.

4.3 House Price Index Estimation

4.3.1 Overview of Estimated Indices

This study constructs two repeated sales indices implementing the methodology established by the empirical framework in Guren (2018). The indices of interest share a monthly temporal frequency, however, they differ in their geographic granularity (area vs. national level) and the fixed-effect specifications used in the regression stage. Throughout the estimation process, all transaction prices from the Statistics Denmark dataset were converted into real terms using the official Consumer Price Index from DST.[14]

4.3.2 Area identification

To ensure the statistical robustness and comparability of the regional analysis, a bottom-up, iterative approach was employed to define the geographic boundaries of the housing market indices. Initially, the process began by mapping the data using the standard, official administrative municipalities (*kommuner*) present in the Statistics Denmark registers. However, an exploratory analysis quickly revealed that this highly granular approach was deeply flawed due to extreme sample heterogeneity and low sample size for most municipalities.

While some major urban municipalities contained a large volume of transactions, others contained very few. Because the methodology required the construction of *monthly* price indices, ensuring a consistent and sufficient number of observations per month in every geographic unit was critical. Smaller municipalities, particularly isolated islands and the sparsely populated regions in the far northwest of Denmark suffered from data sparsity, resulting in noisy monthly index estimates.

This sparsity presented not only an econometric challenge but also a strict institutional barrier. To comply with Statistics Denmark's data privacy and microdata export regulations, any extracted data cell must contain a strict minimum of five observations. Faced with this constraint, there were two potential solutions: either drop the smaller municipalities from the dataset entirely, or merge them. Dropping smaller municipalities was considered a less effective approach; even if only the larger municipalities were kept, the remaining indices would still face vastly different levels of variance and volatility across the remaining indices due to differences in sample size. Therefore, the second path was adopted: aggregating the municipalities into newly defined, standardized geographic areas.

The ultimate goal was to create indices with low enough noise so they could be adequately used during the bubble detection stage of the analysis and therefore the results on that later stage are precise and accurate. To achieve a homogeneous sample size, an ideal benchmark was established of approximately 100,000 observations per newly created area.

[14] See "StatBank Denmark table PRIS9" link in References. Consumer price indices were deflated and standardized to a base year of 2015 to facilitate a direct and consistent comparison across all indices.

4.3.3 Area Creation Rules

The following map visualizes the resulting spatial delineations of the areas and their respective municipality compositions:

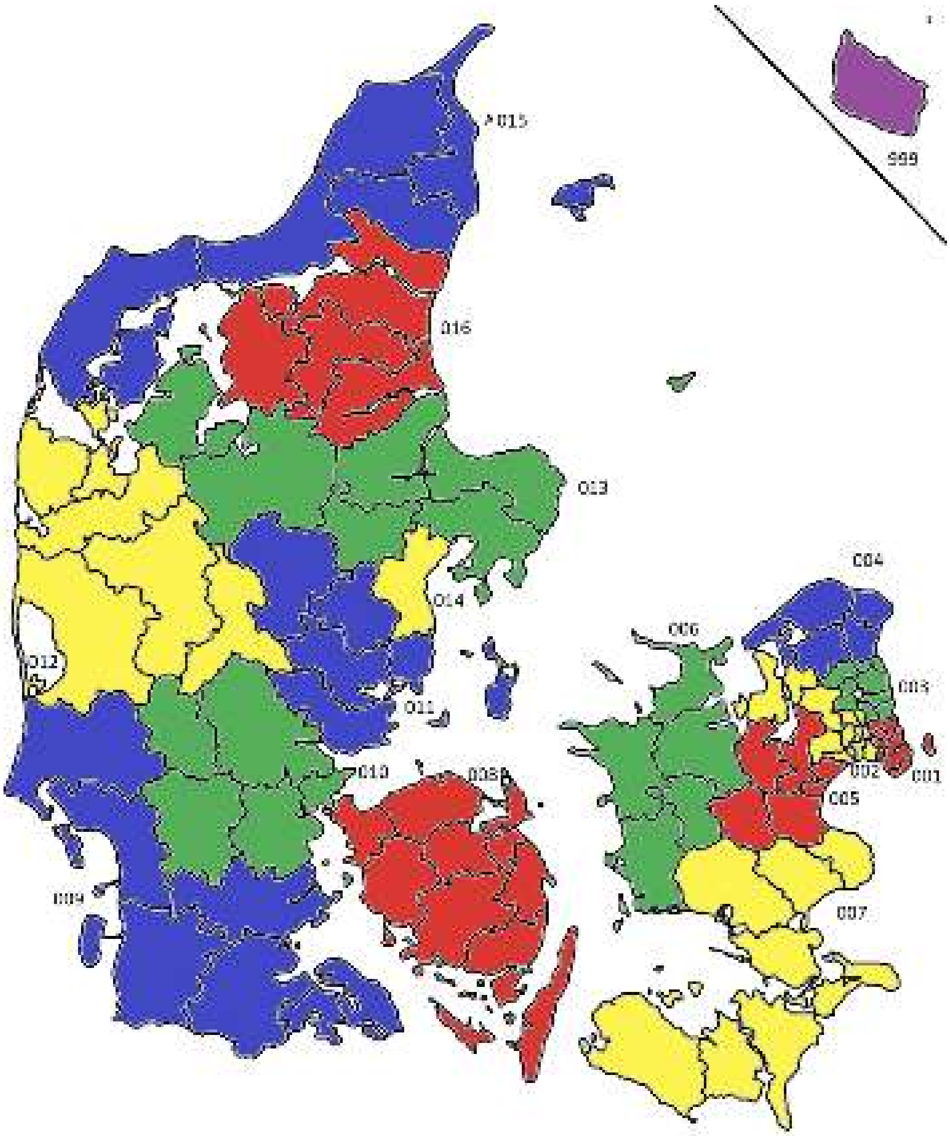


Figure 2: Map illustrating the composition of custom areas. A more detailed description can be found in Denmark area description section in the appendix.

The aggregation process was not random; the areas were constructed so that they strictly met the following rules and requirements:

- Areas should have at least 100,000 observations[15] over the whole sample (from 1992 to 2023).

[15]The Copenhagen index was first examined; it looked very smooth compared to the other communes over the time horizon considered. With an average close to 5000 observations per year over a 30 years time span 100000 observations were used as a reference point by comparing the index with smaller cohorts. Municipalities' codes are the ones used officially by DST.

- All parts of a given *kommune* must be in the same area (a *kommune* cannot be split).
- All *kommunes* in an area must belong to the same official Danish region (Capital, Sjælland, South, Central, North).
- There must be no enclaves in an area. If there are several *kommunes* in an area, all of them must be connected via adjacency so that one could travel from any given *kommune* to another without leaving the area. In the case of *kommunes* consisting of a single island, they must be connected by ferry to the rest of the area.
- Bornholm was entirely removed from the data (area 999 in Figure 2).

Guided by these rules and requirements, the borders of these areas were drawn by emphasizing the geographic and economic closeness between the biggest towns of the *kommunes* comprising the area. Size variability between the areas was actively minimized while simultaneously trying to draw the largest possible number of distinct areas without violating the prior constraints.

4.3.4 Repeated sales: methodology for Area x Month Index

The estimation follows a three-stage procedure to control for property heterogeneity and time-dependent variance. After selecting only the data with repeated sales and valid dates, the analysis applies the following procedure:

Data Pre-processing and Outlier Detection

Prior to the regression analysis, a filtering procedure is implemented to identify and remove idiosyncratic outliers[16]. This filter works by comparing the realized appreciation of an individual property against the average trend of its area.

1. **Individual Property Growth:** Using the actual transaction prices observed for property hl (where h is the individual house while l identifies the area), the analysis calculates its specific capital appreciation between the first sale at time t_1 and the second sale at time t_2:

$$g_{hl} = \frac{P_{hl,t_2}}{P_{hl,t_1}} - 1 \tag{1}$$

2. **Market Benchmark Growth:** Simultaneously, the average price level is computed for the area l in the corresponding years ($\bar{P}_{l,t}$). The market growth serves as the baseline expectation:

$$g_l = \frac{\bar{P}_{l,t_2}}{\bar{P}_{l,t_1}} - 1 \tag{2}$$

3. **Deviation and Exclusion:** The excess volatility is defined as the difference between the

[16]This passage might look in conflict with the last part of the analysis where a similar methodology is used for the identification strategy. The main argument is that the index obtained by removing the outliers had a really high threshold of 100% and that the index constructed using outliers is still almost identical.

individual property's growth and the market benchmark ($\Delta g = g_{hl} - g_l$). An observation is identified as an outlier and removed if the individual price dynamic deviates from the market trend by more than 100 percentage points:

$$\text{Drop if} \quad |g_{hl} - g_l| > 1 \tag{3}$$

This ensures that the dataset excludes non-representative transactions while preserving market volatility.

Stage 1: Fixed Effects Estimation

The log-price of property h in area l at time t is modeled as:

$$\ln(P_{hlt}) = \mu_{hl} + \delta_{lt} + \varepsilon_{hlt} \tag{4}$$

Where μ_{hl} is the property-specific fixed effect and δ_{lt} is the area $\times$ month fixed effect. This interaction is crucial as it allows the price path to evolve independently for every area over time.

Stage 2: Modeling Heteroskedasticity (GLS Weights)

To correct for heteroskedasticity, the GLS weights are calculated[17]. The variance of the error term is assumed to increase with the holding period k. The squared residuals from Stage 1 are grouped into deciles of k. The dispersion variance $\hat{\sigma}^2_{disp}$ for a specific decile d is estimated as:

$$\hat{\sigma}^2_{disp,d} = \frac{1}{N_d} \sum_{j \in d} \hat{\varepsilon}^2_{hlt} \tag{5}$$

Stage 3: Re-estimating the regression in Stage 1 using weights

The final model re-estimates the equation using WLS:

$$w_{hlt} \ln(P_{hlt}) = w_{hlt}\mu_{hl} + w_{hlt}\delta_{lt} + w_{hlt}\varepsilon_{hlt} \qquad \text{where} \qquad w_{hlt} = \frac{1}{\sqrt{\hat{\sigma}^2_{disp,d}}} \tag{6}$$

This index utilizes the raw $\hat{\delta}_{lt}$ coefficients directly. It captures high-frequency price dynamics specific to that area and month.

4.3.5 Repeated sales: methodology for the National Index

The national index employs a fundamentally different specification to generate a robust national benchmark. Unlike the local indices, this model does not allow for location-specific time trends. Instead of the interaction term δ_{lt}, this model utilizes a time only fixed effect (δ_t). The estimating equation

[17]Note that the way weights are computed can substantially differ in the literature; the framework proposed by Guren, inspired by the work of Andersen et al. (2022), was adopted

represents the average price evolution for the entire country, absorbing local heterogeneity solely through the property fixed effects (μ_h):

$$\ln(P_{ht}) = \mu_h + \delta_t + \nu_{ht} \tag{7}$$

By restricting the time dummy to be location-invariant, this index aggregates all transaction pairs across all areas into a single coefficient per month-year. This provides a pure national trend against which the heterogeneous growth of individual areas can be compared. Initially, this aggregate index was computed in nominal terms. However, relying on nominal prices caused the repeated sales model to drop a significantly larger number of outliers, as its outlier detection process does not naturally account for inflation. To maximize the sample size, the final specification employs a real price index. Stage two and three (adapted) are identical to the area x month index.

4.3.6 Index Construction and Normalization

Following the WLS estimation, the raw fixed effects represent the price evolution relative to the initial period or the omitted category. To ensure comparability across areas, the raw estimates are transformed into a standardized index with a common base month x year (Jan 2015).

Step 1: Establishing the Base Period (Jan 2015)

Prior to establishing the baseline, the raw fixed effects $\hat{\delta}_{lt}$ are converted from logarithmic terms to levels by exponentiating the coefficients ($\exp(\hat{\delta}_{lt})$). This ensures that the calculation of the base period average and the index normalization are performed directly on the price levels rather than on the log-linear estimates.

The estimated location-time fixed effects are aggregated to the month x area level. This is achieved by calculating the monthly average FE for each specific area (in a similar way, the aggregate computes the average of all FE values without distinguishing between areas) and it ensures a unique price index value for every location l at time t. Let $\exp(\hat{\delta}_{lt})$ denote the aggregate fixed effect (the coefficient of interest) for location l at time t. Jan 2015 is selected as the reference base period. For each location l, the average value is calculated of the fixed effects observed during the 12 months of 2015[18]. This defines the local reference level $\bar{\delta}^{avg}_{l,2015}$:

$$\bar{\delta}^{avg}_{l,2015} = \frac{1}{N_{2015}} \sum_{t \in 2015} \exp(\hat{\delta}_{lt}) \tag{8}$$

[18]The arithmetic mean is used of all monthly fixed effects in 2015 as the baseline rather than a single month (e.g., January) to ensure that the normalization is not driven by potentially anomalous price levels in any particular month. This averaging procedure increases the stability of the reference point and reduces the mechanical influence of month-specific shocks (e.g., a particularly illiquid January or a temporarily inflated summer market) on the index construction.

Step 2: Index Normalization

The final House Price Index (HPI_{lt}) is obtained by re-scaling the current period's estimate relative to the local 2015 baseline. While the annual index is standardized to the 2015 average, for the monthly index, January 2015 is chosen as the baseline for standardization. The final House Price Index (HPI_{lt}) is obtained by re-scaling the current period's estimate relative to the local 2015 baseline. This ensures that the average index value for every area in Jan 2015 equals 100:

$$HPI_{lt} = 100 \times \frac{\exp(\hat{\delta}_{lt})}{\bar{\delta}^{avg}_{l,2015}} \tag{9}$$

This transformation preserves the relative growth dynamics captured by the regression while anchoring the levels to a common baseline, allowing for a consistent evaluation of price deviations relative to the Jan 2015 benchmark. Same methodology is applied to the national index HPI_t (same formula, without location subscript).

4.3.7 Growth estimation

The growth rate is consistently calculated as a year-over-year percentage change in the index level. In the monthly indices, the current month is compared (t) to the same month in the previous year:

$$Growth_{l,t} = \left(\frac{HPI_{l,t}}{HPI_{l,t-12}} - 1 \right) \tag{10}$$

An analogous methodology is applied to construct the National Index, with the distinction that the estimation relies solely on the time dimension (t) rather than location-by-time interactions (l, t) and that in the denominator HPI has lag one (the previous year is subtracted).

4.3.8 Alternative Specification: Hedonic Approach

A hedonic specification provides an alternative to the repeated-sales framework. Rather than tracking properties sold twice, this approach uses the full transaction sample and controls for observable housing characteristics. The estimating equation for the local area $\times$ month index is:

$$\ln(P_{hlt}) = \mathbf{X}_{hlt}\beta + \delta_{lt} + u_{hlt}, \tag{11}$$

where $\mathbf{X}_{hlt}$ collects the standard hedonic controls[19] entered as linear, squared, and cubic polynomials to allow for non-linear effects, and δ_{lt} is the area-by-month fixed effect, which captures the quality-adjusted price level of a standardized home in area l at month t. An analogous national specification is obtained by replacing the area-by-time fixed effect with a single time fixed effect common to all properties. Recovery of price levels through exponentiation of the time fixed effects, normalization to a January 2015 base, and the computation of year-on-year growth rates follow the same logic as in the repeated-sales index.

[19]Living area, rooms, bathrooms, and building age; these are the same controls used by Guren (2018).

The hedonic series will not be studied throughout the thesis discussion as it diverges from both the repeated-sales estimates and the official Statistics Denmark benchmark, and outlier handling within the cross-sectional regressions is methodologically problematic: trimming requires identifying anomalous properties before the index is estimated. The hedonic specification is therefore reported as a robustness reference and not retained for the bubble-detection and contagion analysis.

4.3.9 Summary

The methodological procedures detailed throughout this section culminated in the construction of two primary monthly indices and three annual indices (used for robustness checks), which serve as the foundation for the subsequent empirical analysis:

- **The Real Monthly x Area Index:** A geographically granular index estimated at the custom area level, constructed strictly utilizing ex-ante real transaction prices with active outlier removal.
- **The Real Monthly x National Index:** An aggregate macroeconomic index, estimated applying the identical specification of ex-ante real transaction prices and active outlier removal.
- **Three Yearly x National indices:** Aggregate macroeconomic indices: one uses hedonic and two use repeated sales methodologies. These specifications are estimated by applying the identical specification of ex-ante transaction prices. The hedonic model does not remove outliers, whereas one version of the repeat-sales model excludes outliers and the other retains them. The same methodology described above is used, but the time variable is specified in years.

4.3.10 Robustness Checks

Comparison with the Official DST Index

To determine which index methodology to retain for the core empirical analysis, a formal evaluation criterion was established based on benchmarking the estimates against official registry data[20].

To ensure perfect comparability with the custom-built indices, two essential transformations were applied to the official DST series. First, because the DST index is standardized to a base year of 2022, it was re-based to 2015=100. Second, as the official index tracks nominal prices, it was deflated into a real price index using the exact same Consumer Price Index (CPI) series applied to the dataset.

The comparative analysis is designed to assess whether the constructed indices can serve as a reliable input for the empirical stages that follow. It is benchmarked against the official Statistics Denmark (DST) series, under the working hypothesis that a valid index should reproduce both the long-term trend and the timing of the main turning points observed in the official figures. The hedonic specification is included alongside the repeat-sales variants as a comparative reference: if it aligns with DST to a similar degree, it represents a viable alternative; if not, the repeat-sales approach becomes the more defensible choice.

[20] The chosen benchmark was the official Statistics Denmark (DST) index, specifically table EJ67: specifically the settings used were: "All Denmark, One family House (more representative) and index". See "StatBank Denmark table EJ67" link in References.

A second diagnostic concerns sensitivity to outlier treatment. The repeat-sales index is re-estimated with and without trimming of extreme price ratios between paired transactions, under the expectation that a robust specification should deliver substantially similar trajectories under both protocols. Outlier handling in the hedonic framework, by contrast, raises a methodological concern that is independent of any empirical outcome: trimming would require identifying anomalous properties prior to the cross-sectional regressions, that is, before the price index is even estimated.

The final choice between specifications rests on the joint outcome of these two diagnostics. The corresponding figures and quantitative comparisons are reported in the Results section.

Deflation Timing and Outlier Sensitivity

To ensure the robustness of the constructed Repeated sales indices and verify its consistency of the data transformations, an additional validation analysis was conducted. For this specific check, the standard data-cleaning procedure that drops outliers was bypassed to observe the raw behavior of the dataset. Two distinct sequences were then compared of adjusting for inflation:

1. **Ex-ante Deflation:** Estimating the index using transaction prices that were converted into real terms *prior* to the regression analysis (the baseline approach).
2. **Ex-post Deflation:** Estimating the index using strictly nominal transaction prices, and subsequently deflating the resulting nominal index values into real terms only *after* the regression stage was complete.

To evaluate the discrepancy between these two methodological paths, the percentage deviation was calculated between the two resulting indices using the following formula:

$$Percentage\_Deviation = \left(\frac{PriceIndex_{Nom\_Adj}}{PriceIndex_{Real}} - 1 \right) \times 100$$

The theoretical expectation (especially when not dropping outliers) is that the timing of the deflation adjustment, whether applied to individual transaction prices beforehand or to the aggregate index afterward, should yield a deviation closely approaching zero. The empirical observations perfectly align with this expectation when utilizing the full, unpruned dataset. Under these baseline conditions, the calculated percentage deviations are negligible across the timeline, confirming that the indices are mathematically robust and structurally consistent in the absence of outlier trimming.

4.4 Bubble detection and date-stamping

After creating the monthly inflation-adjusted repeated-sales price indices, periods are identified in which the real price indices display bubble-like behaviour and delimit their duration. The same procedure is applied to the aggregate repeated-sales index and to each of the 16 area-level repeated-sales indices. The procedure classifies months using a reduced-form rule based on two conditions: statistically significant explosive dynamics and a price level that is above a one-sided long-run trend.

The statistical component is the Phillips-Shi-Yu (PSY) procedure, which uses recursive right-tailed augmented Dickey-Fuller statistics to identify episodes of explosive behaviour. For date-stamping, the Backward Supremum Augmented Dickey-Fuller statistic (BSADF) is used. A month receives a PSY signal when the BSADF statistic for that endpoint exceeds its Monte Carlo simulated 95% critical value. This follows the date-stamping logic in Phillips, Shi and Yu (2015a, 2015b).

This PSY signal alone is not used as the final bubble indicator. In finite samples, recursive tests can occasionally detect local explosiveness during sharp recoveries from depressed levels or during short noisy movements. For this reason, the PSY signal is combined with an above-trend condition based on a one-sided Hodrick-Prescott filter and with a minimum-duration rule. These additional rules do not replace the PSY test, but they restrict the set of detected bubbles to episodes that are both locally explosive and sufficiently elevated relative to a slow-moving trend.

4.4.1 Overview of the implemented procedure

The procedure used in the empirical analysis is summarized in Table 1. Each step is applied separately to each real repeated-sales index. All transformations and filters used in the bubble rule are based on information available up to the month being classified, except for the simulated critical values generated by the testing procedure.

Table 1: Overview of the implemented bubble-detection procedure

Step	Implemented choice	Purpose
1	Take natural logarithms of the real repeated-sales index, defining $y_{i,t} = \log(P_{i,t})$.	Work with proportional price movements rather than index levels.
2	Compute a backward-looking three-month moving average of $y_{i,t}$, denoted $x_{i,t}$.	Reduce short-term index noise without using future observations.
3	Estimate recursive right-tailed ADF regressions on $x_{i,t}$ with zero ADF lags.	Generate the window-specific ADF statistics used by the PSY procedure.
4	Construct the BSADF sequence and the GSADF statistic.	Use GSADF to assess full-sample evidence of explosiveness and BSADF to date-stamp monthly episodes.
5	Compare each $BSADF_{i,t}$ with its simulated 95% critical value.	Define the monthly PSY signal for statistically significant explosive behaviour.
6	Compute a one-sided HP trend on $x_{i,t}$ with $\lambda =$ 1,000,000.	Estimate a slow-moving trend benchmark using only current and past observations.
7	Define an HP signal when $x_{i,t}$ is at least 0.04 log points above the one-sided HP trend.	Require the price level to be high enough.
8	Define a candidate bubble month only when both the PSY and HP signals are equal to one.	Combine the two criteria for a more robust indicator.
9	Keep only candidate episodes lasting at least 12 consecutive months.	Remove short runs that are unlikely to represent persistent exuberance episodes.

4.4.2 Data transformation

Let $P_{i,t}$ denote the real repeated-sales price index for area i in month t. The first transformation done to the index is to take its natural logarithm, $y_{i,t} = \log(P_{i,t})$. Using logarithms matches the way recursive explosive-root tests are commonly applied to asset prices. Phillips, Wu and Yu (2011), for example, work with logarithmic real Nasdaq prices, and Phillips, Shi and Yu (2015a) discuss the use of logarithmic prices under a log-linear asset-pricing approximation.

Testing log prices detects explosive behaviour in the price process. It is not a direct test of overvaluation relative to rents, income, user costs, or any other measured fundamental. The maintained assumption is that the relevant fundamentals are not themselves explosive, which is deemed justifiable at least for rental data. Under that assumption, explosiveness in real prices is difficult to reconcile with a standard no-bubble benchmark. If an omitted fundamental were also explosive, the test would not separate that mechanism from non-fundamental exuberance.

The PSY and HP components are not applied to the raw logged series $y_{i,t}$. They are applied to a short backward-looking moving average of $y_{i,t}$. Denote this transformed series by $x_{i,t}$. From the third month onward, it is defined as $x_{i,t} = \frac{y_{i,t}+y_{i,t-1}+y_{i,t-2}}{3}$.[21]

The purpose of the moving average is to reduce short-term irregularities in the constructed indices. Area-

[21] For the first available observation, the transformed value is the logged index itself; for the second observation, it is the average of the first two logged values.

level repeated-sales indices move unevenly from month to month because the number and composition of transactions varies over time, and it is usually low enough that there is evident measurement error coming from the construction of the indices. Applying the PSY procedure directly to the raw monthly series would make isolated observations more influential, potentially detecting spurious bubbles and failing to detect actual ones, that would get interrupted by the artificial noise. A three-month moving average is short enough to preserve the monthly timing of bubble episodes, but long enough to reduce the effect of a single irregular month.

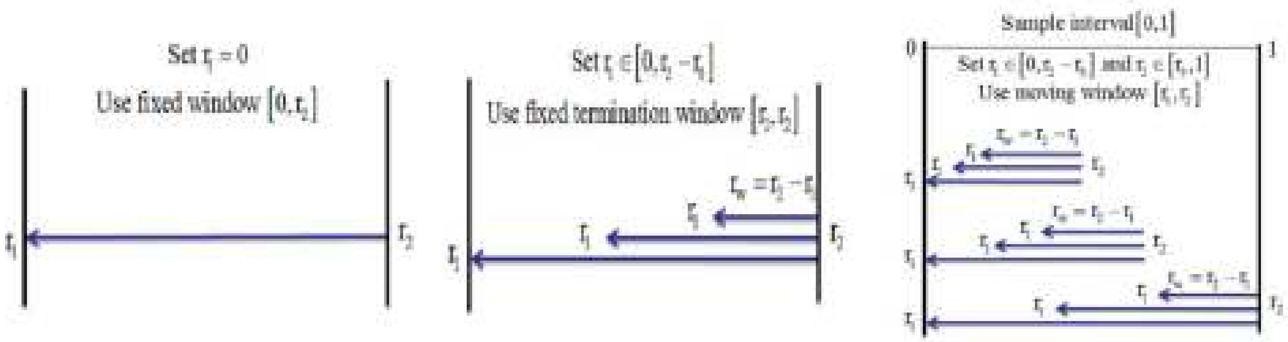
Alternative smoothing choices would affect the results. With no smoothing, the procedure would be more sensitive to isolated spikes. A longer moving average would produce a smoother series, but it would also attenuate abrupt increases and could delay the estimated start and end of episodes, jeopardizing bubble dating too much. A centred moving average is not used because it would include future observations in the value assigned to a given month. That would be inconsistent with the real-time logic of the date-stamping exercise.

4.4.3 PSY test and date-stamping

The PSY procedure (Phillips, Shi and Yu, 2015a) is based on right-tailed unit-root testing. In each regression window, the transformed series $x_{i,t}$ is estimated using an Augmented Dickey-Fuller (ADF) specification of the form:

$$\Delta x_{i,t} = \alpha + \beta x_{i,t-1} + \sum_{j=1}^{k} \phi_j \Delta x_{i,t-j} + \varepsilon_{i,t}. \tag{12}$$

The null hypothesis is $\beta = 0$, corresponding to a unit-root process. Usually, ADF tests are used with a left-tailed alternative hypothesis ($\beta < 0$) to detect stationarity (Dickey and Fuller, 1979). However, in this procedure, the right-tailed alternative ($\beta > 0$) is used instead, interpreting rejection as evidence of explosive behaviour.



(a) ADF test (b) BSADF test (c) GSADF test

Figure 3: Recursive window structures for ADF, BSADF and GSADF tests. From Phillips, Shi and Yu (2015a, Figures 3 and 4).

The test statistic depends on the sample window used to estimate the regression. Let $ADF(r_1, r_2)$ denote the right-tailed ADF statistic computed over the subsample beginning at fraction r_1 of the sample and ending at fraction r_2. The Backward Supremum Augmented Dickey-Fuller statistic (BSADF) statistic at r_2 is the largest ADF statistic (across all backward windows larger than the minimum window r_0) ending at r_2:[22]

$$BSADF_{r_2}(r_0) = \sup_{r_1 \in [0,\, r_2 - r_0]} ADF(r_1, r_2). \tag{13}$$

The Generalized Supremum Augmented Dickey-Fuller (GSADF) statistic consists of the largest of all of the BSADF statistics:

$$GSADF(r_0) = \sup_{r_2 \in [r_0, 1]} BSADF_{r_2}(r_0). \tag{14}$$

While there is a BSADF value for each endpoint r_2, there is only one GSADF statistic per time series. Therefore, the GSADF tests whether the series contains explosive behaviour somewhere in the sample, while the sequence of BSADFs is used to date-stamp the episodes of exuberance.

4.4.4 Lag length in the ADF regressions

In the empirical implementation, $k = 0$ is set in the ADF regressions. This is a parsimonious choice, justified by Phillips, Shi, and Yu (2015a), where they recommend using a fixed, low lag length. It means that the test regression includes the lagged level of the transformed series but does not include additional lagged differences (this amounts to doing non-augmented Dickey-Fuller regressions instead of augmented ones). There are two reasons for this decision. First, the input series has already been smoothed with a short backward-looking moving average, which reduces the high-frequency irregularities that additional ADF lags would partly absorb. Second, recursive tests use many short subsamples. Adding lags consumes observations in every window and can reduce the ability to detect explosive behaviour in short episodes.

The choice of lag length is not neutral. Too few lags can leave serial correlation in the residuals and affect the size of unit-root tests. Too many lags can reduce power and, in recursive procedures, may change both the presence and timing of detected episodes. Schwert (1989) discusses the sensitivity of unit-root tests to lag specification. Phillips, Shi and Yu (2015a) also report that overparameterized lag choices can affect finite-sample behaviour, especially for the GSADF tests. The zero-lag specification is therefore used as a transparent, parsimonious baseline but it is not a claim that no other alternative lag choice would be acceptable.

[22] The minimum window r_0 is set using the following recommended rule: $r_0 = 0.01 + 1.8/\sqrt{T}$ (Phillips, Shi and Yu, 2015a), where T is the number of observations.

4.4.5 Critical values and PSY signal

The BSADF and GSADF statistics have non-standard distributions because they are obtained from repeated right-tailed ADF regressions over recursively selected windows (Phillips, Shi and Yu, 2015a). Standard Dickey-Fuller critical values are therefore not appropriate. The relevant critical values must account for the recursive construction of the statistics, the minimum window size, and the fact that the statistics are defined as suprema over admissible subsamples. In this study, the analysis uses the Stata `radf` command, which, as Baum and Otero (2021) explain, uses precomputed critical values from 2,000 replications using the minimum-window rule suggested in the original PSY paper.[22] Therefore, the same critical values are used for all the indices.

The simulated null process used to obtain these critical values is a non-explosive unit-root process with an asymptotically negligible drift. A simulated path for index i and Monte Carlo replication q can be written as:

$$\widetilde{x}_{i,1}^{(q)} = T^{-1}, \tag{15}$$

$$\widetilde{x}_{i,s}^{(q)} = T^{-1} + \widetilde{x}_{i,s-1}^{(q)} + u_{i,s}^{(q)}, \qquad s = 2, \dots, T, \qquad u_{i,s}^{(q)} \sim N(0,1), \tag{16}$$

where T is the total number of monthly observations in the sample, and $q = 1, \dots, Q$ indices the Monte Carlo replication. The small drift term follows the implementation described by Baum and Otero (2021) for the PSY critical values and is asymptotically negligible.

For each simulated path, the same recursive ADF procedure is applied as in the observed data. For any fixed endpoint t, the ADF regression is estimated repeatedly over all admissible windows ending at t. The largest of these right-tailed ADF statistics gives the simulated backward statistic $\widetilde{BSADF}_{i,t}^{(q)}$. Repeating this for every endpoint produces a full simulated BSADF sequence. The simulated GSADF statistic is then obtained as the largest value of that sequence. The empirical distribution of the Q simulated $\widetilde{GSADF}_i^{(q)}$ values gives the GSADF critical values used to assess whether the full series contains evidence of explosive behaviour somewhere in the sample.

For date-stamping, however, the relevant comparison is endpoint-specific: each observed $BSADF_{i,t}$ is compared with the corresponding simulated distribution of $\widetilde{BSADF}_{i,t}^{(q)}$ across replications. While the GSADF comparison uses one critical value for the full sample, the BSADF date-stamping exercise uses a sequence of critical values indexed by t. For a given endpoint t, the 95% BSADF critical value is the empirical 95th percentile of the simulated BSADF statistics at that endpoint. Formally,

$$I_{i,t}^{(q)}(c) = \begin{cases} 1, & \text{if } \widetilde{BSADF}_{i,t}^{(q)} \leq c, \\ 0, & \text{otherwise.} \end{cases} \tag{17}$$

And then the simulated 95% critical value is:

$$cv_{0.95,i,t}^{BSADF} = \inf\left\{ c : \frac{1}{Q}\sum_{q=1}^{Q} I_{i,t}^{(q)}(c) \geq 0.95 \right\}. \tag{18}$$

This expression states that the critical value is the smallest value for which at least 95% of the simulated BSADF statistics lie below it. The monthly PSY signal is defined by comparing the observed BSADF statistic with the corresponding simulated critical value:

$$PSY\_signal_{i,t} = \begin{cases} 1, & \text{if } BSADF_{i,t} > cv_{0.95,i,t}^{BSADF}, \\ 0, & \text{otherwise.} \end{cases} \tag{19}$$

4.4.6 One-sided HP trend gap condition

The second component of the classification rule is a price level condition. A month is not classified as a bubble just because the PSY statistic indicates local explosive behaviour. The price level must also be high enough that the transformed index is above a slow-moving trend. The trend is estimated using a one-sided Hodrick-Prescott filter applied to the transformed series $x_{i,t}$.

The standard HP filter decomposes a series into a trend component τ_t and a cyclical component by choosing a smooth trend that balances closeness to the data against variation in the trend growth rate. In compact form, the trend is the solution to

$$\min_{\{\tau_t\}_{t=1}^{T}} \sum_{t=1}^{T}(x_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} \left[(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1})\right]^2. \tag{20}$$

The usual HP filter is two-sided: the trend estimated for month t depends on observations after t. This is undesirable in a date-stamping exercise because future observations would influence whether a month is considered above trend. A one-sided HP trend is therefore constructed recursively. For each month t, the HP filter is estimated only on the subsample from the first available observation to t, and the trend value for the final observation of that subsample is stored as the one-sided trend for month t. This is repeated for each month.

$$\text{one-sided trend}_{i,t} = \text{last HP trend estimate from the sample } 1, \ldots, t. \tag{21}$$

The one-sided version avoids incorporation future information, but it must be said that it does not remove all limitations of the HP filter. HP trends depend on the smoothing parameter, and endpoint estimates can be sensitive. Hamilton (2018), for example, is particularly critical of the HP filter for business-cycle applications, including its end-of-sample behaviour. In this thesis, however, the HP filter is not used to infer cyclical dynamics or to estimate a structural fundamental value. It is used only as a transparent trend benchmark for an above-trend screening rule.

The smoothing parameter is set to $\lambda = 1{,}000{,}000$. The value is deliberately high because the trend is intended to capture slow-moving movements in housing prices rather than medium-frequency fluctuations. Goodhart and Hofmann (2008) identify house-price booms using deviations from a one-sided HP trend with a smoothing parameter of 100,000 for quarterly data. For comparison, a common convention in macroeconomic applications is to use $\lambda = 1{,}600$ for quarterly data and $\lambda = 14{,}400$ for monthly data (Hodrick and Prescott, 1997; Rebelo, 2005; Zarnowitz and Ozyildirim, 2006), which corresponds to multiplying the quarterly value by 3^2. Applying the same frequency adjustment to Goodhart and Hofmann's quarterly value gives $100{,}000 \times 3^2 = 900{,}000$, which is close to the value used here. The selected value can therefore be interpreted as a high monthly smoothing parameter in the same order of magnitude as a frequency-adjusted version of Goodhart and Hofmann's quarterly choice.

The choice of λ affects the classification. A lower value would allow the trend to follow the price index more closely, reducing the measured gap and making it harder for a sustained increase to satisfy the HP condition. A higher value would make the trend smoother and would generally increase the gap during prolonged booms, but could also make the rule too permissive if long-run structural changes are interpreted as deviations from trend. The selected value is therefore a compromise: it creates a slow-moving trend while still allowing the trend to adjust gradually over long samples.

The HP gap is defined as the difference between the transformed index and the one-sided HP trend:

$$gap_{i,t} = x_{i,t} - \tau_{i,t}^{1s}. \tag{22}$$

The HP signal is equal to one when this gap is at least 0.04 log points:

$$HP\_signal_{i,t} = \begin{cases} 1, & \text{if } gap_{i,t} \geq 0.04, \\ 0, & \text{otherwise.} \end{cases} \tag{23}$$

Because the variables are in logarithms, 0.04 log points corresponds to roughly 4% above the one-sided trend. This threshold value is close to the one used in Goodhart and Hofmann (2008), but given that in the case it is not the only signal used, a slightly less strict value was chosen instead.

4.4.7 Final bubble indicator

An initial joint signal is created as the intersection of the PSY signal and the HP signal:

$$candidate\_bubble_{i,t} = PSY\_signal_{i,t} \times HP\_signal_{i,t}. \tag{24}$$

A month is therefore a candidate bubble month only if it satisfies both conditions. The PSY component captures local explosiveness in the transformed price process. The HP component requires the price level to be above a slow-moving trend by at least 0.04 log points. The two components serve different purposes and are not interchangeable. A month can be above trend without being explosive, and a month can be explosive without being sufficiently above trend. Only their intersection enters the final classification.

The final step removes candidate episodes shorter than 12 consecutive months. This persistence filter is consistent with Phillips, Shi and Yu (2015a), who impose a minimum duration condition in their date-stamping procedure to avoid classifying very short-lived exceedances as bubble episodes, and with Goodhart and Hofmann (2008), who require housing booms to remain above trend for at least 12 quarters.

Let a candidate episode be a consecutive run of months for which $candidate\_bubble_{i,t} = 1$. If the run lasts fewer than 12 months, all months in that run are reclassified as non-bubble months. If the run lasts exactly 12 months or longer, all months in the run are retained. The final indicator is therefore

$$final\_bubble_{i,t} = \begin{cases} candidate\_bubble_{i,t}, & \text{if its run length } \geq 12, \\ 0, & \text{otherwise.} \end{cases} \tag{25}$$

The purpose of the duration requirement is to remove isolated or very short signals. Recursive tests can produce short exceedances around volatile periods, and the HP gap can also cross the threshold briefly. A twelve-month minimum requires the two conditions to agree for at least a full year. A shorter duration rule would increase the number of detected episodes and would be more sensitive to short price surges. A longer rule would focus only on the most persistent episodes but could remove genuine, shorter-lived episodes. The twelve-month rule is chosen as a balance between excluding temporary movements and retaining meaningful monthly variation in the timing of housing-market episodes.

4.4.8 Consequences of alternative modelling choices

The final classification depends on several modelling choices. The baseline values are fixed before the empirical comparison across areas, so all indices are treated symmetrically. Table 2 summarizes how alternative choices would generally affect the results.

Table 2: Consequences of alternative modelling choices

Choice	Baseline	Likely consequence of alternatives
Price transformation	Use logged real repeated-sales indices.	Raw levels would make changes less comparable across normalized indices. Valuation ratios would require consistent monthly fundamentals and would answer a different question.
Moving average	Backward-looking three-month average.	No smoothing would increase sensitivity to isolated observations. Longer smoothing would reduce noise further but could delay or attenuate detected episodes.
ADF lags	Zero lags.	More lags may reduce residual autocorrelation, but they consume observations in recursive windows and may reduce power or shift estimated dates.
PSY critical-value level	95% BSADF critical value.	A lower threshold would classify more months as explosive. A higher threshold would be more conservative and could shorten or eliminate borderline episodes.
Trend filter	One-sided HP filter.	A two-sided filter would use future observations and could alter historical start and end dates retrospectively.
HP smoothing	$\lambda = 1{,}000{,}000$.	Lower λ would produce a more flexible trend and smaller gaps. Higher λ would produce a smoother trend and generally larger gaps during sustained booms.
HP gap threshold	0.04 log points.	A smaller threshold would admit weaker above-trend movements. A larger threshold would focus on more pronounced deviations from trend.
Duration rule	At least 12 consecutive months.	A shorter rule would retain more temporary signals. A longer rule would retain only more persistent episodes.

The final variable $final\_bubble_{i,t}$ is a binary indicator equal to one in months classified as bubble months by the implemented rule and zero otherwise. It should be interpreted as a model-based classification of explosive, above-trend housing price episodes. It is not direct evidence that prices exceed an observed structural fundamental value. This distinction follows from the design of the procedure: the PSY component tests for explosive dynamics, while the HP component imposes an above-trend price-level condition. Neither component estimates fundamental value.

The advantage of this approach is that it produces comparable dates across all indices using the same rule. The cost is that the classification depends on maintained assumptions and tuning parameters. The results should therefore be interpreted as evidence of bubble-like episodes under the specified reduced-form criterion. The use of common settings across all indices is essential for comparability: differences in detected episodes across areas reflect differences in the behaviour of the transformed indices, not area-specific changes in the detection rule.

4.5 Contagion methodology

This section explains the methodology used to study how bubble conditions move across Danish housing markets after the first-stage detection procedure has already produced monthly BSADF paths for each area. The aim is to show how those BSADF paths are taken into a second-stage contagion model, how the tuning parameters of that model are selected, how Copenhagen and Aarhus are treated as candidate core markets, and how bootstrap inference is then built for the resulting time-varying contagion coefficients. The broad modelling framework follows Greenaway-McGrevy and Phillips (2016), while the bootstrap logic follows Phillips and Shi (2020). In both cases, however, the empirical design is adapted to the data and to the practical limits of this setting.

The section is organized as follows. It begins by defining the measure of explosiveness that enters the contagion model and by setting out the time-varying regression that links a destination area to a candidate core area. It then explains why Copenhagen and Aarhus are treated as the candidate cores. The next part turns to the local Gaussian-kernel estimator, the role of the bandwidth and the lag parameter, and the first-stage selection of those parameters for all ordered area pairs. The final part describes the bootstrap, the way in which bootstrap BSADF paths are constructed, and the way in which those paths are propagated through the contagion model to obtain standard errors and confidence intervals for the time-varying contagion coefficient.

4.5.1 Measure of explosiveness entering the contagion model

Greenaway-McGrevy and Phillips (2016) do not take the original price-rent ratio directly into the second-stage contagion regression. Their starting point is a recursive sequence of slope estimates obtained from a right-tailed AR(1) equation for each region. For area i, they begin from the regression

$$\Delta y_{i,t} = \alpha_i + \beta_i y_{i,t-1} + e_{i,t}, \qquad t = 1, \dots, T. \tag{26}$$

In their setting, equation (26) is estimated repeatedly over recursive subsamples, so the output is not one full-sample coefficient but a time path of recursive estimates, denoted $\hat{\beta}_{i,s}$. The subscript s indices the endpoint of the recursive estimation window. These $\hat{\beta}_{i,s}$ paths are then treated as summary measures of local explosiveness in each region.

The second-stage contagion equation in Greenaway-McGrevy and Phillips (2016) is therefore written in terms of those recursive slope estimates:

$$\hat{\beta}_{j,s} = \delta_{1j} + \delta_{2j}\left(\frac{s}{T-S+1}\right)\hat{\beta}_{\text{core},s-d} + u_{j,s}, \qquad s = S, \dots, T, \quad j \neq \text{core}. \tag{27}$$

Equation (27) is the starting point for the contagion framework. The subscript j denotes the destination region, while core denotes the region taken as the possible source of contagion. The coefficient δ_{1j} is an intercept, and $\delta_{2j}(\cdot)$ is the time-varying response of region j to the lagged core-market measure of explosiveness. The argument $s/(T - S + 1)$ rescales calendar time to the unit interval. The delay parameter d allows the effect of the core market on region j to appear with a lag rather than contemporaneously.

The empirical design keeps the broad logic of equation (27), but changes the variable that enters it. Instead of using recursive $\hat{\beta}$-paths, the monthly BSADF paths are used already produced in the first-stage bubble-detection procedure. This change needs to be made explicit from the start, because it affects the meaning of every later equation. The BSADF path is the chosen object here for three reasons. First, it is the recursive measure of explosiveness already used to date bubble episodes. Second, it is already available month by month for every area from the first-stage exercise, so no further recursive AR(1) path needs to be constructed. Third, it is the object for which the implementation can produce bootstrap replications in a direct and coherent way.

From now on in this section, for a given area i and month t, $b_{i,t} = BSADF_{i,t}$.

The contagion model in this thesis therefore asks whether the BSADF path in a destination area tends to move with the lagged BSADF path in a candidate core area, allowing that relation to vary over time.

In Greenaway-McGrevy and Phillips (2016), $\hat{\beta}_{i,s}$ is a recursively estimated slope coefficient from an AR(1) equation. In this setting, $b_{i,t}$ is the recursively constructed BSADF statistic itself. The two objects are related, because both come from recursive right-tailed testing, but they are not the same object and should not be confused.

4.5.2 Time-varying contagion equation

Let c denote the candidate core region and j a destination region. Let t index the months that remain once the largest admissible lag has been taken into account. The second-stage equation is written as a time-varying regression in which the destination-area BSADF path depends on a lagged BSADF path from the core area:

$$b_{j,t} = \alpha_j(t) + \delta_{2j}(t) b_{c,t-d_j} + u_{j,t}. \tag{28}$$

In equation (28) the coefficient $\alpha_j(t)$ is a time-varying intercept that captures slow changes in the average level of the destination-area BSADF path, while the coefficient $\delta_{2j}(t)$ is the main object of interest. A positive value for $\delta_{2j}(t)$ means that months in which the core market exhibits stronger explosiveness are associated with stronger explosiveness in region j. A value close to zero means that the destination-area BSADF path is locally unrelated to the lagged BSADF path of the core market. A negative value means that the two move in opposite directions locally.

In Greenaway-McGrevy and Phillips (2016), this equation is fed with recursive $\hat{\beta}$-paths. In the implementation, the same idea is applied to the BSADF paths $b_{j,t}$ and $b_{c,t-d_j}$ instead. Before estimating the time-varying slope, both series are first demeaned by subtracting their means over the whole sample:

$$\tilde{b}_{j,t} = b_{j,t} - \frac{1}{N}\sum_{s=1}^{N} b_{j,s}, \tag{29}$$

$$\tilde{b}_{c,t-d} = b_{c,t-d} - \frac{1}{N}\sum_{s=1}^{N} b_{c,s-d}. \tag{30}$$

Here $\tilde{b}_{i,t}$ denotes the demeaned BSADF path for area i at month t. Notice that the effective estimation sample is shorter than the full monthly sample whenever positive lags are allowed. In the implementation, as in Greenaway-McGrevy and Phillips (2016), the lag parameter is searched over the integers $d \in \{0, 1, \ldots, 12\}$. The estimation sample therefore starts after the twelfth available lag, so that all candidate delays are evaluated on the same set of destination-month observations. With T denoting the total number of monthly observations in the BSADF path and $d_{\max} = 12$ being the maximum admissible lag, $N = T - d_{\max}$ is defined.

This detail is important. If different lag candidates were compared on different sample lengths, the choice of lag would partly reflect changing sample size rather than only the fit of the local contagion equation.

4.5.3 Choice of candidate core markets

Greenaway-McGrevy and Phillips (2016) treat Auckland as the single core market because it is the country's largest urban centre and because their first-stage evidence suggests that exuberance there appears earlier and more strongly than elsewhere. The Danish data do not point as clearly to one unique leading market. As shown in the results of the bubble detection phase, Aarhus sometimes appears to lead other areas, but the evidence for it is not strong enough to justify treating Aarhus alone as the undisputed source market.

For that reason, the contagion analysis is not organised around a single core region. Instead, Copenhagen and Aarhus are treated as two candidate cores. Copenhagen is the largest and most economically important city in the country, while Aarhus is the second largest urban centre and the strongest alternative candidate on observational grounds. The methodology therefore estimates separate contagion paths from each of these two cities to every other area. This produces 30 ordered pairs: Copenhagen to the other 15 areas, and Aarhus to the other 15 areas.

This should be understood as a hypothesis about possible centres of geographic propagation, made before looking at the second-stage contagion results.

4.5.4 Local Gaussian-kernel estimation of the contagion path

The coefficient $\delta_{2j}(t)$ is allowed to vary smoothly over time rather than jump between a small number of regimes. Greenaway-McGrevy and Phillips (2016) estimate this coefficient by local level kernel regression. The implementation uses the same estimator, with the destination and source variables replaced by the demeaned BSADF paths defined in equations (29) and (30). The time-varying estimation procedure described in this section and in the next one is based fully on Annex 5.1.2 of Greenaway-McGrevy and Phillips (2016).

Let $r_t = t/N$ denote rescaled time. For a given bandwidth h and lag d, the local estimate of the contagion coefficient at rescaled time r is

$$\hat{\delta}_{2j}(r; h, d) = \frac{\sum_{s=1}^{N} K_{hs}(r)\tilde{b}_{j,s}\tilde{b}_{c,s-d}}{\sum_{s=1}^{N} K_{hs}(r)\tilde{b}_{c,s-d}^2}. \tag{31}$$

Equation (31) is a weighted least-squares slope estimator computed locally around the date represented by r. The numerator is the weighted local covariance between the destination-area and lagged core-area BSADF paths. The denominator is the weighted local sum of squares of the lagged core-area path. Their ratio therefore gives the local slope that best fits the destination-area demeaned BSADF path to the lagged core-area demeaned BSADF path around that date. The estimate is local because observations close to r receive more weight than observations further away.

The weights are generated by a kernel function. In the implementation, as in Greenaway-McGrevy and Phillips (2016), a Gaussian kernel is used:

$$K_{hs}(r) = \frac{1}{h}K\left(\frac{s/N - r}{h}\right), \tag{32}$$

$$K(u) = (2\pi)^{-1/2} \exp\left(-\frac{u^2}{2}\right). \tag{33}$$

The kernel determines how much influence each monthly observation s has on the estimate at date r. Observations with s/N close to r receive the largest weights. Observations further away still receive positive weight under the Gaussian kernel, but their contribution becomes very small. The bandwidth h controls the width of this local neighbourhood. A small h produces a very local estimate that reacts sharply to short-lived changes but can be noisy. A large h produces a smoother coefficient path but at the cost of averaging over more distant periods and therefore masking short-lived changes in contagion.

The denominator in equation (31) also matters. If the lagged core-area BSADF path were locally almost constant, the denominator could become very small and the local slope could become unstable. The implementation therefore applies a simple numerical safeguard and only computes the slope where the local denominator is sufficiently far from zero[23]. However, this edge case never occurs in practice

[23]The chosen threshold is 10^{-12}.

in the data.

A useful way to read equation (31) is to compare it with an ordinary regression slope. If every observation had the same weight, the estimator would collapse to a full-sample slope using the demeaned BSADF paths. The kernel just changes the way information is weighted across time. The resulting coefficient path $\hat{\delta}_{2j}(r; h, d)$ can rise, fall, flatten, or change sign over time, so that it can be measured if the strength of contagion is stronger in some periods than in others.

4.5.5 Selection of the lag and bandwidth parameters

The local estimator in equation (31) depends on two tuning parameters: the delay d and the bandwidth h. Greenaway-McGrevy and Phillips (2016) choose them jointly in a two-step way. First, for each admissible lag d, they choose the bandwidth by cross-validation. Second, they choose the delay that gives the best fit once the bandwidth has been chosen conditional on that lag. The same logic is used here, although for the BSADF paths instead of their recursive $\hat{\beta}$-paths.

For each destination region j and candidate delay d, the bandwidth is searched over the interval

$$H_T = \left[N^{-1/2}, N^{-1/10}\right]. \tag{34}$$

This interval shrinks with the sample size, but not too quickly. The lower bound $N^{-1/2}$ prevents the bandwidth from collapsing to a value that would make the estimate depend on very few points. The upper bound $N^{-1/10}$ prevents the bandwidth from becoming so large that the estimate is almost global. In the implementation, this interval is discretised into a finite grid of candidate bandwidths. More specifically, the search is carried out over 40 equally spaced values between the two bounds.

For each candidate bandwidth h in H_T and each candidate delay d, the bandwidth is evaluated by leave-one-out cross-validation. The idea is to estimate the local slope at each time point using all observations except the one currently being predicted, and then measure how well that local slope predicts the omitted destination-area BSADF value. Denoting the leave-one-out estimator by $\check{\delta}_{2j}(r; h, d)$, the cross-validation problem is

$$\check{h}_{jT}(d) = \arg\min_{h \in H_T} \sum_{s=1}^{N} \left[\tilde{b}_{j,s} - \check{\delta}_{2j}(r_s; h, d)\tilde{b}_{c,s-d}\right]^2. \tag{35}$$

The breve in $\check{h}_{jT}(d)$ indicates that this is a bandwidth selected conditional on a given lag d. It is not yet the final bandwidth used in the point estimate. The corresponding leave-one-out local slope is

$$\check{\delta}_{2j}(r_s; h, d) = \frac{\sum_{p=1,\, p\neq s}^{N} K_{hp}(r_s)\tilde{b}_{j,p}\tilde{b}_{c,p-d}}{\sum_{p=1,\, p\neq s}^{N} K_{hp}(r_s)\tilde{b}_{c,p-d}^2}. \tag{36}$$

Equation (36) is the same local estimator as equation (31), except that the observation being evaluated is omitted from both the numerator and the denominator. Without this omission, the criterion would partly reward bandwidths that fit each point simply because the same point enters its own local neighbourhood. Leave-one-out cross-validation instead asks which bandwidth predicts each observation best from the rest of the sample.

Once the best bandwidth has been found for each candidate lag, the lag is selected by comparing the fit across all admissible delays $d \in \{0, 1, \ldots, 12\}$. In the implementation, this comparison is done through the associated sum of squared prediction errors,

$$SSE_j(d) = \sum_{s=1}^{N} \left[\tilde{b}_{j,s} - \check{\delta}_{2j}(r_s; \check{h}_{jT}(d), d)\tilde{b}_{c,s-d} \right]^2, \tag{37}$$

and, equivalently, through the corresponding local-regression R-squared,

$$R_j^2(d) = 1 - \frac{SSE_j(d)}{TSS_j}, \qquad \text{where} \qquad TSS_j = \sum_{s=1}^{N} \tilde{b}_{j,s}^2. \tag{38}$$

Because TSS_j is fixed within a given ordered pair, maximizing $R_j^2(d)$ is equivalent to minimizing $SSE_j(d)$. The selected lag is therefore

$$\check{d}_j = \arg \max_{d \in \{0,1,\ldots,12\}} R_j^2(d). \tag{39}$$

Once $\check{d}_j$ has been chosen, the final bandwidth is the one that had previously minimized the cross-validation criterion at that lag:

$$\hat{h}_j = \check{h}_{jT}(\check{d}_j). \tag{40}$$

Note that $\check{h}_{jT}(d)$ and $\hat{h}_j$ are not the same object. The first is the bandwidth that minimizes the cross-validation criterion conditional on a particular lag d. The second is the final bandwidth retained after the lag has been chosen. However, $\hat{d}_j = \check{d}_j$.

The point estimate of the contagion path is obtained by returning to the full local estimator in equation (31) and evaluating it with these selected values:

$$\hat{\delta}_{2j}(r) = \hat{\delta}_{2j}(r; \hat{h}_j, \hat{d}_j). \tag{41}$$

This stage delivers two outputs at once: the point estimate of the time-varying contagion path and the pair-specific fixed values of $\hat{d}_j$ and $\hat{h}_j$.

The restriction of the lag search to at most 12 months is a modelling decision. The advantage is a reduction in computation time. The cost is that the procedure rules out slower forms of transmission that would take more than a year to appear.

4.5.6 Bootstrap construction of BSADF paths

Greenaway-McGrevy and Phillips (2016) report descriptive time-varying contagion paths, but do not provide formal statistical inference for them. In this setting, inference is introduced by bootstrap. The reason for bootstrapping can be stated briefly. The object entering the contagion model is itself recursive and non-standard, namely the BSADF path. As a result, simple analytical standard errors for the second-stage coefficient are not readily available. Phillips and Shi (2020) propose a bootstrap designed for the PSY procedure. That logic is adopted here and then extended to the second-stage contagion regression.

The bootstrap begins area by area, not pair by pair. Let $x_{i,t}$ denote the transformed series for area i that enters the first-stage PSY procedure. With the ADF lag length fixed at zero, the null model used for bootstrap generation is:

$$\Delta x_{i,t} = \mu_i + e_{i,t}, \qquad t = 2, \ldots, T. \tag{42}$$

Equation (42) is the no-explosiveness null used in the bootstrap data-generating step. In the implementation, it is estimated separately for each area on the transformed monthly series, and the fitted residuals $\hat{e}_{i,t}$ are stored. For each bootstrap replication q, a bootstrap innovation is then generated as

$$e_{i,t}^{(q)} = w_{i,t}^{(q)} \hat{e}_{i,\ell_t}. \tag{43}$$

In equation (43), $w_{i,t}^{(q)}$ is a random weight drawn from the standard normal distribution and $\hat{e}_{i,\ell_t}$ is a residual drawn with replacement from the residual pool of area i. Drawing a residual with replacement preserves the empirical scale of the shocks under the null, while multiplying by a random weight avoids forcing homoskedasticity. The bootstrap shock therefore remains centred at zero but can reproduce changing volatility more flexibly than a simple i.i.d. residual resample.

A point that matters for the interpretation of the second stage is that this bootstrap is implemented separately for each area. Each area has its own residual pool, its own random-weight sequence, and its own resampled innovation path. The same random weights are not imposed across areas, and a given bootstrap replication does not force the same time pattern of resampled shocks on different areas. In other words, the implementation does not construct a joint cross-sectional bootstrap under the null. Instead, it bootstraps each area-specific transformed index on its own, treating it as its own random-walk-type process under the null and reproducing uncertainty in that area's BSADF path conditional on that area's own innovation structure.

Given the bootstrap innovation, the bootstrap transformed series is generated recursively from the first observed level:

$$x_{i,1}^{(q)} = x_{i,1}, \tag{44}$$

$$x_{i,t}^{(q)} = x_{i,t-1}^{(q)} + e_{i,t}^{(q)}, \qquad t = 2, \dots, T. \tag{45}$$

Equation (45) imposes the null of no explosiveness while keeping the bootstrap sample on the same scale as the observed transformed series.[24] Once $x_{i,t}^{(q)}$ has been generated, the full BSADF sequence is recomputed for that bootstrap sample using exactly the same procedure as in the original data. Denoting the resulting path by $b_{i,t}^{(q)}$, this can be written as

$$b_{i,t}^{(q)} = BSADF(x_{i,\cdot}^{(q)})_t, \qquad q = 1, \dots, Q, \qquad i = 1, \dots, 16. \tag{46}$$

The original BSADF path is stored as replication $q = 0$, and the bootstrap replications are indexed by $q = 1, \dots, Q$, with $Q = 999$ in the implementation. The output of this area-by-area step is therefore a full collection of BSADF paths for every area: one observed path and 999 bootstrap paths. These replicated BSADF paths are then carried into the second-stage contagion model.

This is also the point at which the present methodology departs from Phillips and Shi (2020) in an important way. They use the bootstrap just to obtain critical values for the recursive test itself. Here the same resampling logic is used first to obtain the full distribution of BSADF paths, but then extended to the second step, to obtain a distribution for the $\hat{\delta}_{2j}(r)$ contagion coefficient.

4.5.7 Bootstrap contagion paths, standard errors and confidence intervals

Once the bootstrap BSADF paths have been constructed for every area, they are carried into the second-stage contagion model. For a given ordered pair $c \to j$ and a given bootstrap replication q, define the demeaned bootstrap paths in the same fashion as before:

$$\tilde{b}_{j,t}^{(q)} = b_{j,t}^{(q)} - \frac{1}{N} \sum_{s=1}^{N} b_{j,s}^{(q)}, \tag{47}$$

$$\tilde{b}_{c,t-\hat{d}_j}^{(q)} = b_{c,t-\hat{d}_j}^{(q)} - \frac{1}{N} \sum_{s=1}^{N} b_{c,s-\hat{d}_j}^{(q)}. \tag{48}$$

The bootstrap contagion coefficient is then obtained by applying the same local-kernel formula as in equation (31), but now using the fixed lag and bandwidth selected from the original data:

[24]Notice how, unlike in equation (42), there is no drift term here. Following Phillips and Shi (2020), the bootstrap recursion is generated under the null of no explosiveness, where any deterministic drift is treated as asymptotically negligible.

$$\hat{\delta}_{2j}^{(q)}(r) = \frac{\sum_{s=1}^{N} K_{\hat{h}_j s}(r)\tilde{b}_{j,s}^{(q)}\tilde{b}_{c,s-\hat{d}_j}^{(q)}}{\sum_{s=1}^{N} K_{\hat{h}_j s}(r)\left(\tilde{b}_{c,s-\hat{d}_j}^{(q)}\right)^2}. \tag{49}$$

Equation (49) is much more straightforward than the first-stage selection problem because $\hat{d}_j$ and $\hat{h}_j$ are now treated as fixed. The bootstrap therefore propagates uncertainty in the BSADF paths into uncertainty in the contagion coefficient, conditional on the lag and bandwidth selected from the original sample.

Re-estimating the optimal values of d and h for each bootstrap replication to better simulate the whole procedure would be preferable, but it would also be hugely impractical given the computation power available for this thesis. For ease of computation, the implementation therefore selects $\hat{d}_j$ and $\hat{h}_j$ once from the original data and then holds them fixed throughout the bootstrap stage. The consequence is that the bootstrap distribution does not include the extra uncertainty that would arise if the tuning parameters themselves were re-selected in every replication. The resulting standard errors and intervals are therefore conditional on the selected values of $\hat{d}_j$ and $\hat{h}_j$. This is a limitation: the intervals quantify uncertainty around the time-varying contagion path given the chosen lag and bandwidth, not around the full tuning-parameter selection procedure.

For each month t in each ordered pair, the bootstrap standard error is computed as the sample standard deviation of the bootstrap coefficient across replications $q = 1, \dots, Q$:

$$\widehat{se}\Big(\hat{\delta}_{2j}(t)\Big) = \left[\frac{1}{Q-1}\sum_{q=1}^{Q}\Big(\hat{\delta}_{2j}^{(q)}(t) - \bar{\delta}_{2j}(t)\Big)^2\right]^{1/2}, \tag{50}$$

$$\bar{\delta}_{2j}(t) = \frac{1}{Q}\sum_{q=1}^{Q}\hat{\delta}_{2j}^{(q)}(t). \tag{51}$$

The point estimate itself is the coefficient path obtained from the original BSADF data. The reported 95% confidence interval is then constructed in symmetric form around that point estimate, with 1.96 being the standard normal critical value associated with a two-sided 95% confidence interval:

$$CI_{95}\Big(\hat{\delta}_{2j}(t)\Big) = \Big[\hat{\delta}_{2j}(t) - 1.96\,\widehat{se}\Big(\hat{\delta}_{2j}(t)\Big),\, \hat{\delta}_{2j}(t) + 1.96\,\widehat{se}\Big(\hat{\delta}_{2j}(t)\Big)\Big]. \tag{52}$$

If the interval in equation (52) excludes zero at month t, the estimated contagion effect at that month is treated as statistically different from zero at the 5% level. If the whole interval lies above zero, the evidence points to positive contagion from the core market to the destination market at that time. If the whole interval lies below zero, the local relation is negative. If zero lies inside the interval, the point estimate may still be of interest, but the data does not provide enough evidence to distinguish it statistically from no contagion at that month.

The confidence interval used here is a standard-error interval rather than a percentile interval. This is another modelling choice. It keeps the interpretation simple and ties the interval directly to the variability of the bootstrap coefficient around the original point estimate. The cost is that any asymmetry in the bootstrap distribution is not built directly into the interval. Since the main concern is determining whether the coefficient is distinguishable from zero, that trade-off is acceptable.

4.5.8 Advantages and pitfalls of this methodology

The final output of the contagion procedure is therefore a monthly path $\hat{\delta}_{2j}(t)$ for each of the 30 ordered pairs, together with a bootstrap standard error and confidence interval at each month. The coefficient path can be interpreted as a local, time-varying measure of how strongly explosiveness in the chosen core market is associated with later explosiveness in another area. Because the coefficient is estimated nonparametrically in time, it is free to strengthen during a broad bubble, weaken during quiet periods, and even change sign if the local co-movement changes direction.

The methodology has several advantages. It matches the recursive nature of the first-stage BSADF measure. It allows contagion to vary over time instead of imposing one average effect for the whole sample. It avoids having to estimate a separate structural bubble model for each area. Moreover, through the bootstrap, it adds formal inference to a framework that had originally only been used in descriptive form.

At the same time, the methodology tries to detect whether bubble spillovers take place, but does not identify the economic driver of such contagion. It is also limited in that its inference is conditioned on fixed, previously selected lag and bandwidth values. And it assumes that Copenhagen and Aarhus are the only possible candidate cores on economic and observational grounds instead of proving that they are. All these limits should be kept in mind when interpreting the results of this thesis.

Within those limits, however, the procedure is coherent and defensible. The first-stage bubble-detection exercise produces monthly BSADF paths. The second-stage local-kernel regression turns those paths into time-varying contagion coefficients. The bootstrap then maps uncertainty in the BSADF paths into uncertainty in the contagion coefficients. For the purposes of this thesis, that sequence provides a practical way to move from recursive bubble detection to assessing their geographic propagation.

4.6 Open-market loss transactions during the 2008 crisis

4.6.1 Objective and empirical object

This section studies whether post-purchase household balance-sheet characteristics predict the timing of loss-making housing sales. The analysis focuses on market transactions that are not classified as foreclosures, and asks whether households that entered the crisis with different combinations of leverage and liquidity were more or less likely to sell at a loss before the market trough.

In this section, an open-market loss sale is defined as a registered market sale, excluding foreclosures, that satisfies the loss and underperformance conditions defined in the section "Loss-sale event and explanatory variables". The expression is useful because it separates the object of interest from bank repossessions: the sale takes place in the market, but it may still be informative about how balance-sheet constraints shape realised loss sales. In this setting the word “open-market” describes the institutional form of the transaction, not the absence of constraints.

The motivation comes from several different findings. Genesove and Mayer (2001) show that homeowners facing nominal losses tend to set higher asking prices and have lower sale hazards, consistent with nominal loss aversion[25]. This literature implies that the original purchase price is a salient reference point in housing decisions. A completed sale below that reference point is therefore used here as a condition for the classification of a housing loss. Stein (1995) and Ferreira et al. (2010) show that negative equity can reduce sales and mobility because selling may require additional liquidity at closing and because moving often requires financing a new down payment. This points towards the possibility that financially constrained households are less likely to realise losses through open-market transactions.

The design proposed here differs from the standard foreclosure and distress-study settings because foreclosures are excluded from the data and, instead of analysing financial shocks, financial conditions post-purchase are observed[26] and used as predictors of later sale outcomes. This is a weaker design than one based on contemporaneous shocks, and the estimates should be read as conditional associations; however, this empirical choice allows the analysis to exploit a large-scale registry of market transactions.

The empirical strategy requires us to carefully identify loss sales and distinguish them from general market downturns. Transaction timing and the effects of right-censoring should be taken into account, given that many households hold their properties past the trough. A Cox proportional hazards model is ideal for this setting because it estimates how explanatory variables shift the conditional Hazard Rate without assuming a specific parametric form for the underlying time profile. This is the same model used by Genesove and Mayer (2001), although with a different identification strategy.

The magnitude of any sale discount is not estimated. The local-index comparison is used only to

[25] Genesove and Mayer (2001) estimate the anchoring effect at the listing stage. The present analysis does not observe listings and therefore does not test their mechanism directly; it uses their result only to motivate why selling below the purchase price is an economically meaningful event rather than an arbitrary price cutoff. A similar listing-stage pattern is documented by Andersen et al. (2022).

[26] All financial covariates (mortgage balance, liquidity, wealth, DTI, income) are taken from the administrative IND-registers, which record household balance-sheet positions at year-end (31 December). This means that for each household, the variables used in the Cox model are those measured after the house purchase.

distinguish property-level underperformance from broad local market decline. Since the complete choice set for each household is not observed, no attempt is made to estimate a model of mobility. Instead, a more pragmatic approach is adopted. A transaction that both realizes a nominal loss and underperforms the local benchmark is treated as a conservative measure of a loss-sale event. A survival model is then used to estimate which post-purchase financial positions predict this outcome.

The next section follows by presenting the Cox model and the two regression equations that are used as the core of this section. Next, the event of interest and the variables that enter these equations are defined. The six time windows and the diagnostic tools applied to determine which windows are suitable for inference are then described. Finally, the limitations are explained.

4.6.2 Theory foundation: Cox model and baseline specification

Let T_i represent the time until household i undertakes a loss-generating sale in the open market, and let t denote continuous elapsed time in years from the model-specific time origin. The hazard function is:

$$h_i(t) = \lim_{\Delta t \downarrow 0} \frac{\Pr(t \leq T_i < t + \Delta t \mid T_i \geq t)}{\Delta t}. \tag{53}$$

It is the instantaneous rate at which the event occurs at time t, conditional on the household not having experienced the event before t. In this application, the event is a sale that meets the open-market loss definition given in the next subsection. A household that does not sell, or sells without satisfying the loss definition, is not treated as having failed under this event definition before the censoring date.

The Cox proportional hazards model writes this conditional rate as:

$$h_i(t \mid X_i) = h_{0,s(i)}(t) \exp(X_i' \beta), \tag{54}$$

where X_i is the vector of observed controls measured at the end of the year of purchase, and $h_{0,s(i)}(t)$ is the baseline hazard for the municipality stratum $s(i)$. Stratifying by municipality allows the common time profile of risk to differ across local markets. This is useful because the level and speed of the downturn were not identical across Denmark. The stratification does not estimate a separate coefficient vector for each municipality; it allows different baseline hazards while keeping a common set of controls.

The model does not estimate the baseline probability that a Danish household will sell at a loss in a given month like a probability model would do. It estimates whether households with one set of characteristics have a higher or lower conditional rate of a qualifying sale than households in the reference category. The coefficient of interest is the cross-sectional difference in sale timing within that downturn.

The main advantage of the Cox model is that it leaves the baseline hazard $h_{0,s(i)}(t)$ completely unspecified. The model does not require an assumption of a parametric duration distribution (e.g., Weibull or log-normal) and does not attempt to estimate $h_{0,s(i)}(t)$. Instead, the baseline hazard absorbs all common time-varying macroeconomic shocks (e.g. interest rate fluctuations or aggregate market downturns). The household-level coefficients are then identified. For uncensored observations with event indicator $d_i = 1$, the partial likelihood is:

$$L(\beta) = \prod_{i:\, d_i=1} \frac{\exp(X_i'\beta)}{\sum_{j\in R(Y_i)} \exp(X_j'\beta)}, \tag{55}$$

where Y_i is the observed duration and $R(Y_i)$ is the set of households still at risk immediately before Y_i. The common crisis profile can be flexible, while the controls describe relative differences between households that are comparable within each risk set. Within any given stratum, the local baseline hazard $h_{0,s(i)}(t)$ is shared by everyone in the risk set, meaning it mathematically factors out of the denominator and cancels with the numerator. While this cancellation happens locally within each municipality, a single, common vector of coefficients is estimated jointly by maximizing the partial likelihood across all strata simultaneously.

The coefficients are interpreted through Hazard Ratios (HR) instead of ordinary regression coefficients. A Hazard Ratio is a multiplicative effect on the conditional event rate at each point in event time. For a categorical regressor entered as a set of indicator variables, the Hazard Ratio associated with a category measures the conditional event rate relative to the omitted reference category, holding the remaining covariates fixed:

$$\mathrm{HR}_r = \exp(\beta_r), \tag{56}$$

Where r indexes the category of interest within a given categorical regressor. A Hazard Ratio above one means a higher conditional rate of an open-market loss sale relative to the reference group. A Hazard Ratio below one means a lower conditional rate[27]. For categorical variables, every reported coefficient is relative to an omitted category. In the main tables, the omitted group is chosen to represent the least financially exposed position: low leverage and high financial capacity, with the relevant reference category omitted from each set of indicators.

The proportional hazards specification and the partial likelihood derivation follow Cox (1972).

4.6.3 The two specifications

Two specifications were chosen rather than one because the baseline and the interaction specifications answer different questions. The baseline model evaluates whether leverage and liquidity independently predict an open market loss sale. Then, motivated by the double-trigger hypothesis, which suggests that the combination of negative equity and illiquidity is particularly critical in the foreclosure setting, an interaction specification is introduced to test whether the association between leverage and liquidity also matters in open-market situations.

The baseline specification is:

$$\begin{aligned} h_i(t \mid X_i) = h_{0,s(i)}(t) \exp(&\beta'\mathrm{LTV}_i^{cat} + \theta'\mathrm{Liq}_i^{dec} + \phi'\mathrm{Wealth}_i^{dec} \\ &+ \psi'\mathrm{DTI}_i^{dec} + \gamma' Z_i + \delta'\mathrm{Year}_i), \end{aligned} \tag{57}$$

[27] For example, a Hazard Ratio of 0.70 for a leverage category would mean that households in that category have a conditional hazard 30 percent lower than the omitted leverage group, conditional on the rest of the model

where LTV_i^{cat} is the categorical leverage measure, Liq_i^{dec} is the liquidity decile, Wealth_i^{dec} and DTI_i^{dec} are financial controls deciles. Z_i contains property and demographic controls[28], and year_i denotes purchase-year fixed effects when the cohort restriction includes more than one purchase year[29].

The interaction specification is:

$$\begin{aligned} h_i(t \mid X_i) = h_{0,s(i)}(t) \exp(&\beta' \text{LTV}_i^{cat} + \theta' \text{Liq}_i^{dec} + \Lambda' (\text{LTV}_i^{cat} \otimes \text{Liq}_i^{dec}) \\ &+ \phi' \text{Wealth}_i^{dec} + \psi' \text{DTI}_i^{dec} + \gamma' Z_i + \delta' \text{Year}_i), \end{aligned} \tag{58}$$

where $\text{LTV}_i^{cat} \otimes \text{Liq}_i^{dec}$ is the full set of interactions between leverage categories and liquidity deciles. This specification investigates whether the association between leverage and sale timing depends on liquidity. A high-LTV household with enough liquid assets may be able to realise a loss and move, while a high-LTV household without liquid assets may remain locked in and appear in the censored part of the sample. The interaction model is designed to distinguish these cases.

All financial variables in both the equations above are measured after the time of purchase, more precisely at the end of the year of purchase. This timing is central to the interpretation. The coefficients do not show how a contemporaneous income shock changes sale risk. They show whether balance-sheet positions after the purchase predict later open-market loss sales during the downturn.

In both equations, the full set of categorical indicators is estimated with one category omitted. For the interaction model, the omitted cell is the combination of the omitted LTV category and the omitted liquidity category. These interaction terms capture the hazard of facing both constraints simultaneously, relative to the baseline group.

4.6.4 Loss-sale event and explanatory variables

The event indicator is constructed from three conditions:

The first is a timing condition. The property must have been sold before the market trough. The trough is set at January 2012, based on the house-price index constructed earlier in the thesis and aligned with the national price-index evidence[30]. Households still observed without a sale at that date are right-censored, but not removed from the data. This timing restriction isolates the downturn studied in this thesis and it also aligns with standard survival-analysis practice (Therneau and Grambsch, 2000) by limiting the observation window to a period where the explanatory variables remain recent and relevant. The trough cutoff keeps the analysis focused on the period when loss sales can be found

[28]Property controls follow the hedonic structure used in Guren's hedonic regression: number of rooms, building age, square meters, number of bathrooms, including squared and cubic transformations. Demographic controls include age (and its squared and cubic transformations) and education indicators. The vector also contains an unoccupied-property dummy and an indicator for the number of properties owned.

[29]Year-of-purchase fixed effects are included only in the multi-year specifications (Models 1 and 2). In the single-year cohort models they are not separately identified and are omitted.

[30]The identification of January 2012 as the absolute trough of the Danish housing cycle is derived directly from the dataset's price index. This empirical minimum aligns with the official property price indices published by Statistics Denmark (see "StatBank Denmark table EJ121" in References), settings: Unit = "Price index, seasonally adj."; Region = "All Denmark"; Property Category = "One-family houses").

and the initial explanatory variables are still informative about the financial situation of the owner. The event should be read as just an open-market loss-sale indicator, not as a measure of seller distress.

The second condition is a nominal loss. Let $P^N_{p,i}$ be the nominal purchase price and $P^N_{s,i}$ the nominal second-sale price. A sold property satisfies the nominal loss condition if:

$$P^N_{s,i} < P^N_{p,i}. \tag{59}$$

This condition captures the nominal anchoring behavior mentioned in the loss-aversion literature[25]. It is not enough on its own, because the magnitude of a general housing crash can vary highly by area. Furthermore, if deflation is present, it could happen that a gain in real-terms is classified as a loss. For this reason a third condition was added.

The third condition is underperformance relative to the local benchmark[31]. Let $P^R_{p,i}$ and $P^R_{s,i}$ be the real purchase and sale prices, and let $I_{a(i),p}$ and $I_{a(i),s}$ be the area-level price index for the same area $a(i)$ at purchase and sale. Define the holding period in years as:

$$\Delta_i = \frac{\tau_{s,i} - \tau_{p,i}}{12}, \tag{60}$$

where $\tau_{p,i}$ and $\tau_{s,i}$ are measured in calendar months. The individual and area annualised real growth rates are:

$$g_i^{\text{real}} = \left(\frac{P^R_{s,i}}{P^R_{p,i}}\right)^{1/\Delta_i} - 1, \qquad g_i^{\text{area}} = \left(\frac{I_{a(i),s}}{I_{a(i),p}}\right)^{1/\Delta_i} - 1. \tag{61}$$

The underperformance condition is:

$$g_i^{\text{real}} < g_i^{\text{area}}. \tag{62}$$

A sale is coded as an open-market loss sale only if the timing condition, the nominal loss condition and the underperformance condition hold simultaneously. The nominal loss condition captures the behavioural threshold while the area-benchmark condition reduces the risk of coding ordinary market depreciation as household-level distress, while also adjusting for area-level price movements. This benchmark isolates the empirical target (an open-market loss that underperforms the contemporaneous local price trend).

While an extensive literature already examines the dynamics of foreclosures, this thesis deliberately takes a different path. Foreclosures are excluded from the event definition so that the analysis isolates open-market loss sales.

Defining the main variables

The first trigger is leverage. The property-specific loan-to-value ratio at the 31 of December of the purchase year is measured as:

$$\text{LTV}_{i,0} = \frac{\text{mortgagetotal}_{i,0}}{P^N_{p,i}}. \tag{63}$$

[31] This area-index benchmark serves as a municipality-level proxy for the local discount identification strategy.

The numerator uses the available administrative bond-debt measure as a proxy for outstanding mortgage debt[32]. The denominator is the nominal purchase price, because the purchase price is the relevant reference value for a nominal loss and for the equity position shortly after entry[33]. This differs from a current-market-value LTV measure. The current value would describe the household's updated equity position during the crisis, while the measure used here captures the initial leverage position that predicts later outcomes. The sale decision is anchored to the property being sold, so the first trigger should measure leverage attached to that asset rather than the household's entire debt position[34]. The broader debt burden is not ignored; it enters separately through the debt-to-income and wealth controls. LTV captures the equity position tied to the house, while the other financial controls capture the household's wider capacity to absorb shocks. LTV is entered as four categories rather than linearly[35]. This avoids imposing a single constant log-linear slope over the whole leverage distribution and allows the hazard to change discretely at economically relevant thresholds. The categories are shown here:

Category	**Threshold**	**Interpretation**
1 (reference)	$\text{LTV} \leq 0.60$	Households with a large equity buffer on 31 December of the purchase year. This group is used as the low-leverage **reference category**.
2	$0.60 < \text{LTV} \leq 0.80$	Households close to the standard upper range of low-cost mortgage borrowing in the Danish system (statutory maximum for covered-bond mortgages).
3	$0.80 < \text{LTV} \leq 1.00$	Highly leveraged households. They are more exposed to moderate price declines and may rely on additional debt.
4	$\text{LTV} > 1.00$	Households with debt above the purchase price. This category is used to capture strict negative equity or near-equivalent balance-sheet pressure under the available debt measure.

Table 3: Leverage categories used in the survival models

The second trigger is liquidity. The liquidity variable is taken from the administrative registers and ranks households into deciles based on liquid asset holdings[36]. The top liquidity decile (decile 10) is the reference category in the regression output design. Liquidity is meant to proxy the household's capacity to cover a shortfall, absorb moving costs, or continue servicing debt during the downturn.

[32] The preferred mortgage-balance variable from Danmarks Nationalbank used by Andersen et al. (2022) is only available from 2009 onwards and is therefore incompatible with the sample period. The administrative bond-debt variable is used `OBLGAELD` from the Statistics Denmark IND registers as a proxy. The official documentation acknowledges that this measure may not perfectly reflect the outstanding principal at any given date due to refinancing dynamics, fluctuations in bond prices, and the reporting structure of Danish mortgage banks. See "TIMES variable OBLGAELD" link to official stata website in References. The implied measurement noise is discussed in the limitations.

[33] Genesove and Mayer (2001). Ferreira et al. (2010) use the owner's self-reported current value in a survey.

[34] Cf. Elul et al. (2010) and Ferreira et al. (2010), who aggregate first mortgages with HELOCs and other home-secured second liens.

[35] Categorical specification follows Elul et al. (2010) and Ferreira et al. (2010), who break LTV at the standard U.S. underwriting thresholds. The cutoffs are anchored instead to Danish institutional thresholds; see Therneau and Grambsch (2000) on the use of step-function covariates as a way to relax log-linearity in the Cox model.

[36] Andersen et al. (2022) define liquidity narrowly as bank-deposit holdings, treating stocks and bonds as wealth rather than liquidity; this distinction is preserved. The variable is the recorded balance at 31 December of the year of purchase.

In the standard default literature, low liquidity can raise default risk because the household cannot keep paying after a shock. In a market-sale setting, however, very low liquidity may also prevent the household from selling an underwater property, because the household cannot finance the closing shortfall. High liquidity can then be associated with a higher observed hazard of loss sale among leveraged households, not because high liquidity is distressing in itself, but because it makes a loss-realising exit feasible. The interaction specification is included precisely because the expected sign of liquidity depends on the leverage position. Mortgage total, liquidity and all the variables used to compute DTI and Wealth measures are measured on 31 December of the purchase year.

Two additional balance-sheet variables are included as controls. Debt-to-income is defined as:

$$\mathrm{DTI}_{i,0} = \frac{\mathrm{mortgagetotal}_{i,0} + \mathrm{bankdebttotal}_{i,0}}{\mathrm{incometotal}_{i,0}}. \tag{64}$$

Net financial wealth is constructed as:

$$\begin{aligned}\mathrm{Wealth}_{i,0} = {} & (\mathrm{cash}_{i,0} + \mathrm{stocks}_{i,0} + \mathrm{bonds}_{i,0} + \mathrm{panktakt}_{i,0}) \\ & - (\mathrm{bankdebt}_{i,0} + \mathrm{mortgagetotal}_{i,0} + \mathrm{pantgaeld}_{i,0}).\end{aligned} \tag{65}$$

The subscript 0 denotes the 31 December of the year of purchase.

Here, pantakt represents the market value of mortgage deeds (pantebreve) held in custody as receivables and pantgaeld represents outstanding mortgage-deed debt.

Both variables are entered in deciles. Deciles are used for the same reason as the LTV categories: in the specification, the financial variables are allowed to have non-linear effects, rather than forcing them into a single linear slope, relaxing the log-linearity assumption of the Cox model[37]. Wealth is included because liquidity alone may miss households that hold non-deposit financial assets or have a stronger net financial position. DTI is included because two households with similar LTVs may face different repayment burdens relative to income.

The vector Z_i contains the remaining controls. Property controls follow the hedonic structure used by Guren (2018): number of rooms, building age, square meters and number of bathrooms, including polynomial terms where specified. Demographic controls include age and education indicators for the household, with age entered flexibly. The specification also includes an unoccupied-property indicator and an indicator for the number of properties owned. These controls do not remove all selection concerns, but they reduce the risk that the leverage-liquidity coefficients merely reflect observable differences in property type, age profile or household composition.

Purchase-year fixed effects are included only where there is within-model variation in purchase year. They absorb differences between entry cohorts inside the same sample window, such as buying closer to the peak or entering under different credit and price conditions. In single-year cohort models, these fixed effects are not separately identified and are omitted.

[37]See model assumptions

4.6.5 Time origin and six model progression

The survival model is estimated under six related time-window designs. The purchase cohort chosen defines the sample used in the Cox regression in a given model: only homes that have been bought during this period enter the sample. On the other hand, two different time origin frameworks are used, and two different measures of duration are considered for different models. These duration measures are the ones that enter the proportional hazards regression, instead of just regular time.

The purpose for the creation of different models is to find whether the estimated Cox structure is stable across plausible definitions of the at-risk period. A household that bought in April 2005 and sells in late 2011 may have experienced six years of income changes, refinancing, family changes and asset accumulation. A household that bought in 2006 is observed closer to the peak and has a shorter gap between measured financial conditions and crisis exposure. The cohort design is a practical way to see where the purchase-date variables still provide a credible description of the household's relevant financial position.

The trough date is the same in all models and, as explained above, is also used as one condition in the event definition. This decision was made to ensure that the explanatory variables remain informative throughout the study period. This right-censors the event, which is something to take into account when interpreting the results of the model. Let's define some timing variables:

$$\tau_{p,i} = \text{calendar month of purchase,} \tag{66}$$

$$\tau_{s,i} = \text{calendar month of second sale, if observed,} \tag{67}$$

$$T_{peak} = \text{August 2007} \rightarrow \text{detected using the constructed index and compared to DST} \tag{68}$$

$$T_{trough} = \text{January 2012} \rightarrow \text{detected using the constructed index and compared to DST.} \tag{69}$$

Two time origins are used. In the purchase-origin design, the household becomes at risk at its own purchase date. The observed duration is:

$$k_{years} = \frac{\min(\tau_{s,i}, T_{trough}) - \tau_{p,i}}{12}. \tag{70}$$

In the peak-synchronised design, the household becomes at risk at the aggregate market peak. The observed duration is:

$$Duration_{crisis} = \frac{\min(\tau_{s,i}, T_{trough}) - T_{peak}}{12}. \tag{71}$$

This last design isolates the downturn phase more directly. It removes pre-peak holding time from the risk clock and places households on the same crisis timeline. Observations with a second sale before T_{peak} are dropped in the peak-synchronised specifications, because they are not observed as being at risk during the post-peak downturn[38].

[38] Observations with a second sale before the peak are dropped from Models 2, 4, and 6. Removing these pre-peak exits ensures that all retained households are observed as being at risk during the post-peak phase.

The two different time specifications answer related but not identical questions. The purchase-origin definition measures how long after buying a household executes a qualifying sale. It keeps the full holding-period interpretation but mixes pre-peak and post-peak months. The peak-synchronised specification asks how quickly households sell once the aggregate downturn begins. It is closer to the crisis mechanism, but it is less informative about total holding time. Reporting both versions allows the results to distinguish between these two notions of exposure.

The six models come from crossing three purchase-cohort restrictions with the two time origins. The unrestricted bubble window covers purchases from April 2005 to January 2007[39]. The two narrower cohorts split the bubble-period purchases into 2005 and 2006 entry groups. In summary:

Model	**Purchase cohort**	**Time origin**	**Duration**
Model 1	April 2005–January 2007	Purchase month ($\tau_{p,i}$)	k_{years}
Model 2	April 2005–January 2007	Market peak (T_{peak})	$Duration_{crisis}$
Model 3	April 2005–December 2005	Purchase month ($\tau_{p,i}$)	k_{years}
Model 4	April 2005–December 2005	Market peak (T_{peak})	$Duration_{crisis}$
Model 5	January 2006–December 2006	Purchase month ($\tau_{p,i}$)	k_{years}
Model 6	January 2006–December 2006	Market peak (T_{peak})	$Duration_{crisis}$

Table 4: Time windows and survival-time definitions

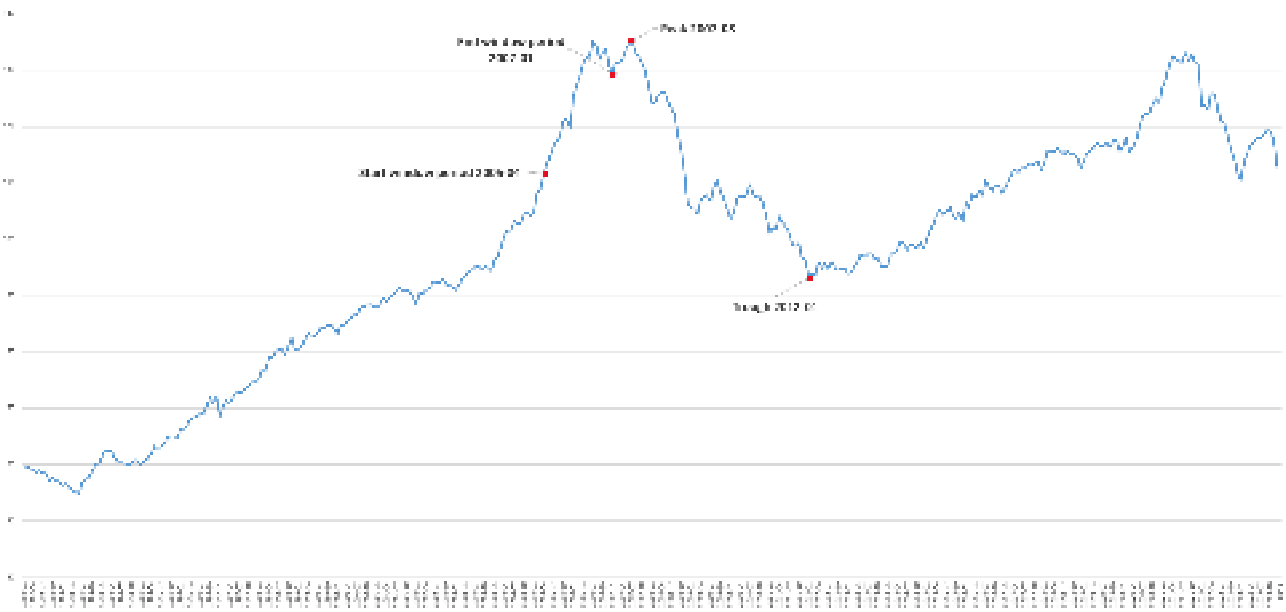


Figure 4: Time window, peak and trough definitions in the repeated sales monthly national index.

For each of the six models, two specifications are estimated: the baseline model and the leverage-

[39]The bubble window is identified using PSY tests and an HP filter on the national-monthly price index.

liquidity interaction model described earlier. This gives twelve Cox estimates. The full set is useful because the main modelling issue is time dependence. If the earliest purchase cohort is followed for a long period before the trough, financial variables measured post-purchase may no longer describe the household's relevant condition at the time of sale. If the peak-synchronised clock is used, the model focuses on the crisis phase, but it also removes sales that occurred before the peak. The six-model design makes this trade-off explicit.

Models 1 and 2 are the broadest designs. They use the full bubble-purchase window and therefore maximise sample size, but they also combine households that entered the market at different points of the run-up. Models 3 and 4 isolate the 2005 cohort (from April to December), which gives a cleaner entry group but has the longest distance between purchase conditions and the trough. Models 5 and 6 isolate the 2006 cohort, which is closest to the peak and is therefore the cohort for which financial variables are most likely to remain informative during the crash. This is the economic reason why the diagnostic tests are expected to matter: failure of the earlier cohorts is not only a statistical inconvenience, but also consistent with information decay in the controls.

The results section uses the diagnostic tests described below to determine which of these windows is suitable for interpretation. The retained models are selected by whether the specification satisfies the proportional-hazards and functional-form checks. The non-retained models are still going to be informative because they show where the Cox assumptions become less credible.

4.6.6 Model assumptions, diagnostics and robustness checks

The Cox design relies on three assumptions that matter for this application. The first is proportional hazards. Conditional on the included covariates and municipality strata, the Hazard Ratio associated with a covariate should be approximately constant over the observation window. If the effect of liquidity becomes weaker over time, a single coefficient may average over different effects and become hard to interpret. The short crisis window and the cohort restrictions are intended to reduce this problem. The key point is that proportionality is an assumption about relative hazards, not about the baseline risk itself. The baseline hazard may rise sharply in the worst months of the crisis and fall later. The model requires that the ratio between two households with different covariates remains reasonably stable over that time scale. Because the crisis changed the role of financial constraints over time, this assumption is not taken for granted and is tested explicitly.

The second assumption concerns functional form. In a Cox model, the covariates enter the log-hazard linearly unless transformed or grouped. For this reason, the main financial variables are grouped into categories or deciles[40]. This replaces a strong log-linear assumption, required by the cox model, with a weaker piecewise-constant assumption. The main problem of using this approach is that households within the same bin are treated as having the same coefficient, even if their underlying financial positions differ.

[40] Elul et al. (2010) partition LTV into narrow ten-percentage-point bins, treating it as a step function. Four broader categories are used tied to the institutional thresholds; Therneau and Grambsch (2000) discuss the trade-off between within-bin homogeneity and bias from imposed log-linearity.

The third assumption is non-informative censoring. Conditional on observed covariates, the reason an observation is censored should not contain additional information about the unobserved event process. In this setting, censoring mainly means that the household is not observed completing a qualifying open-market loss sale before January 2012. This is the most fragile assumption in the design. Some households may remain censored not because they are unaffected by the downturn, but because they are locked in and cannot realise a loss through an open-market transaction. This matters especially for high-LTV, low-liquidity households, for whom the absence of an observed loss sale may reflect an inability to transact rather than an absence of financial pressure.

This concern does not imply that the model necessarily understates the true hazard of open-market loss sales. Rather, it means that a low estimated loss-sale hazard may understate broader balance-sheet stress. Lower estimated hazards for high-LTV, low-liquidity households should therefore not be read as evidence that they are financially unaffected, but as evidence that they are less likely to realise a market sale at a loss within the observation window.

Several checks are used to assess whether the specifications are credible enough for inference:

Multicollinearity is assessed using variance inflation factors. Since standard VIF diagnostics are based on the regressor matrix rather than on the Cox likelihood itself, they are computed from an OLS regression that uses the same control variables as the baseline Cox specification[41] and cluster standard errors on kommunes (analogously to the Cox model). The dependent variable in the auxiliary regression is the open-market loss indicator, but the VIF values are interpreted only as diagnostics for linear dependence among regressors. For each regressor X_j, the variance inflation factor is:

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2}, \tag{72}$$

where R_j^2 is the coefficient of determination of the auxiliary regression of X_j on the remaining regressors. Because R_j^2 is a function of the regressor matrix alone, the choice of dependent variable does not change the outcome, which is what justifies substituting the failure indicator for the Cox response. The interaction specifications are not included in the VIF analysis, because interaction terms are related to their main effects by construction.

All Cox regressions are stratified at the municipality level, and standard errors are clustered by municipality. For the broader multi-year models, clustering also accounts for year-level common shocks where specified. The stratification handles differences in local baseline hazards, while clustered standard errors reduce the risk that spatially correlated shocks overstate precision. These adjustments align the inference with the spatial structure of the data.

The proportional-hazards assumption is checked graphically and formally. The graphical check uses log-log survival plots (see appendix) and the related Schoenfeld-residual diagnostics to see whether

[41] Stata does not allow to run this test directly to the Cox model; however, the VIF depends only on the regressor matrix and not on the response (Belsley et al. 1980), which justifies recovering the values from an auxiliary OLS that substitutes the failure indicator for the Cox response.

the hazards appear to evolve proportionally over time. The formal check is the test of Grambsch and Therneau (1994), implemented in Stata as `phtest`[42]. The idea is that, under proportional hazards, the effect of each control on the hazard should not depend on time. The test is built on the Schoenfeld residuals: at every failure time, each residual records the gap between the control value of the subject who fails and the average value of the same control among the subjects still at risk. Under correct specification these residuals fluctuate around zero at every t_k; any systematic drift with time is evidence against proportionality. Grambsch and Therneau (1994) formalise the alternative by letting the true coefficient drift over time:

$$\beta_j(t) \equiv \beta_j + \theta_j\, g_j(t), \tag{73}$$

for a chosen function $g_j(t)$. Proportional hazards corresponds to $H_0 : \theta = 0$, and the resulting test statistic, derived from a generalised least-squares fit of the scaled residuals on $g(t)$, is:

$$T(G) = \left(\sum_k G_k\, \hat{r}_k\right)' D^{-1} \left(\sum_k G_k\, \hat{r}_k\right), \tag{74}$$

asymptotically χ^2 with p degrees of freedom under H_0, where G_k and D are defined in equations (4) and (10) of the paper. A rejection indicates a Hazard Ratio that drifts with time. A technical caveat applies. The derivation of $T(G)$ assumes homogeneity of variance across risk sets, which is what allows the consistent estimation of the asymptotic variance by D. Grambsch and Therneau (1994) note that the test is fairly robust to departures from this assumption. Since the final models stratify by municipality, the global `phtest` is read as a diagnostic indicator rather than as a definitive proof of proportionality.

Functional-form adequacy is assessed with the Stata `linktest`[43]. The test is based on the idea that, if a regression-like equation is properly specified, no additional independent variable should be significant in that equation except by chance. One kind of specification error occurs when the dependent variable needs a transformation to relate properly to the independent variables. The link test is built around an auxiliary regressor that is especially likely to be significant when the link is wrong. Let $\mathbf{y} = f(\mathbf{X}\boldsymbol{\beta})$ be the original model and $\hat{\boldsymbol{\beta}}$ the parameter estimates, the test calculates:

$$\texttt{\_hat} = \mathbf{X}\hat{\boldsymbol{\beta}} \tag{75}$$

and

$$\texttt{\_hatsq} = \texttt{\_hat}^2. \tag{76}$$

The model is then refitted with these two variables as regressors. If the original specification is correctly specified, the linear index already captures the systematic part of the relationship, so `_hatsq` should add nothing significant; the reading consistent with correct specification is therefore `_hat` significant alongside `_hatsq` insignificant. A significant `_hatsq` signals that part of the systematic relationship is not absorbed by the chosen specification, typically because of an omitted nonlinearity, an omitted

[42] See "stcox PH-assumption tests" in References.

[43] See "Link test for model specification" in References . The notation and the description below follow the Stata reference.

interaction, or a missing covariate.

The six-model progression itself is a robustness check. It tests whether the findings depend on using the whole bubble purchase window, the 2005 cohort, or the 2006 cohort, and whether they depend on using purchase time or peak-synchronised time as the origin. In the empirical results, only the models that pass the diagnostic criteria are used for interpretation (the other specifications are reported in the appendix).

Finally, descriptive Kaplan-Meier estimates are used to display non-parametric differences in failure patterns across selected financial groups. These curves are not substitutes for the Cox estimates, because they do not adjust for the full control set. They are included to check whether the adjusted results are consistent with the broad, unadjusted survival patterns.

4.6.7 Limitations

The design has five main limitations.

First, the financial covariates are not measured at the exact purchase date but at the 31 December of the year of purchase, since the IND administrative registers report household balance-sheet positions only at year-end; the duration clock, by contrast, starts at the actual purchase date. The covariates thus capture the household's balance sheet shortly after entry into homeownership. They are suitable for asking whether after-purchase conditions predict loss sales, but they do not observe the balance sheet at the moment of sale. This matters because income, debt, assets and family circumstances can change substantially during a downturn. The cohort restrictions partly address this by shortening the relevant interval, but they cannot remove the problem[44]

Second, the leverage measure is constrained by data availability. The preferred mortgage-balance measure used in some Danish studies is not available over the full period required here, so the analysis uses the available administrative bond-debt variable as a proxy for outstanding mortgage debt. This introduces measurement error into the LTV categories. If that error is classical, it may attenuate the estimated leverage effects; if it is related to refinancing behaviour, the direction is less clear.

Third, liquidity and wealth deciles are only proxies for financial resilience. They do not capture informal transfers, family support, unreported assets, medical shocks, divorce, unemployment spells within the year, or other events that may force a sale. The liquidity trigger therefore measures post-entry capacity, not the exact trigger event. This is the main difference between this study and papers that observe contemporaneous unemployment or credit-card utilisation.

Fourth, excluding foreclosures is necessary for the research question but narrows the scope of the conclusion. The estimates describe the market-sale segment. They do not describe the most severe bank-led exits, and they should not be read as the total effect of the crisis on distressed households. If the most constrained households are more likely to enter foreclosure or never sell, the estimated

[44]Purchase dates span less than two years (April 2005–January 2007), so inflation differences are negligible. The main financial variables enter the model as ratios (LTV, DTI) or as deciles within each sample, both of which are invariant to a uniform price-level rescaling.

Hazard Ratios for open-market loss sales may be a lower-bound view of balance-sheet stress in the broader population.

Another limitation concerns the event definition itself. The local underperformance rule is designed to avoid confusing ordinary market decline with a loss sale, but it cannot observe property condition, renovations, maintenance, bargaining power or private information at the time of sale. Some transactions may be coded as failures because the house had deteriorated relative to the market; others may fail to be coded because a constrained seller happened to sell in a strong local submarket.

For these reasons, the results should be interpreted as association-based evidence on market-sale timing, not as causal estimates of the effect of leverage or liquidity. The value of the design is that it applies a consistent event definition, a clear survival-time structure, and explicit diagnostics to a setting where the usual foreclosure-based double-trigger framework cannot be transferred directly. The strongest conclusions are therefore restricted to the model windows that satisfy the diagnostics, and to the market-sale segment defined in this section.

Table 5: Design choices and limitations: likely effect on the estimated Hazard Ratios

Source	**Baseline choice**	**Likely effect on estimates**
Post-purchase covariates	Balance sheet after purchase.	Misses contemporaneous shocks; attenuation, worse for early cohorts.
Mortgage-balance proxy	`OBLGAELD` register variable.	Classical measurement error; attenuation of LTV effects.
Foreclosure exclusion	Voluntary open-market sales only.	Most distressed households exit outside the event; lower-bound estimates.
Endogenous censoring	Lock-in keeps illiquid borrowers censored.	Downward bias concentrated on high-LTV / low-liquidity cells.
Local-benchmark rule	Real CAGR < area-index CAGR.	Unobserved property quality; two-sided noise, no systematic direction.
Categorical binning	LTV in four categories; financials in deciles.	Within-bin homogeneity; mild attenuation around thresholds.

Several directions remain for enhancing the robustness of this analysis. Specifically, accounting for time-on-market via listing dates would refine the distinction between behavioral and search-related delays, while controlling for heterogeneity in risk preferences and mortgage contracts would provide a more granular view of household-level financial responses. Reproducing this setting using a margin around the area discount, rather than a strict inequality, could better isolate genuinely large loss sales from marginal underperformers and could be used as a robustness test. Finally, incorporating interest rates would allow for a clearer identification of household-specific constraints, rather than allowing these aggregate dynamics to be absorbed by the baseline hazard.

5 Results

5.1 Index Validation

5.1.1 Index comparison to DST

Figure 5: Annual performance of the constructed national index compared to the DST benchmark

Before proceeding to bubble detection and contagion analysis, the repeat-sales index is validated constructed from the registry transactions against the official Statistics Denmark (DST) benchmark[45]. The figure above reports four panels: (i) the hedonic index (yellow), included only to motivate its exclusion; (ii) the real repeat-sales index with outlier trimming (blue); (iii) the same index without trimming (orange); and (iv) the DST benchmark (grey). The bottom-right panel reports the percentage deviation induced by applying the nominal-to-real conversion before versus after the trimming procedure in the repeat-sales methodology. Since DST publishes yearly observations, monthly fixed effects are replaced with yearly fixed effects for comparability; all other steps of the methodology are unchanged.

In the top left corner the constructed yearly indices are compared to the DST series over the period spanning from 1992 to 2023. The performances of all the repeated sales models (with and without outliers) are very close to the benchmark. The hedonic index, however, shows very different trends.

In the top right the year-on-year growth rates constructed from level differences are shown. They confirm close co-movement of the repeated sales index with DST throughout the whole sample, including the timing and depth of the 2008 crisis.

[45]DST uses the SPAR method to create the index as explained in "Statistics Denmark. (n.d.). StatBank Denmark statistical processing description" link in References

In the bottom left, the annual counts of observations for the same indices over the same years are displayed. DST aggregates all property transactions, whereas the sample is restricted by the filters detailed in Section 3. The year-on-year variation in the two counts is, however, aligned, indicating that the filters do not introduce cyclical selection relative to the aggregate market[46]. The trend movements of the repeated sales index and of the DST appear to be similar.

Finally, in the bottom right, the nominal-to-real sensitivity, when dropping outliers, is measured. Deviations between the two conversion timings are negligible in the central years and widen at the extremes. By construction, no deviation arises when outliers are retained: the discrepancy comes from the trimming criterion identifying different extreme observations when applied to nominal versus real prices.

5.1.2 Limits and extensions

The repeated sales index was chosen because it showed behaviour and trends that were very similar to those of the benchmark. This index is considered to be a well-designed proxy for the purposes of the thesis. Nonetheless, it is recognized that the analysis could have been extended by integrating the hedonic and repeat-sales approaches and/or by extending the methodology to the untrimmed repeat-sales index.

The outlier-trimmed real index is adopted for the whole analysis. The choice is based on three considerations: First, repeat-sales estimators are sensitive to extreme price ratios between paired transactions. Secondly, extreme ratios typically reflect non-market events (major renovations, intra-family transfers) rather than the price signal the index is designed to capture. Finally, the trimmed series tracks the DST benchmark closely, preserving macroeconomic validity while reducing idiosyncratic noise in the downstream analysis.

As discussed in Section 2, the repeat-sales framework is not immune to selection bias, temporal instability, and censoring at the sample boundaries. Its alignment with the DST benchmark nonetheless supports its use as a valid input for the empirical analysis that follows.

[46]The larger gaps at the beginning and end of the sample reflect the standard right- and left-censoring of repeat-sales designs (see Section 2).

5.2 Bubble Detection and Contagion

5.2.1 Bubble detection

The main takeaway from the bubble detection results is clear. Exuberance is found in much of the country, but timing varies slightly between areas. The main episode occurs in the mid-2000s, and exuberance is clearly detected in all but one of the areas.[47] Other bubble activity is much narrower and localized and should not be read as a second nationwide cycle of the same type.

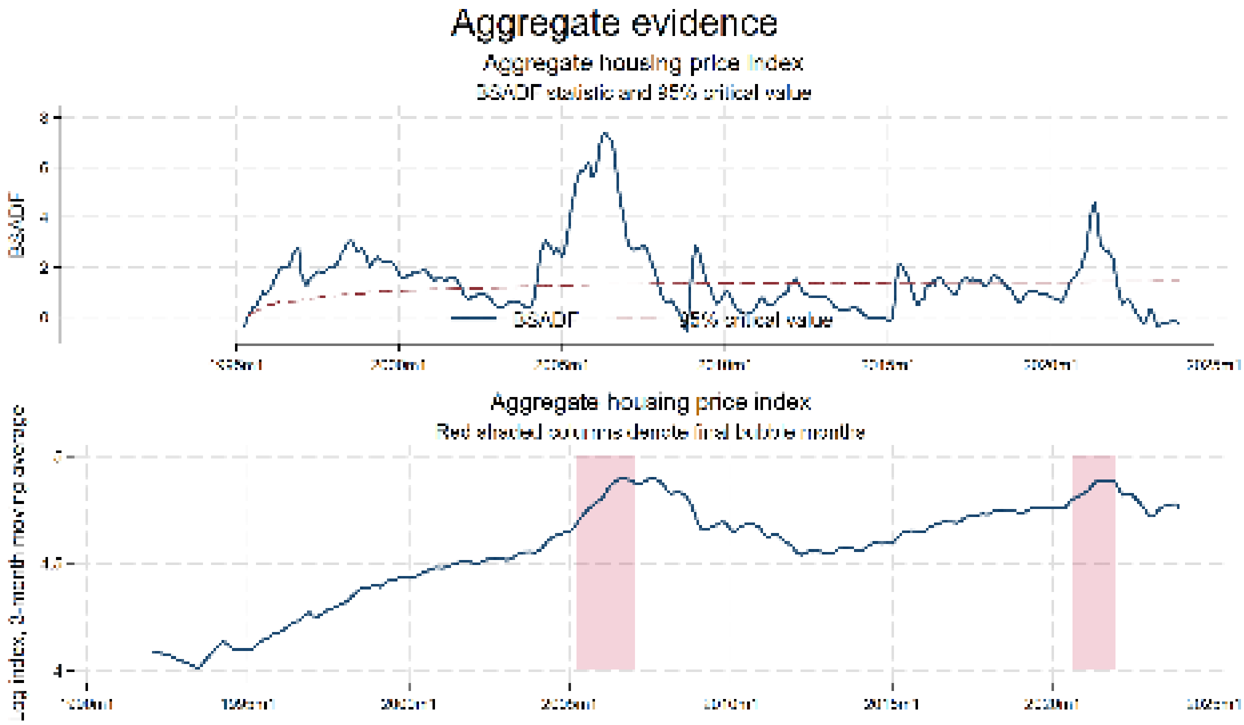


Figure 6: Evidence from the national aggregate index. The upper panel reports the aggregate BSADF path and its 95% critical value. The lower panel reports the transformed index, with shaded periods denoting months classified as bubble months under the final composite detection rule.

Figure 6 shows the result for the whole of Denmark. At this level, the final detection rule identifies two episodes: April 2005 to January 2007, and September 2020 to December 2021. The first episode corresponds to the well-known boom before the global financial crisis. The second takes place much later, after the COVID-19 pandemic, and it is also shorter in duration. Taken on its own, the aggregate picture suggests two broad national bubble episodes. However, as mentioned before, the area-level results show that this is only partly correct.

[47] As explained shortly, in area 015, GSADF is still significant, but the composite bubble date-stamping procedure does not find any bubble.

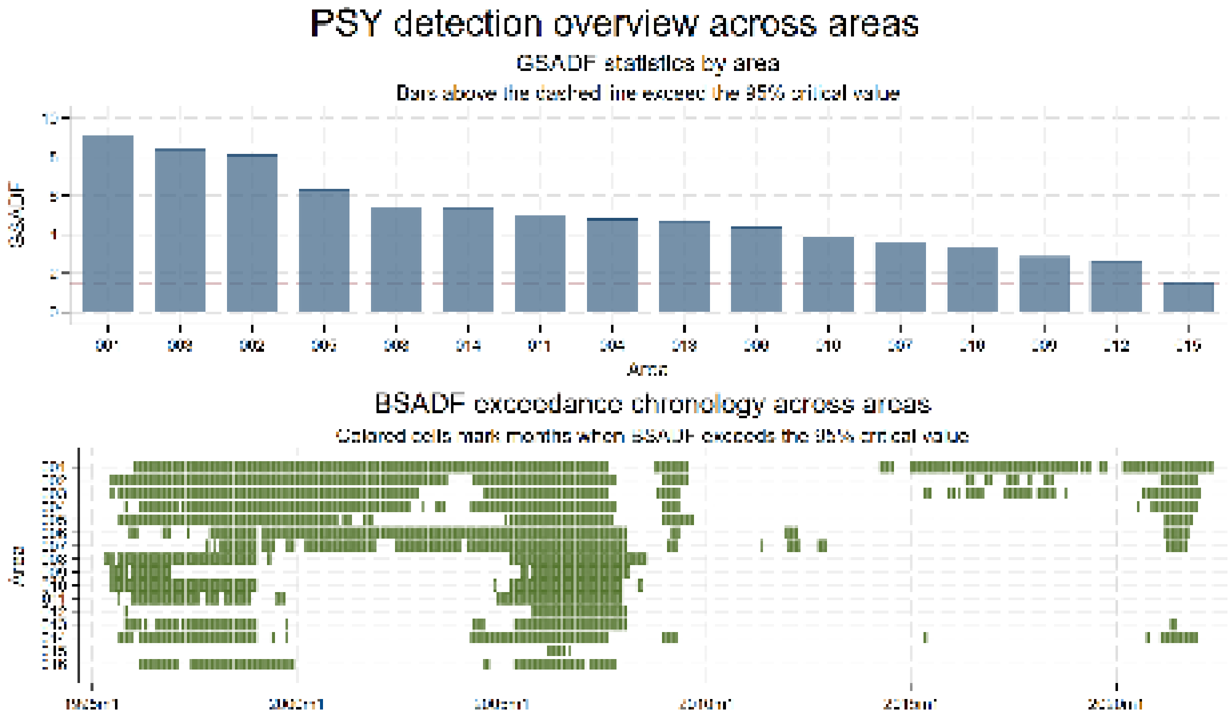


Figure 7: PSY-stage detection across the 16 areas. The upper panel reports GSADF statistics by area. The lower panel shows the months in which the BSADF path exceeds its corresponding 95% critical value.

Statistical explosiveness, purely in the PSY sense, is widespread across the sample. As shown in the upper panel of Figure 7, all 16 areas have GSADF statistics above the 95% critical value. Copenhagen (001) is the strongest case, with a GSADF statistic of 9.13. Areas 003 and 002, which are mainly composed of Copenhagen suburbs, follow with values of 8.36 and 8.11. At the other end of the ranking, area 015, located in Northern Jutland, has the weakest statistic, 1.53, but it still lies above the common 95% critical value of 1.42. Therefore, at the recursive-test stage, every area displays at least some explosive behaviour somewhere in the sample.

The lower panel of Figure 7 also shows a similar result: all areas have at least some period of explosiveness, and in many areas BSADF is larger than the critical value for a large number of months, including many where visual inspection of the price index would not necessarily suggest a bubble. However, it is also worth noting that only around half of the areas show explosiveness around the 2020s episode found in the aggregate index, which already points towards this episode being a localized one.

That broad PSY result does not carry over into the composite bubble classification. Comparing Figure 8 with Figure 7 shows that adding the Hodrick-Prescott and minimum duration conditions to the bubble date stamping process significantly decreases the totsl length and number of bubbles episodes detected. This is exactly what the combined procedure was designed to do. The one-sided HP filter and the minimum-duration rule manage to narrow the PSY signal down to a smaller set of episodes that are long enough and occur at periods where the price level is actually high. The final classification is therefore more selective than the PSY step not only in theory, but also in practice: the

HP filter and duration conditions definitely have an effect in the bubble dating process.

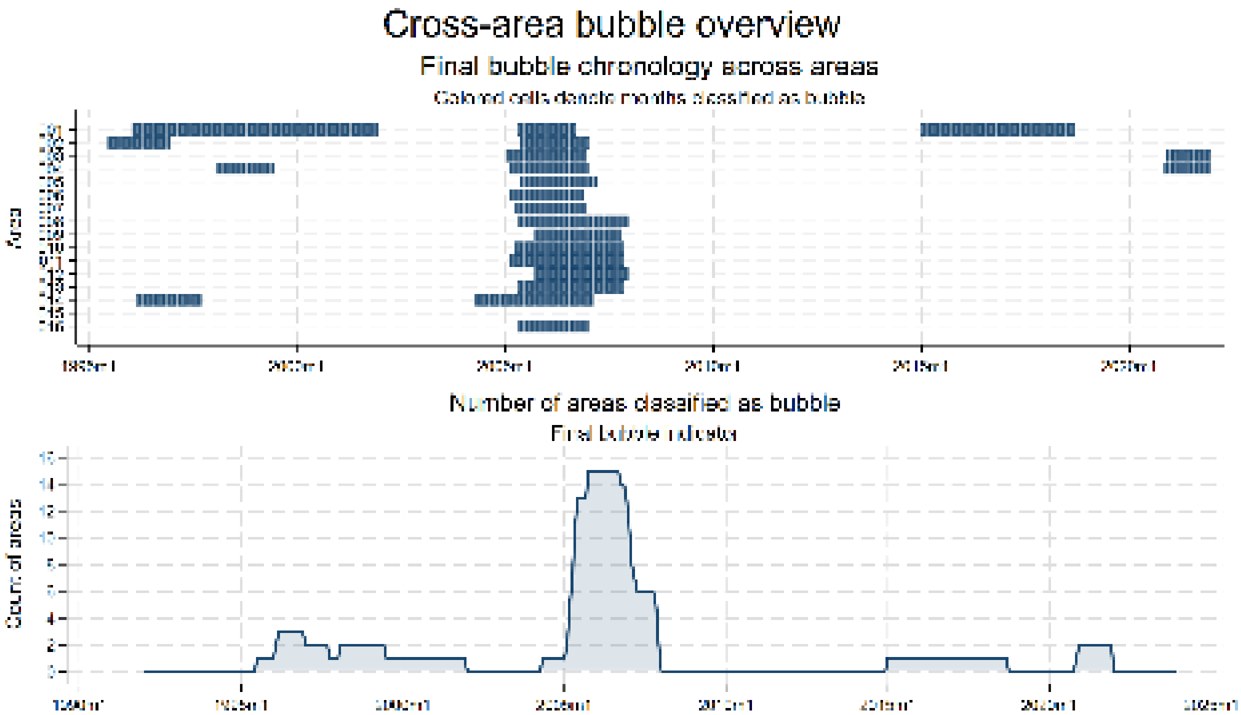


Figure 8: Cross-area final bubble indicator chronology. The upper panel shows the final bubble months for each area. The lower panel shows the number of areas simultaneously classified as bubble months.

In Figure 8 for most areas the only bubble episode that remains is the mid-2000s one. Notable exceptions to that pattern include Copenhagen and the other areas in the Capital Region, as well as Aarhus, for which other exuberant periods are also flagged. Between October 2005 and September 2006, 15 of the 16 areas are simultaneously classified as bubble months. Area 015 however, which was already the one with the lowest GSADF, now shows no exuberance anywhere in the sample.

In general, these results are consistent with the ones related to the aggregate index, showing evidence of a broadly shared national boom during 2005-2007. However, they disagree with the aggregate analysis regarding the second 2020s bubble, where it is only detected in areas 003 and 004, which are located north of Copenhagen. In other words, the post-pandemic episode is not broad-based in the way the mid-2000s episode was, instead it just reflects developments in a narrow part of the country. The aggregate series makes the later episode look national, while the area-level chronology shows that it is not.

Table 6: Key detection results for the aggregate index and selected areas

Series / area	GSADF	Bubble months	Episodes	Final bubble episode(s)
Aggregate index	7.36	38	2	2005m4–2007m1; 2020m9–2021m12
001 (Copenhagen)	9.13	133	3	1996m2–2001m12; 2005m5–2006m9; 2015m1–2018m9
014 (Aarhus)	5.34	53	2	1996m3–1997m9; 2004m5–2007m2
003	8.36	36	2	2005m2–2006m12; 2020m12–2021m12
004	4.81	54	3	1998m2–1999m6; 2005m3–2007m1; 2020m11–2021m12
015	1.53	0	0	None

Table 6 summarizes the bubble detection results for some of the most relevant indices. Copenhagen is clearly the most bubble-prone area in the sample: it has the highest GSADF statistic, the largest number of bubble months, and three distinct final episodes. Aarhus also has more than one episode, and notably its main bubble episode starts in May 2004, slightly before other areas, as it can also be observed in Figure 8. This result prompted the consideration of Aarhus as a candidate core market in the contagion phase of the analysis. Areas 003 and 004, located north of Copenhagen, are the only areas that generate the later 2020-2021 combined bubble signal. Area 015 shows statistically significant PSY explosiveness, but none of it survives to the final classification.

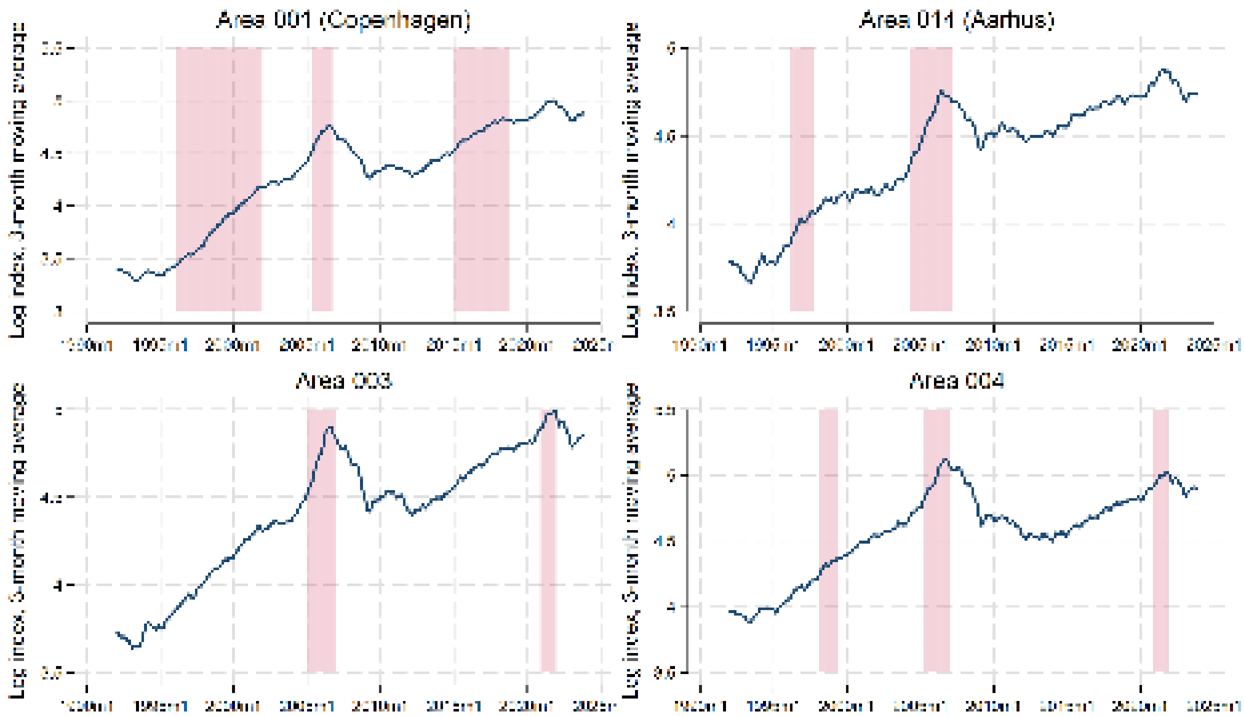


Figure 9: Selected area-level bubble histories. The figure reports the transformed indices for Copenhagen (001), Aarhus (014), and the two areas (003 and 004) where an episode in the 2020s is detected. Shaded periods denote bubble periods detected by the composite procedure.

Figure 9 graphically showcases the detection results for most of the areas where more than one bubble is detected. While the indices look very similar when compared visually, the bubble dating procedure detects exuberance in different periods in each index. Copenhagen's later episode occurs in 2015-2018, and not in the post-pandemic window, like in areas 003 and 004. Aarhus shows no bubble in the 2010s or the 2020s. On the other hand, Copenhagen and Aarhus, along with Areas 002 (not shown in this figure) and 004, show signs of exuberance before or around the turn of the century, although those periods are not followed by a correction.

This variety of results might seem puzzling at first, taking into account the apparent similarity between the indices. However, this effect can be partly explained by the additional rules imposed during the combined bubble detection procedure, such as the minimum duration rule. For example, if the lower panel in Figure 6 is closely observed, it can be observed that, while still not ubiquitous, post-pandemic bubbles are detected by BSADF in many more areas, but those events are often short in duration and therefore get filtered out by the rule, exacerbating the differences between areas with otherwise similar behaving indices. While the minimum duration condition is helpful to remove the many short, likely spurious, bubbles that abound in the lower panel of Figure 6, and make the results more straightforward, it can also lead to the underdetection of short but relevant bubble episodes, and it introduces arbitrariness into the process.

Taken together, the detection results show five main findings. First, as expected, the most prominent episode of exuberance took place between 2005 and 2007, with duration differing slightly by area. Second, evidence of statistically explosive behaviour is stronger in Copenhagen and its surroundings, but is far from being confined to there. Third, post-pandemic exuberance is much shorter and more localized than the main mid-2000s episode. Fourth, the use of additional conditions aside from the BSADF statistic is useful to filter out very short, spurious or implausible bubbles, but it also risks failing to detect short but relevant events, while adding arbitrariness to the procedure and artificially creating or amplifying differences between areas. Finally, due to evidence of exuberance appearing there before in other areas during the bubble event in the period, the detection part of this thesis points to Aarhus as a potential candidate core region for the contagion analysis.

5.2.2 Bubble contagion

The contagion results are much weaker than the detection results. The question in the second stage is whether the destination-area BSADF path moves significantly with the (sometimes lagged)[48] BSADF path of a candidate core market. The answer is, for the present data, in most cases, no. Once bootstrap-derived uncertainty is taken into account, statistically significant pairwise contagion is scarce.

Only Copenhagen and Aarhus were considered as candidate core markets in this analysis, and all possible pairs of combinations from a candidate core area to the rest of the areas were tested. In the case of Copenhagen, the results are clear. While the $\hat{\delta}_{2j}(r)$ point estimates are often positive, they are

[48] As explained in the methodology section, the implementation admits delays from 0 to 12 months, differing in each regression, so the explanatory variable is not necessarily always lagged, and not always lagged by the same amount. This is something that must be taken into account when interpreting the $\hat{\delta}_{2j}(r)$ coefficients.

never significantly different from zero, for any area and any point in time. Therefore, contrary to what economic intuition may suggest, it seems that housing market exuberance in Copenhagen does not directly spill over to other parts of the country.

This is an important negative finding. If a dominant source market had to be nominated before looking at the data, Copenhagen would be the obvious candidate. It is the country's capital, its largest market, it has the highest GSADF statistic, and it has the highest number of months flagged as bubble out of all the areas. Yet, under the implemented contagion procedure, it produces no statistically significant $\hat{\delta}_{2j}(r)$ transmission path at all. That sharply limits any interpretation based on a Copenhagen-led ripple effect.

Table 7: Summary of all statistically significant positive contagion results

Core → Destination	**Selected lag**	**Share of significant months**
014 → 002	0 months	89.5%
014 → 003	5 months	67.5%
014 → 005	1 month	100%
014 → 008	0 months	10.2%
014 → 010	0 months	7.5%

Table 7 and Figure 10 show that, unlike with Copenhagen, with Aarhus as core there are some statistically significant results. Positive and significant coefficients appear for a small set of destinations, especially areas 002, 003, and 005, and with a much shorter duration for areas 008 and 010. For the rest of the destinations, however, no evidence of bubble spillovers coming from Aarhus is found.

However, this result does not lend itself to a clean diffusion narrative. The most significant destination areas (002, 003 and 005) are not part of Jutland or geographically nearby to Aarhus, as would be expected if Aarhus were acting as a clear core market. Instead, those areas are part of the Capital Region, consisting mainly of the suburbs of Copenhagen and other nearby towns. In addition, Table 7 shows that 3 out of the 5 statistically significant regressions use 0-month delay, so those cannot be interpreted as explosive behaviour in Aarhus preceding that in other areas.

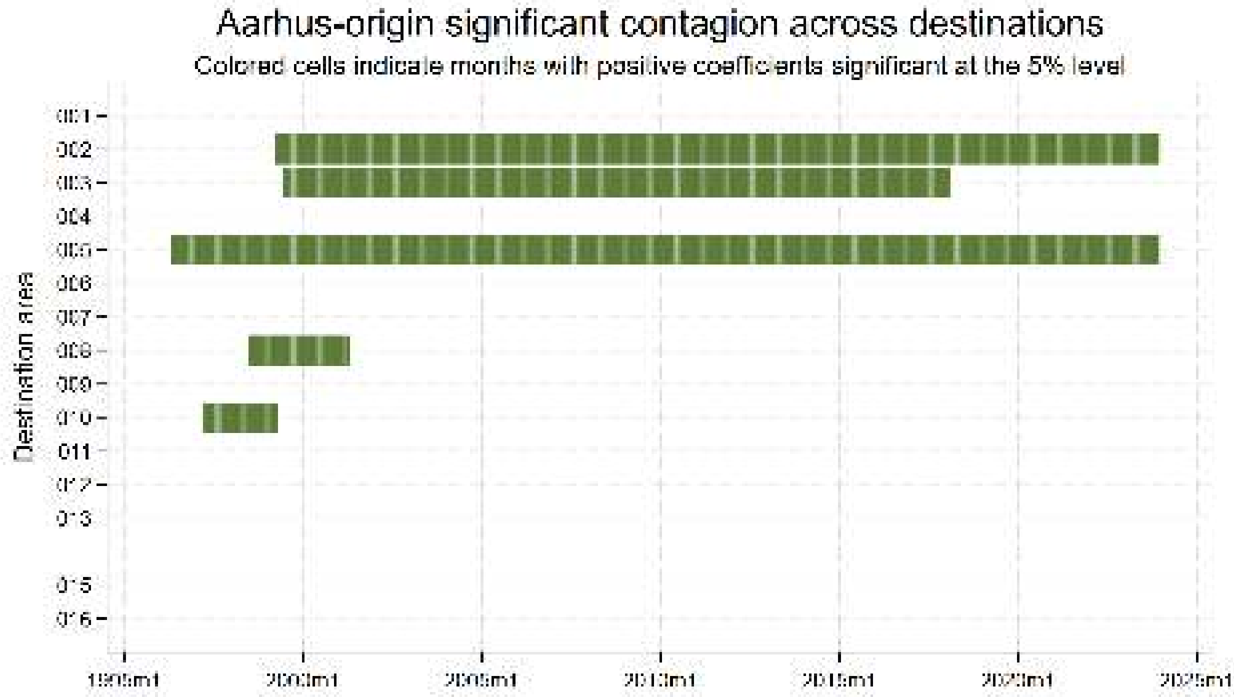


Figure 10: Chronology of statistically significant positive contagion for pairs with Aarhus as origin. Colored cells denote months in which the estimated $\hat{\delta}_{2j}(r)$ coefficient is positive and statistically significant at the 5% level.

Furthermore, as Figure 11 shows, there does not seem to be any evidence of this diffusion relationship being stronger during times of exuberance. The three main pairs, 014 → 002, 014 → 003, and 014 → 005, are statistically significant for long stretches of the sample, in a smooth and persistent manner. The 014 → 005 $\hat{\delta}_{2j}(r)$ path is the clearest example. It is positive and significant almost everywhere, but it is also nearly flat. This also puts into question, at least for the present data, the benefit of the time-varying procedure followed in this thesis, when perhaps a more parsimonious approach would output similar results with the added benefits of easier interpretation and lower computing complexity.

All of that does not make the estimates invalid, but it does make the economic story less persuasive. That is difficult to read as the transmission of a speculative episode at a particular point in time. Put differently, the significant results look more like long-run positive co-movement in BSADF paths than like a wave of speculative activity moving outward from Aarhus.

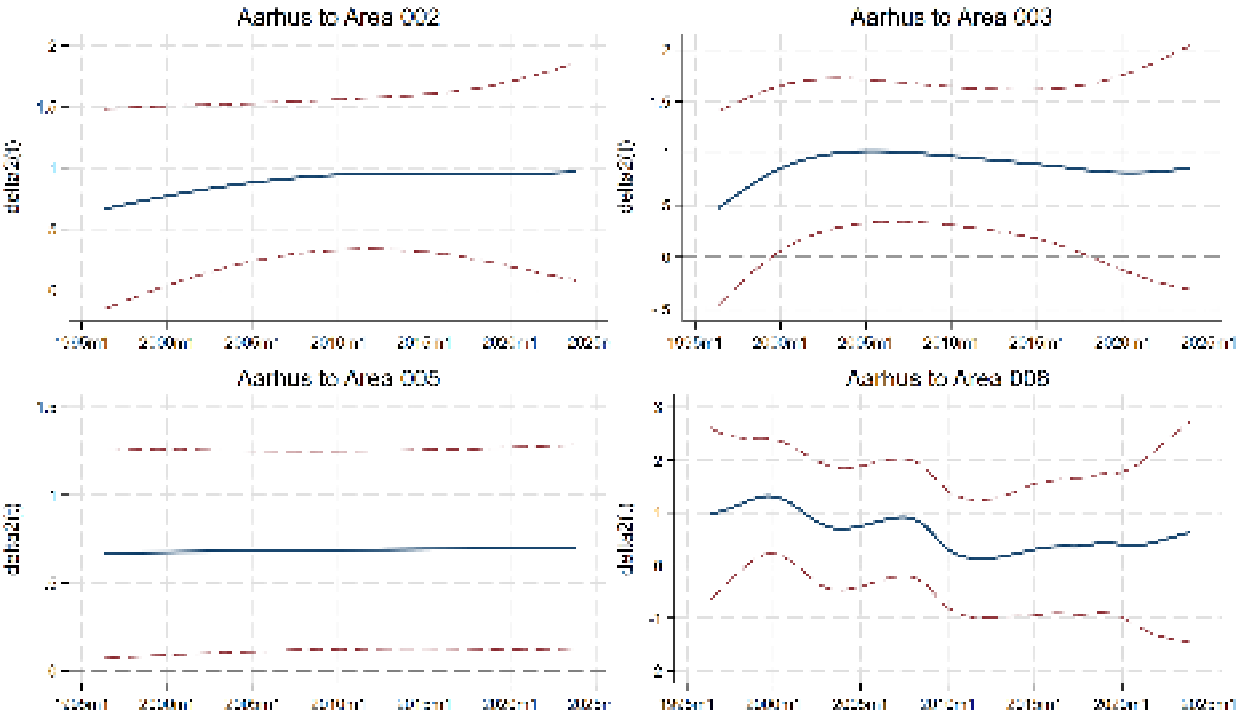


Figure 11: Selected Aarhus-origin contagion paths. The figure reports the three strongest positive cases, 014 → 002, 014 → 003, and 014 → 005, together with 014 → 008 as a weaker comparison case. Dashed lines denote the 95% confidence interval.

The relation between the two largest cities is also unconvincing. Copenhagen does not significantly transmit to Aarhus, and Aarhus does not significantly transmit to Copenhagen. Economic intuition could suggest a link between the two main cities, but the results do not support that reading in any robust statistical sense.

Overall, the contagion stage shows a narrow and rather unconvincing set of positive results. Copenhagen produces no statistically significant paths. Aarhus produces a few, but those cases are not sufficient evidence to argue for a case of strong bubble transmission.

5.2.3 Joint reading of detection and contagion results

Read together, the two stages of the analysis point to a sharp contrast. Bubble detection results are strong, while contagion results are weak.

The detection stage shows that Denmark experienced a broad housing bubble during the mid-2000s. The date stamping procedure for the aggregate data identifies a national episode of exuberance from April 2005 to January 2007, while the area-based analysis classifies 15 of the 16 areas as bubble months simultaneously from October 2005 to September 2006. The later aggregate episode in the 2020s is, by contrast, narrower, and only found in some areas.

The contagion stage does not turn that chronology into a clear leader-follower structure. Under the implemented specification, pairwise transmission from Copenhagen is absent, and pairwise transmission from Aarhus is limited to a few cases that are not especially persuasive as bubble contagion. This means that the widespread bubble chronology of the mid-2000s should not be read too quickly as a sequence in which one core area clearly preceded the rest of the country.

A more convincing interpretation is that Danish housing dynamics contain a strong common component. In the mid-2000s, that common component was large enough to generate overlapping bubble episodes across almost the entire country. In the 2020s, bubble behaviour reappeared briefly in some areas, especially in the Capital Region. In neither case does the second-stage procedure provide strong evidence that one local core market transmitted bubble conditions to the others in a stable, statistically significant way.

5.3 Open-Market Loss Transactions during the 2008 Crisis

Before interpreting the survival coefficients, a series of methodological checks were conducted to assess whether the assumptions of the Cox proportional hazards model hold and to identify, among the six estimated specifications, the windows that are consistent with the requirements of the framework. These checks are applied to twelve estimates, and the model-selection step is reported before any economic interpretation. The unconditional incidence of strict open-market loss sales is comparable across the three samples (approximately 5.5%) and the conditional distributions across LTV, liquidity, DTI, and wealth groupings exhibit consistent patterns, supporting the comparability of the three windows (see Appendix "Summary statistics discounted sales").

5.3.1 Multicollinearity Check

Standard post-estimation collinearity diagnostics are not directly applicable after Cox regressions; an auxiliary OLS regression was estimated using the strict open-market loss-sale indicator as the dependent variable and retaining the full set of regressors. The corresponding Variance Inflation Factor (VIF) was computed. Most of the financial regressors of interest exhibit VIF values below 5, although the wealth deciles 1, 2 and 3 record values between 5.0 and 6.4. This is interpreted as moderate but acceptable, since the deciles are mechanically correlated by construction. The only variables exhibiting high VIF values ($\gg 10$) are the squared and cubic transformations of property age, square meters, number of bathrooms, and household-head age. This collinearity arises by construction from the polynomial expansion and it does not affect the financial regressors. The detailed VIF table is reported in the Appendix (see "VIF statistic").

5.3.2 Proportional Hazards Test and Final Selection

The Schoenfeld global test (or "phtest" in Stata) was performed and compared with the functional-form misspecification test (in Stata "linktest"). Of the twelve specifications analysed in this thesis, only those associated with the 2006 purchase cohort (Models 5 and 6 in the methodology numbering) delivered estimates that simultaneously satisfy the global proportional hazards test and the link test, in both the additive and the full-interaction forms. The remaining specifications are reported in the Appendix. They failed at least one of the two structural tests (see Figure 12) so their interpretation is less reliable; for this reason, this section focuses on models 5 and 6. For the four retained specifications associated with the 2006 cohort, the global Schoenfeld test confirmed the proportional hazards assumption at conventional levels (p-value > 0.05)[49]. The link test indicated correct functional form: the squared term `_hatsq` was non-significant in all four retained specifications ($p > 0.35$ in every case)[50]. It is noted that some individual controls show statistically detectable departures from proportionality on the Schoenfeld test, but the global test is the criterion relied on at the model level. With several

[49]"Rejection of the null hypothesis of a zero slope indicates deviation from the proportional-hazards assumption". See "PH assumption test" link to official stata website in References.

[50]If the model is correctly specified, then regressing on the prediction and its square should not reveal additional explanatory power.
See "Link test for model specification" link to official stata website in References.

thousand failure events, the test is highly powered and minor deviations become statistically detectable even when they are economically small.

The cohort-2005 specifications (Models 3 and 4) deliver only a partial validation, with Model 4 satisfying the proportional-hazards test in the full-interaction form ($p = 0.281$) but failing the link test ($p = 0.007$). The unrestricted-window specifications (Models 1 and 2) fail the Schoenfeld test in both functional forms but satisfy the link test, suggesting that the binding violation is the proportionality of the hazards rather than the functional form. The following tables show the values of the statistics in each model. In red, the reason why the model was excluded based on the statistics is shown. In marked yellow, the models that passed the ph test are shown.

Model 1 (first specification)		
Phtest		0,0000
Linkstat	0,000	0,361
N subjects		93768
N failures		5163
Model 1 (second specification)		
Phtest		0,0000
Linkstat	0,000	0,850
N subjects		93768
N failures		5163

Model 2 (first specification)		
Phtest		0,0028
Linkstat	0,000	0,511
N subjects		88496
N failures		4319
Model 2 (second specification)		
Phtest		0,0072
Linkstat	0,000	0,460
N subjects		88496
N failures		4319

Model 3 (first specification)		
Phtest		0,0000
Linkstat	0,000	0,009
N subjects		43988
N failures		2477
Model 3 (second specification)		
Phtest		0,0000
Linkstat	0,000	0,018
N subjects		43988
N failures		2477

Model 4 (first specification)		
Phtest		0,0014
Linkstat	0,001	0,082
N subjects		39987
N failures		1891
Model 4 (second specification)		
Phtest		**0,1038**
Linkstat	0,001	0,007
N subjects		39987
N failures		1891

Model 5 (first specification)		
Phtest		**0,2807**
Linkstat	0,000	0,409
N subjects		46694
N failures		2542
Model 5 (second specification)		
Phtest		**0,0929**
Linkstat	0,000	0,620
N subjects		46694
N failures		2542

Model 6 (first specification)		
Phtest		**0,3979**
Linkstat	0,000	0,371
N subjects		45422
N failures		2284
Model 6 (second specification)		
Phtest		**0,0967**
Linkstat	0,000	0,453
N subjects		45422
N failures		2284

Figure 12: Diagnostic statistics and model-selection outcomes.

Model	**Buy period**	**Origin**	**Market regime**	**Reg. 1**	**Reg. 2**
Model 1	2005m4–2007m1	$\tau_{p,i}$	Mixed (boom & bust)	Failed	Failed
Model 2	2005m4–2007m1	T_{peak}	Pure deflationary	Failed	Failed
Model 3	2005m4–2005m12	$\tau_{p,i}$	Late-boom focus	Failed	Failed
Model 4	2005m4–2005m12	T_{peak}	Pure deflationary	Failed	Failed
Model 5	2006m1–2006m12	$\tau_{p,i}$	Pre-peak / crisis	**Passed**	**Passed**
Model 6	2006m1–2006m12	T_{peak}	Pure deflationary	**Passed**	**Passed**

Table 8: Summary of empirical models and diagnostic outcomes.

5.3.3 Economic Rationale for the Exclusion of Models 1–4

Models 1 and 2 fail the Schoenfeld test for an economically interpretable reason. The market regimes over which the effect of the same control variable fundamentally change. [51] The Cox model assumes that the ratio of a control on the hazard of a loss sale remains approximately constant over time; this requirement is violated whenever a single specification mixes the boom phase, with the crash phase. Notably, Models 1 and 2 satisfy the link test, suggesting that the functional form of the model is correct and that the binding violation is regime-mixing, not misspecification.

The cohort-2005 specifications (Models 3 and 4) reduce the entry window but still capture households that bought early in the bubble. By the time the crash hits, these households have already accumulated more than two years of price appreciation, which gives them an equity buffer. The same coefficient means different things at different points in the observation window, and this is exactly the kind of time-varying behavior that the proportional hazard test and the linktest pick up. The test statistics flag the issue, so these models were excluded.

The two retained models (Models 5 and 6) isolate a consistent regime. Model 5 captures households that purchased throughout 2006, immediately exposing them to the depreciation that began in late 2007 and continued until the macroeconomic trough. Model 6 considers the same 2006 cohort but synchronizes the timeline to the macroeconomic peak, restricting the survival window to the deflationary phase (August 2007 to January 2012). By isolating a single market regime dominated by systemic deflation, the Hazard Ratios across the leverage and liquidity categories remain statistically proportional over time[52].

5.3.4 Interpretation of the Cohort-Selection Outcome

In conclusion, the diagnostic outcome can be seen as evidence about the empirical conditions under which controls remain informative for the entire crisis horizon. The 2006 cohort is the unique entry window in which households purchased close enough to the peak to make their post-purchase financial position a meaningful predictor of behavior throughout the deflationary phase. Households that entered the bubble in 2005 had time to accumulate equity gains that altered their effective leverage trajectory and fragmented the mapping between post-purchase conditions and observed survival.

5.3.5 Descriptive Evidence: Kaplan-Meier cumulative Loss-Sale Estimates

Before turning to the parametric estimates, non-parametric Kaplan–Meier failure curves[53] stratified by leverage category and by liquidity decile and computed on the cohort-2006 sample are presented.

[51] Since this time window aligns with the expansionary bubble phase detected by the PSY test, is rational to assume that the market regime was very different in the expansionary phase compared to the contraction phase (very likely the effect is close to being the opposite); same argument can be applied to models 3 and 4.

[52] Unfortunately, the bubble-detection procedure detected the expansion phase only until January 2007, so the 2007 cohort was not examined. However, that would have been a useful extension of the results and it could have been used to validate the model findings.

[53] Illustrates the cumulative probability, defined as the "Proportion of items in the population dying (or experiencing the event)" by time t, which is calculated as $1 - S(t)$, where $S(t)$ is the probability of survival at time t. See Kaplan and Meier (1958).

These estimates impose no functional structure beyond the survival framework itself, and therefore provide a visual assessment of the central pattern that the Cox specifications formalize.

Loss-sale event rates. Figure 13 reports the cumulative failure probability of the loss-sale event stratified by the four LTV categories defined in the methodology. The pattern is monotonic and inverse to the prediction of the standard financial-distress literature since the cumulative probability of a strict open-market loss sale declines as the leverage category rises at the end of the observation window. Households in LTV category 1 (LTV ≤ 0.6) exhibit the highest cumulative failure probability; the curve flattens systematically as one moves to category 2 ($0.6 < \text{LTV} \leq 0.8$), category 3 ($0.8 < \text{LTV} \leq 1.0$), and finally category 4 (LTV > 1.0), which records the lowest cumulative incidence. *Higher leverage is associated with a lower probability of executing a market-loss sale.*

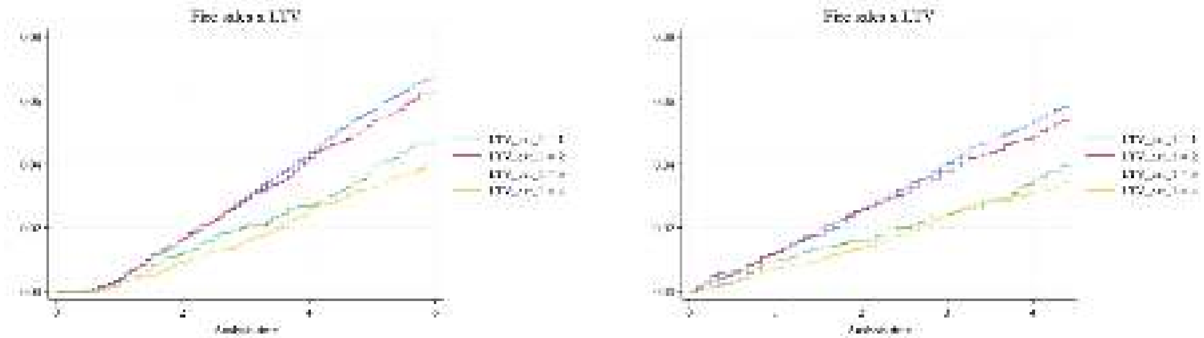


Figure 13: Kaplan–Meier failure estimates by LTV category, model 5-6

Why low-LTV households also show up among the loss sales

A reader may notice something that seems counter-intuitive in the results: households that had a low LTV at 31 December of the purchase year (category 1, LTV ≤ 0.6) are the most represented among the realized loss sales. This finding is, in principle, consistent with the equity constraint hypothesis: high-LTV households may face binding equity limits that prevent them from voluntarily executing loss-making sales, leaving the observed losses concentrated among low-LTV households who retain the option to transact. Before stating this as a finding, however, liquidity constraint should be analyzed jointly with LTV.

First, low-LTV after the purchase does not automatically mean wealthy as it could identify households that sold a previous property and reinvested the equity into the new one. These transactions could reflect older households relocating in response to exogenous demographic or occupational shocks. The underlying driver in this cases is mobility, not financial pressure. Second, two distinct frictions reduce the propensity to sell at a loss: a behavioural one (nominal loss aversion, Genesove and Mayer 2001) and a financial one (the equity constraint emphasised by Stein 1995 and Ferreira, Gyourko and Tracy 2010). For high-LTV households both bind simultaneously. For low-LTV households the second channel does not bind (the equity cushion can easily absorb the loss). Third, foreclosures are excluded by construction. During the crash, part of the high-LTV households ended up in foreclosure rather than in voluntary sales, and those observations leave the sample. Low-LTV households almost never go into foreclosure; they stay in the sample and they can execute a loss sale. The cross-section of observed loss-sale events is tilted toward low-LTV households.

Loss-sale event rates by liquidity decile The corresponding stratification by liquidity decile (Figure 14, comparing deciles 1, 5, and 10) shows a different picture. The three curves are essentially overlapping over the observation window, suggesting that liquidity *in isolation* does not produce a clear monotonic pattern in the failure probability. This result already anticipates one of the central findings of the parametric analysis given that the role of liquidity is not as an independent driver of market exit, but it acts as a moderator of the leverage effect, an asymmetry that is more clear only once the LTV × liquidity interaction is explicitly modelled.

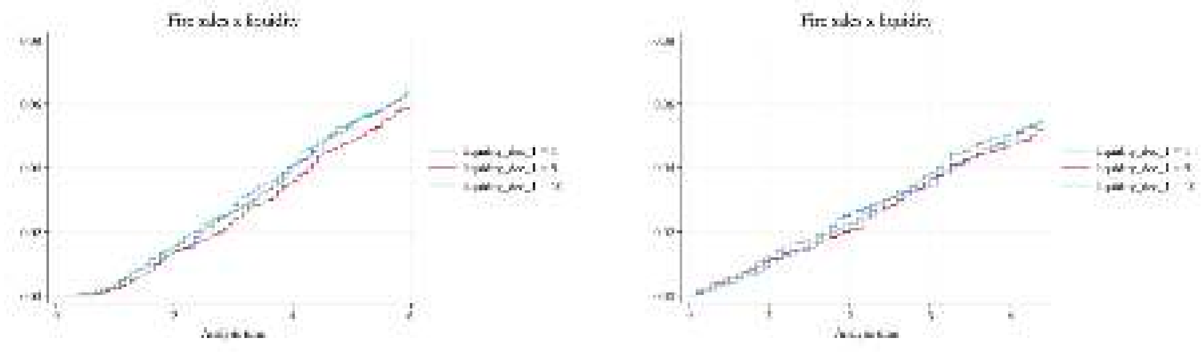


Figure 14: Kaplan–Meier failure estimates by liquidity decile (1, 5, 10), model 5-6

Hazard graphs. Hazard graphs[54] are also estimated. The curves are almost flat and they overlap over the whole sample for the liquidity deciles. In the LTV case, however, they start with different magnitudes; LTV cat 1 and 2 are above, on average, compared to LTV cat 3 and 4 (cat 1 and 2 almost overlap as categories 3 and 4).

[54] Displays the age-specific failure rate, representing the instantaneous risk of an event occurring at time t conditional on survival up to that point, as defined by Cox (1972).

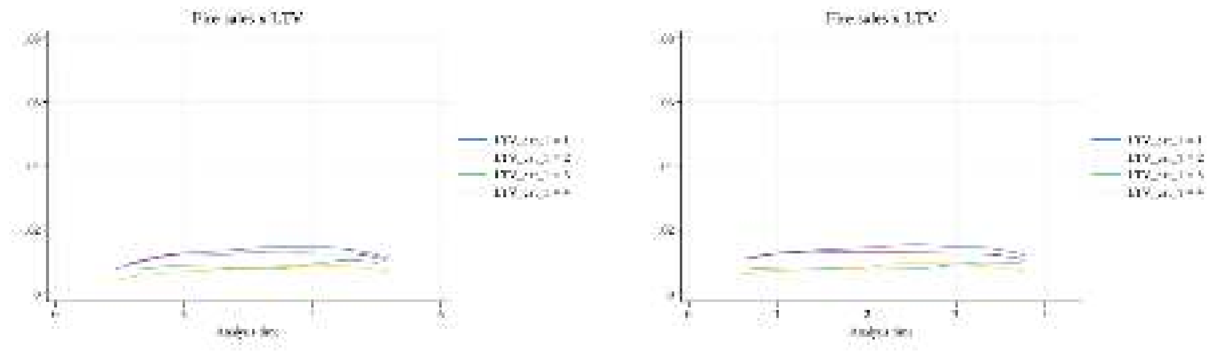


Figure 15: Hazard graphs estimates by LTV category, model 5-6

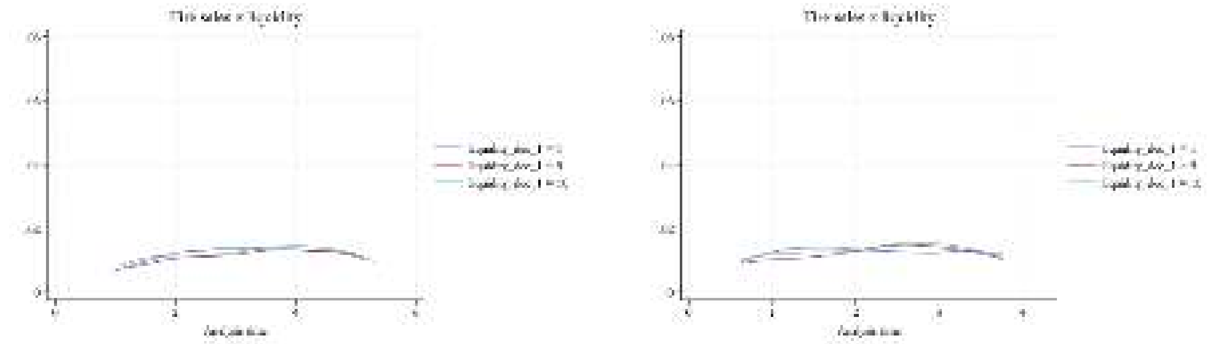


Figure 16: Hazard graphs estimates by liquidity decile (1, 5, 10), model 5-6

The hazard patterns confirm the cumulative failure results since liquidity alone does not stratify the failure rate, while LTV generates a clear and persistent ranking. This ranking aligns with the equity constraint hypothesis and its interaction with leverage. These plots provide an initial assessment of which variables drive separation in failure rates before conditioning on covariates; the Cox specifications below will test whether these patterns persist after controlling for financial situation post-purchase, household and property characteristics.

5.3.6 Parametric Results: The Cox Estimates

This section presents the empirical findings from the four retained specifications associated with the 2006 cohort: the baselines and the full-interaction specifications with both time-origin definitions [55]. Because the failure event strictly isolates voluntary market transactions executed at a loss relative to the regional baseline, the Hazard Ratios represent the cross-sectional relationship between initial financial conditions and the realized loss outcome. The discussion is organized in three steps: the baseline for the 2006 cohort with the purchase-date origin of time (model 5, Reg. 1), the full-interaction

[55] The full coefficient tables of all models are reported in the Appendix; the relevant tables for this section are Regression 1 results 5-6 and Regression 2 results 5-6.

specification on the same sample (model 5, reg. 2), and then these results are compared with the peak-to-trough specifications (model 6 reg 1 and reg 2).

Model 5 - Baseline specification

The baseline specification evaluates the independent associations between leverage, liquidity, and the hazard of an open-market loss sale, conditional on the structural and demographic controls. The estimation is run on $N = 46{,}694$ observations with $F = 2{,}542$ failures.

Leverage effects. Moving into higher leverage categories is associated with a lower hazard of completing a strict open-market loss sale. Relative to the LTV ≤ 0.6 baseline, LTV category 3 exhibits a Hazard Ratio of 0.637, and LTV category 4 a Hazard Ratio of 0.518, both significant at the 1 % level. This pattern seems to be monotonic. Each step into a higher leverage class is associated with a reduction in the hazard. This finding is consistent with the equity-constraint mechanism formalised by Stein (1995): selling below the outstanding mortgage requires the household to cover the equity gap at closing, which reduces the feasibility of loss-realising sales for the most leveraged households.

Liquidity effects. Liquidity deciles operate similarly: the lowest liquidity deciles exhibit Hazard Ratios significantly below the top-decile baseline. Hazard Ratios for the lower deciles range broadly between 0.74 and 0.86 (e.g. decile 1: HR = 0.800, decile 3: HR = 0.745, decile 5: HR = 0.763, all significant at the 5% level).

Wealth and DTI effects. The wealth gradient runs in the opposite direction to the liquidity one. The lowest wealth deciles exhibit Hazard Ratios significantly *above* one (decile 1: HR = 1.384; decile 5: HR = 1.346, both significant at the 5% level), and the gradient flattens as wealth rises. Less wealthy households at $t = 0$ are therefore associated with a higher hazard of a loss sale. Among the DTI deciles, decile 1 (the most stretched income-servicing class) attains marginal significance (HR = 1.259, significant at the 10% level), consistent with the idea that flow constraints contribute to the hazard but with smaller magnitude than stock imbalances[56].

Economic interpretation: the equity-constraint mechanism.

The leverage gradient is in line with the equity-constraint mechanism. Highly leveraged households face difficulty in selling, since (in the absence of refinancing) the transaction would require them to cover the gap between the sale price and the outstanding mortgage at closing. The liquidity gradient is harder to interpret. The lower hazard for illiquid households does not mean that liquid households are the ones *causing* sales; it more likely reflects a selection effect at origination. Banks tend to grant mortgages to illiquid borrowers only when they have other forms of protection that are not observed in the data, such as family support[57] or non-financial wealth[58] not recorded in the registers. The separate effects of leverage and liquidity are both consistent with the framework and with equity-constraint hypothesis, but the full interaction is needed to capture the behavioral mechanism behind the results.

[56]As shown in the following discussion, DTI measures are not very good at predicting a loss-sale event as they do not absorb a lot of variance and are often statistically not significant

[57]Or other forms of debt assumptions

[58]Illiquid assets such as other properties or assets

Model 5 - Full-interaction specification

The full-interaction specification reveals an asymmetry that remains invisible in the additive model: the role of liquidity *depends* on the leverage category to which the household belongs. The discussion is restricted to the cells that are statistically identified at conventional levels. All Hazard Ratios described in this section is significant at the 1% level.

Sign and magnitude of the interaction. Within the highest leverage category (LTV > 1.0), being in one of the bottom-to-middle liquidity deciles suppresses the hazard of selling. The third liquidity decile combined with category 4 LTV gives HR = 0.288, meaning these households are roughly 70% less likely to sell than the reference group (individuals in the most liquid decile and with the lowest mortgage). Other significant cells in the same column include the second decile (HR = 0.252), the fifth decile (HR = 0.469) and the sixth decile (HR = 0.420). The pattern is similar but more selective in category 2: for example, the fifth decile combined with category 2 LTV yields HR = 0.409, and the sixth decile combined with category 2 LTV yields HR = 0.438. The economic interpretation is that having a massive debt is not enough to lock a household in. What really matters is the combination of negative equity and the lack of liquid resources to cover the shortfall at closing. This pattern is identified at the bottom-to-middle of the liquidity distribution and at the top of the leverage distribution; it is not a uniform effect across all liquidity deciles.

Economic interpretation: wealthy deleveraging vs. poor paralysis. The financial-distress narrative predicts that the combination of high leverage and low liquidity should constitute the most vulnerable cohort, with the highest probability of a forced sale. In the voluntary-sale setting, the same combination is the least represented among the realized failure events. Households that entered the market with high leverage and low initial liquidity are doubly constrained, either because they do not tolerate the loss psychologically or because they lack the liquidity needed to clear the mortgage[59]. The hazard rises only when high initial leverage is combined with the top liquidity deciles, that is, with households whose post-purchase financial position was compatible with absorbing a nominal loss. Within the open-market framework, the realized loss sales are concentrated among the liquid households; whether this reflects an active deleveraging choice or a selection on post purchase conditions correlated with the willingness to sell cannot be settled within this design.

Table 9 summarizes the comparison between the additive and the full-interaction specifications of Model 5. The transition between the two specifications shows how the leverage main effects, sharply estimated in the additive model, are absorbed into the interaction terms once the joint distribution of LTV and liquidity is allowed to vary.

[59]The controls are measured after the purchase; it is not observed how the household's liquidity position evolved through the crisis; for this reason the interpretation here refers to ex-post conditions.

Table 9: Model 5: Baseline vs. Full-Interaction Specification

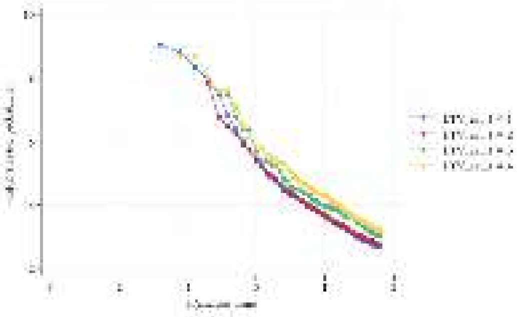
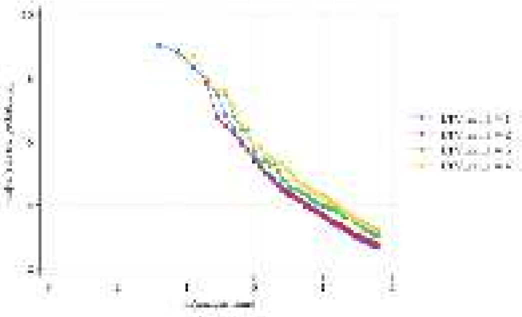
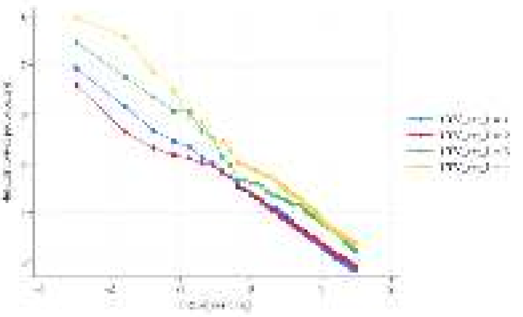
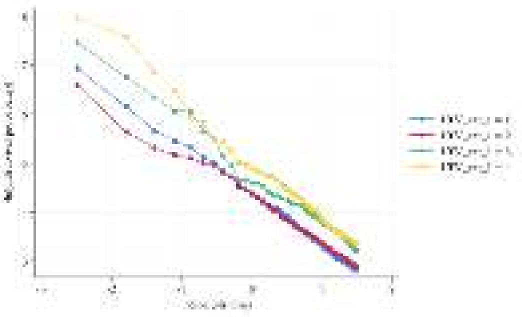
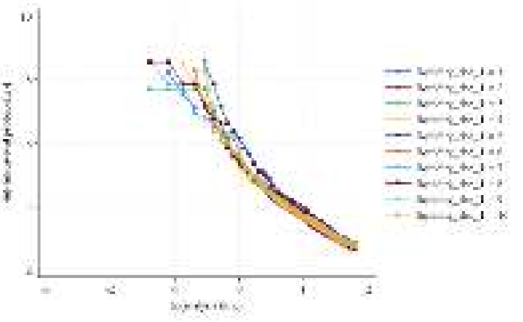
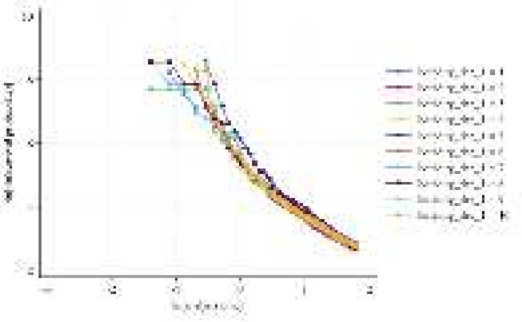
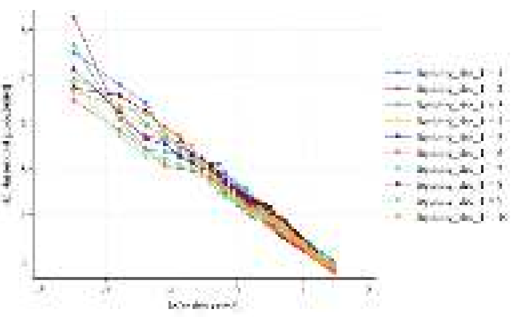
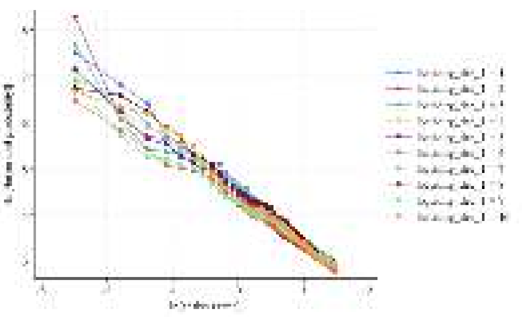
	Reg. 1 (Baseline)	Reg. 2 (Interaction)
Leverage (ref: LTV ≤ 0.6**)**		
LTV cat 2 ($0.6 < \text{LTV} \leq 0.8$)	0.904	1.440*
LTV cat 3 ($0.8 < \text{LTV} \leq 1.0$)	0.637***	1.125
LTV cat 4 ($\text{LTV} > 1.0$)	0.518***	0.934
Liquidity (ref: decile 10)		
Decile 1 (lowest)	0.800**	0.967
Decile 3	0.745***	0.917
Decile 5	0.763***	0.999
Decile 7	0.826**	1.034
Selected interactions (Liq $\times$ **LTV)**		
Liq 2 $\times$ LTV 4	—	0.252***
Liq 3 $\times$ LTV 4	—	0.288***
Liq 5 $\times$ LTV 4	—	0.469***
Liq 6 $\times$ LTV 4	—	0.420***
Liq 5 $\times$ LTV 2	—	0.409***
Liq 6 $\times$ LTV 2	—	0.438***
Other financial controls		
Wealth dec 1 (lowest)	1.384**	1.214
DTI dec 1 (highest)	1.259*	1.227
Housing controls	Yes	Yes
Demographic controls	Yes	Yes
Municipality stratification	Yes	Yes
Cluster-robust SE	Yes	Yes
Observations	46,694	46,694
Failures	2,542	2,542

Notes: Hazard Ratios reported. Standard errors clustered at the municipality level. *** p<0.01, ** p<0.05, * p<0.1. In the interaction model, the reference cell is LTV ≤ 0.6 (LTV category 1) combined with liquidity decile 10. Some interactions, deciles and LTVs are reported to show how liquidity and LTV interact.

Model 6 - Changing the time setting

The design of model 6 strips out the noise of the late expansionary phase and isolates the systemic shock from any residual heterogeneity in pre-peak holding behavior. The estimation is run on $N = 45{,}422$ observations with $F = 2{,}284$ failures. It is useful to look at the log-log survival plot to visually see the difference in the PH assumption of the two estimated Cox models[60].

[60]Note, usually these graphs are interpreted as follows: if the plotted curves for different categories run parallel, the ph assumption holds; if they intersect, the assumption is violated. However, the models were filtered based on the proportional hazard statistical test. These plots can be used as a proxy to understand what is going on behind the model but they are not perfectly representing it since the models are both stratified.

Log-log survival diagnostics. The top four panels compare LTV categories across models 5 and 6: the first regression (left column) and second regression (right column) of each model show the same pattern: category 1 (lowest LTV) at the top, categories 3 and 4 (highest LTV) at the bottom. Models 5 and 6 can be easily distinguished by the shape of these curves since model 6 looks more linear compared to model 5 thanks to peak-synchronised design (images 3-4-6-8). The bottom four panels show liquidity deciles: all ten curves overlap and converge by the end of the period, with the same pattern across both regressions of models 5 and 6. These plots pool observations across municipalities and do not reflect Cox stratification, however they show how the design of model 5 and 6 changes the survival probability. Same graphs for DTI and Wealth can be found in the appendix.

Comparison: Model 5 vs. Model 6, Baseline Specifications.

The two specifications produce very similar estimates. The LTV gradient is almost identical, and the liquidity deciles preserve their direction and individual significance in both models. The only noticeable difference is an attenuation in the significance of the wealth and DTI gradients (e.g. wealth decile 1 moves from $p \approx 0.04$ in Model 5 to $p \approx 0.07$ in Model 6), which is plausibly the consequence of dropping the pre-peak observations from the at-risk window. The stability of the LTV and liquidity coefficients across the two timeline definitions reinforces the interpretation that the financial gradients are not an artifact of the survival origin.

Table 10 reports the four retained specifications side by side. The stability of the LTV gradient and of the interaction terms across the two timeline definitions is the central robustness pattern of the analysis.

Table 10: Model 5 vs. Model 6: Main Coefficients Across Specifications

	Baseline (Reg. 1)		Interaction (Reg. 2)	
	Model 5	Model 6	Model 5	Model 6
Leverage (ref: LTV ≤ 0.6)				
LTV cat 2	0.904	0.911	1.440*	1.541**
LTV cat 3	0.637***	0.619***	1.125	1.145
LTV cat 4	0.518***	0.523***	0.934	0.985
Liquidity (ref: decile 10)				
Decile 1 (lowest)	0.800**	0.782**	0.967	0.990
Decile 3	0.745***	0.745**	0.917	0.939
Decile 5	0.763***	0.792**	0.999	1.055
Decile 7	0.826**	0.816**	1.034	1.036
Key interactions (Liq $\times$ LTV)				
Liq 2 $\times$ LTV 4	—	—	0.252***	0.277***
Liq 3 $\times$ LTV 4	—	—	0.288***	0.263***
Liq 5 $\times$ LTV 4	—	—	0.469***	0.405***
Liq 6 $\times$ LTV 4	—	—	0.420***	0.401***
Wealth and DTI				
Wealth dec 1	1.384**	1.349*	1.214	1.179
DTI dec 1	1.259*	1.218	1.227	1.186
Housing & demographic controls	Yes	Yes	Yes	Yes
Municipality stratification	Yes	Yes	Yes	Yes
Cluster-robust SE	Yes	Yes	Yes	Yes
Observations	46,694	45,422	46,694	45,422
Failures	2,542	2,284	2,542	2,284

Notes: Hazard Ratios reported. Standard errors clustered at the municipality level. *** p<0.01, ** p<0.05, * p<0.1. Reference cell in interaction models: LTV ≤ 0.6 combined with liquidity decile 10.

Stability of the interaction across temporal designs. As in Regression 1, the interaction terms also show very similar coefficients and significance levels. As before, the suppressing effect fades as the liquidity distribution moves upward and disappears at the top deciles. The fact that this asymmetric pattern shows up in both specifications, despite their different temporal structures, suggests that the dual-trigger mechanism is a real feature of the deflationary phase and not just an artifact[61].

Marginal effects of the LTV main term. In the full-interaction specification, the LTV main effects represent the leverage gradient evaluated at the omitted liquidity decile. Within the interaction terms, several cells combining high leverage with the lower-to-middle liquidity deciles show Hazard Ratios significantly below one. The economically interpretable variation is therefore concentrated in the interaction terms, which capture the joint mapping of leverage and liquidity.

[61] By aligning the timeline with the macroeconomic peak and following households until the trough, Model 6 follows the suggestion in Therneau and Grambsch (2000) of stabilizing the coefficients through a more homogeneous time window.

Economic interpretation. Once the noise of the expansionary phase is removed, Model 6 points to an almost identical reading of the data: among the 2006 cohort, the loss sales observed during the crash are concentrated among households that possessed higher liquidity at the end of the year, while those that started the period with constrained liquidity are markedly under-represented in the failure events. LTV categories also seem to be a very strong predictor of below-benchmark sales.

The asymmetric pattern is summarized visually in Figure 17, which reports the Hazard Ratios from the LTV $\times$ liquidity interaction for both Model 5 and Model 6. The deeper green cells in the LTV-4 column (rightmost in each panel), combined with low-to-middle liquidity deciles, identify the structural heart of the result: households with negative equity but limited after-purchase liquidity are markedly under-represented among the realized loss transactions, in both timeline specifications.

Model 5 (purchase origin)

	LTV 2	LTV 3	LTV 4
Liq 1	0.72	0.11**	0.77
Liq 2	0.77	0.36*	0.25***
Liq 3	0.80	0.49	0.29***
Liq 4	0.51**	0.32**	0.61
Liq 5	0.41***	0.50	0.47***
Liq 6	0.44***	0.65	0.42***
Liq 7	0.60	0.34**	0.54**
Liq 8	0.56	1.01	0.50**
Liq 9	0.75	0.60	0.68

Model 6 (peak origin)

	LTV 2	LTV 3	LTV 4
Liq 1	0.63	0.12**	0.60
Liq 2	0.72	0.41	0.28***
Liq 3	0.76	0.40	0.26***
Liq 4	0.55**	0.27**	0.59
Liq 5	0.41***	0.52	0.41***
Liq 6	0.42***	0.64	0.40***
Liq 7	0.56	0.38	0.54**
Liq 8	0.47**	0.89	0.48**
Liq 9	0.64	0.52	0.69

Figure 17: Hazard ratios from the LTV $\times$ liquidity interaction (Reg. 2) for Model 5 and Model 6. Greener cells indicate stronger hazard suppression relative to the reference cell. Reference cell: LTV ≤ 0.6 combined with liquidity decile 10 (HR = 1). The asymmetric pattern in the LTV-4 column (low-to-middle liquidity combined with negative equity) is the central empirical result, and it replicates almost identically across the two timeline designs. *** p<0.01, ** p<0.05, * p<0.1.

5.3.7 Interpretation of Control Variables

Across the retained specifications, comments are made only on the controls that show individual statistical significance and behave consistently with the empirical literature.

- **Wealth deciles.** Wealth deciles 1 and 5 in the first regression exhibit Hazard Ratios significantly above one in both models, consistent with a structural fragility channel. The intuition is that being in a lower wealth decile (less wealthy) after purchase increases the probability of selling with a nominal loss.

- **Debt-to-Income (DTI).** The first DTI decile, which captures the highest debt-service burden relative to income, exhibits a Razard Ratio above one but is statistically significant only at the 10% level in the first regression of model 5. Being in a lower DTI decile (having a lower debt-to-income ratio) at $t = 0$ is associated with a higher probability of an open-market loss sale.

- **Demographic controls.** Education dummy 1 attains significance in both models (HR > 1.18, $p \approx 0.015$), suggesting that lower educational attainment is associated with a higher hazard of an open-market loss sale[62]. Household-head age enters with HR $\approx$ 0.920 on the linear term, consistent with a lower hazard for older households[63].
- **Multiple-property indicator.** Households owning more than one property exhibit HR $\approx$ 1.15 in both models, supporting the interpretation that investor-type and multi-property households were more willing to execute deleveraging sales than single-residence owners[64].
- **Housing structural controls.** Interestingly, among the structural controls, the bathrooms count attains significance in both models (and in all the linear, quadratic and cubic polynomials). Bathrooms can be seen as a proxy for property size and price segment: larger and more expensive houses sell less frequently, trade in a thinner market, and tend to transact at larger discounts, which raises the probability of a sale below the regional benchmark[65]

All the control results are summarized in the table below:

	Model 5		**Model 6**	
Variable	**Reg. 1**	**Reg. 2**	**Reg. 1**	**Reg. 2**
Wealth dec. 1	1.384**	1.214	1.349*	1.179
Wealth dec. 2	1.339**	1.181	1.256	1.102
Wealth dec. 5	1.346**	1.183	1.361**	1.188
DTI dec. 1	1.259*	1.227	1.218	1.186
Age (linear)	0.920**	0.921**	0.926*	0.926*
Age (squared)	1.001*	1.001*	1.001	1.001
Education dummy 1	1.211**	1.211**	1.185**	1.185**
Multiple properties	1.145**	1.166**	1.172**	1.195**
Bathrooms (linear)	7.435***	7.503***	10.04***	10.08***
Bathrooms (squared)	0.332***	0.331***	0.284***	0.284***
Bathrooms (cubic)	1.184***	1.185***	1.212***	1.212***
Observations	46,694	46,694	45,422	45,422

Table 11: Hazard Ratios of control variables across models 5 and 6, both regression specifications

Notes: Hazard Ratios from stratified Cox regressions. Robust standard errors clustered at the municipality level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Reference categories: wealth dec. 10, DTI dec. 10, educ. dummy 5. Reg. 1: baseline specification. Reg. 2: with LTV × liquidity interaction.

[62] Definitions of education dummies are discussed and defined in the Appendix.

[63] The reason could be that older people have a lower propensity to change their house because they do not need to move for work or they are simply more experienced.

[64] Plausibly because the principal-residence anchoring effect that drives the standard equity constraint hypothesis is weaker for non-primary holdings.

[65] The magnitude of the *Bathrooms* coefficient should be interpreted with caution, since the linear, quadratic, and cubic terms partially offset each other.

5.3.8 Final Economic Interpretation

The synthesis of the two retained specifications leads to a coherent reading of the Danish housing crash, conditional on the sample of voluntary open-market sales (foreclosures excluded) and on the fact that all the controls are measured after purchase.

The data suggest that high leverage on its own is not associated with a higher hazard of an open-market loss sale; the elevated LTV categories show a lower hazard compared to the category 1 and 2. This pattern is consistent with the equity constraint effect found in previous studies: households with negative or low equity are less likely to sell at a loss because they lack the funds to cover the mortgage gap. The most financially trapped households simply do not sell during the studied period, remaining in the non-transacting (censored) group. This baseline appears to be moderated by liquidity. Among households entering with high leverage, those who also entered with stronger after-purchase liquidity are more often present among the realized loss transactions, while the symmetric cell, high leverage combined with low liquidity, is less represented. The cross-section of households that ended up bearing a voluntary loss appears tilted toward the liquid, although the underlying behavioral mechanism is not directly observable in the framework[66]. By construction, the foreclosure-based dual-trigger mechanism cannot be tested in the sample; what is observed is that within the voluntary-sale segment the symmetric pattern does not emerge.

This asymmetry has direct implications for the interpretation of the standard double-trigger hypothesis. In the foreclosure literature, the joint occurrence of negative equity and a liquidity shock is identified as the canonical trigger of forced sale. In the open-market segment examined here, the same joint occurrence is the cell that is most empirically absent from the failure events. The data identify a cross-sectional mapping between post-purchase conditions and realized outcome, not the dynamic mechanism through which households arrived at the sale decision.

5.3.9 Robustness Checks Summary

The convergence of the retained specifications across distinct temporal windows (Model 5 vs. Model 6) and functional forms (additive vs. full-interaction) supports the stability of the estimates with respect to the timeline specification. The asymmetric interaction between leverage and liquidity persists across the cohort-restricted purchase-date design and the peak-synchronized design. Three statistical tests support these results. First, the global proportional hazards test passes for all retained specifications, supporting the interpretability of the Hazard Ratios as approximately constant proportional effects over the observation window. Second, the link test does not detect functional misspecification, supporting the adequacy of the categorical discretization for leverage and the decile partition for liquidity, wealth, and DTI. Third, the geographical stratification of the baseline hazard at the municipality level, combined with cluster-robust standard errors at the same level of aggregation, controls for unobserved spatial heterogeneity and partially absorbs intra-cluster correlation, providing a conservative inference

[66]This pattern can be read as complementary to the macroeconomic perspective of Mian and Sufi (2010), who document that high household leverage amplifies the depth of housing downturns; the results add that, within the voluntary-sale segment of one such downturn, the realized loss transactions are over-represented among households whose after-purchase liquidity was stronger.

framework. Two residual limitations are acknowledged that condition the interpretation of these findings. First, the controls are measured post-purchase, while the crisis unfolds over years; the resulting attenuation bias likely produces conservative estimates of the true dual-trigger effect, as the predictive content of post-purchase conditions decays as the household's financial trajectory diverges from its starting point. Second, by excluding foreclosures, inference is restricted to the voluntary-sale segment; households whose liquidity constraint became binding through a forced foreclosure are absent from the sample. The combination of these two channels suggests that the estimates should be read as a conservative, association-based assessment of the joint role of leverage and liquidity in the timing of voluntary loss sales during the Danish housing market during the 2008 crash, rather than as a precise quantification of its full causal magnitude. Finally, a breakdown of the limitations:

Table 12: Methodological Limitations: Manifestation and Visibility in Results

Limitation	**Expected Effect**	**Evidence in Results**	**Visible?**
Post-purchase covariates	Information decay with time	Covariates measured at year-end of purchase. Only the 2006 cohort passes the proportional-hazards and link-test diagnostics, consistent with attenuation worsening for earlier cohorts.	Yes
OBLGAELD (LTV proxy)	Attenuation of LTV effect	LTV Hazard Ratios remain robust, but classical measurement error in the bond-debt proxy alters the coefficient.	No
Foreclosure exclusion	Voluntary sales only	Risk set restricted to open-market transactions; forced sales absent. Findings describe the voluntary-sale segment and are framed as such.	Yes
Endogenous censoring	Downward bias in high-LTV / low-liq cells	Constrained households are more likely to remain censored through lock-in or foreclosure.	*Implicit*
Local benchmark rule	Measurement error (property quality)	Some structural controls show large coefficients (e.g. bathroom HR ≈ 7), consistent with the rule absorbing unobserved quality. Reported but not attributed.	*Partial*
Categorical binning	Mild attenuation around thresholds	LTV in four categories, financials in deciles. Within-bin homogeneity smooths threshold effects; signs preserved, magnitudes compressed.	*Implicit*
Liq./Wealth (proxies)	Selection confounding	Liquidity and wealth deciles capture post-entry capacity, not the contemporaneous trigger.	No

Note: Seven methodological limitations from Section 4.6.7 mapped to their empirical manifestations in Section 5.3. "Yes" = explicitly acknowledged; "Implicit" = visible but not attributed; "Partial" = reported but not linked to its source.

5.3.10 Non-Retained Specifications

Models 1 and 2 fail the Schoenfeld test in both functional forms but satisfy the link test, indicating that the binding violation is the proportionality of the hazards rather than functional-form misspecification. Model 3 (cohort 2005, purchase-date origin) fails the Schoenfeld test in both forms. Model 4 (cohort 2005, peak-to-trough origin) satisfies the Schoenfeld test in the full-interaction form but fails the link test. The point estimates for these four models qualitatively replicate the asymmetric LTV $\times$ liquidity interaction recovered in the retained specifications, but the diagnostic failures preclude their use for inference. They are included in the documentation as a sensitivity benchmark and as evidence that the cohort-2006 selection emerges from a structural property of the data rather than from arbitrary specification choices. All regression tables can be found in the "Open-market Loss Sales" section in the appendix.

6 Conclusion

A crucial first step of the framework involved constructing a price index that supports all the subsequent econometric work. A repeat-sales estimator was adopted, with paired transactions trimmed at the extremes. A hedonic specification was estimated in parallel as a robustness check and is retained only to motivate its exclusion. Indices are produced at the area-by-month level for the contagion and detection work and at the national level for cross-checking. Validation against the Statistics Denmark benchmark indicates close co-movement in both levels and year-on-year growth rates over the 1992–2023 window, including around the 2008 inflection point, and the year-on-year variation in transaction counts is aligned with the aggregate market. The trimmed real index is therefore treated as a well-behaved proxy for the empirical analysis that follows.

With that input in place, the thesis turned to the three questions opened in the introduction: how speculative episodes in housing markets can be detected, whether they propagate geographically once ignited, and which households end up bearing the cost when the cycle turns in an open-market setting. The Danish data offer an answer to each, and in two cases an answer that runs against the initial prior.

On the first question, the results of applying the composite bubble dating analysis on the aggregate national index show evidence of two periods of exuberance. The first is the clearest and it takes place between April 2005 and January 2007. The area-based analysis shows that this episode is present across the whole country, with the most evidence for statistical explosiveness found in Copenhagen and the least in Northern Jutland. The second takes place from September 2020 to December 2021. Unlike the prior one, this episode is not ubiquitous to the whole country. In fact, the area-based composite methodology shows that, after taking into account the Hodrick-Prescott filter and the minimum-duration condition, only two areas, north of Copenhagen, show evidence of exuberance. Even if this limited result is due to the bubble being short enough in most areas to be filtered out by this rule, the pure BSADF results still only show exuberance in about half of the areas, making it safe to conclude that this bubble was shorter and more localized than the first.

On the second question, the contagion evidence is much weaker than the detection evidence. None of the fifteen ordered pairs originating in Copenhagen produces a statistically significant transmission path, despite Copenhagen being the largest city and the location with the most evidence of exuberance. Aarhus generates some significant pairs, but the geography of those connections does not point towards a narrative of Aarhus as a core area: the significant destination areas are not located close to Aarhus, in the periphery of Aarhus, but rather in the periphery of Copenhagen, which is not the pattern a centre-periphery diffusion story would predict. The Copenhagen-to-Aarhus pair is also not significant in either direction. The data suggest that the common pattern visible in the bubble chronologies is probably better read as the response of many local markets to broadly shared national forces instead of as a sequential diffusion story from one identifiable origin.

On the third question, results call for more cautious interpretation. A stratified Cox proportional-hazards model is estimated on the cohort of households purchasing in 2006, the unique window for which both the Schoenfeld test and the link test are passed across the retained specifications. Foreclosures are excluded by construction, and the failure event is defined by nominal loss, relative underperformance compared to the local area and timing of sale. Within this cohort, the data suggest that elevated loan-to-value categories are associated with a lower hazard of an open-market loss sale. The realised loss transactions appear tilted toward initially liquid households, and the cell combining high leverage with low liquidity appears under-represented. One possible reading is that the most constrained households did not transact at a loss because if they did, they would perhaps be unable to purchase another home afterwards due to not having enough liquidity to afford the downpayment, or even in some cases to close their mortgage. Instead they remained in the right-censored portion of the sample in a manner consistent with equity lock-in. The evidence is suggestive rather than confirmatory. This reading is conditional on several modelling assumptions. These assumptions include: proportional hazards, right-censoring at the trough, assumptions on the identification strategy and right use of post-purchase controls.

Each of these answers carries nuances and limitations. The bubble indicator detects explosive, above-trend behaviour, not overvaluation relative to a measured fundamental. The contagion bootstrap holds the lag and bandwidth parameters fixed at their original-sample values, a transparent simplification that future work with greater computing capacity could relax. The survival analysis relies on financial conditions measured after the purchase, whose informational content decays as the crisis unfolds.

7 Further Research

Although this thesis provides valuable insights into housing bubbles, many aspects remain open for improvement and further investigation.

First of all, the price-index stage could be extended by combining the repeat-sales and hedonic approaches in the spirit of Case and Quigley (1991). The repeat-sales estimator tracks the official DST benchmark closely, but it discards every single-sale transaction and weights paired sales of very different quality equally, using a hybrid estimator instead could recover the cross-sectional information held in those single sales and re-weight the pairs by the closeness of their hedonic match. Adopting methods that require fewer observations to construct a well-behaved index would allow analysis of this subject at a smaller-scale, perhaps within submarkets in Copenhagen and Aarhus.

On the detection side, future work could apply the PSY procedure not only to real house price indices, but also to valuation ratios such as price-to-rent or price-to-income. The detection strategy used in this thesis identifies explosive and above-trend price dynamics, but it does not directly estimate whether prices exceed a measured fundamental value. Its interpretation therefore relies on the maintained assumption that the relevant fundamentals are not themselves explosive. Testing price-to-rent or price-to-income ratios would reduce this limitation, because the explosive component would then be measured relative to an observable proxy for fundamentals rather than in prices alone. If the same episodes remained visible in those ratios, the evidence for non-fundamental exuberance would be stronger. If, instead, the explosive behaviour disappeared once prices were normalised by rents or income, the results would be more consistent with rapidly growing fundamentals than with a bubble interpretation. This extension would, however, require harmonised monthly measures of rents or income at the same geographical level as the constructed house-price indices. In the Danish setting, this is not a trivial requirement, since rent regulation, compositional differences between owner-occupied and rental housing, and the frequency at which income is observed could all affect the construction and interpretation of the ratios.

On the contagion side, future work could relax the conditional nature of the bootstrap inference. In the present thesis, the lag and bandwidth parameters are selected in the original sample and then held fixed across bootstrap replications for computational reasons. This makes the bootstrap feasible, but it means that the reported confidence intervals are conditional on the selected tuning parameters. A more complete procedure would re-select both the lag and the bandwidth inside each bootstrap replication. This would propagate the full uncertainty generated by the tuning stage into the final confidence intervals. Such an extension would be especially relevant for the marginally significant Aarhus transmission paths, since those results may be sensitive to the particular lag and smoothing window selected in the original sample. If the Aarhus effects survived a bootstrap procedure that re-optimised the tuning parameters in every replication, the evidence for contagion from Aarhus would be more convincing. If they disappeared, the current results would be better interpreted as fragile local correlations in the BSADF paths rather than as robust transmission effects.

A further extension would be to move beyond contagion models that require one candidate core area to be specified in advance. The current design follows a core-to-destination logic, where Copenhagen and Aarhus are treated as possible origins of transmission. This is transparent and easy to interpret, but it imposes a directional structure before the data are analysed. Future research could instead develop a more symmetric framework in which all areas are allowed to transmit explosive dynamics to one another. For example, an all-to-all pairwise contagion model, a network-based specification, or a time-varying spatial model could be used to identify whether some areas become temporarily central during specific episodes rather than being assumed to be permanent core markets. Such approaches would also make it possible to distinguish more clearly between true sequential diffusion and a common national shock that raises explosiveness across many areas at roughly the same time. This distinction is particularly important given the results of this thesis, where the detected bubble chronologies often appear similar across areas, while statistically significant contagion paths are limited once bootstrap inference is applied. The hazard analysis rests on assumptions whose violation would materially alter the interpretation: the proportional-hazards form, the post-purchase measurement of financial covariates, foreclosure exclusion as a sample-selection device, classical measurement error in the LTV proxy, and the local-benchmark event rule. Future research could explore each of these in turn rather than treating them as a single block. It would be informative to investigate alternative survival specifications that relax the proportional-hazards assumption such as accelerated-failure-time models with parametric or spline-based baselines, or fully non-parametric estimators to determine whether their results survive outside the 2006 cohort window. Further work might examine continuous LTV specifications at the cost of reintroducing a log-linearity assumption that the categorical specification was designed to relax; a comparison between the two would be informative. The more substantive extension concerns the failure event itself: rather than excluding foreclosures, future work could model voluntary loss sales and foreclosures jointly through a competing-risks framework, which would re-introduce the most constrained households into the risk set.

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Appendix

Data description

In this Appendix section, the main variables used in the code are presented, the filters applied to the datasets and indicate where they can be sourced from Statistics Denmark. Most of these variables are constructed using the Stata code developed by Andersen et al.(2022)

Table 13: Statistics Denmark registry modules used in the data compilation.

Category	Module	Content
Keys	`BOL ‘i’`	Identifiers linking individuals to residential units.
Household Characteristics	`BEF ‘i’`	Demographics, family structure, age, and gender.
	`IND ‘i’`	Individual and household income, assets, and debt.
	`INDSLUT ‘i’`	Tax deductions for home improvements and services.
	`UDDA ‘i’`	Highest completed formal education for adult members.
Owner’s Characteristics	`EJER ‘i’`	Historical real estate ownership, start dates, and transfers.
Property Characteristics	`nbpf2013`	Crosswalk key harmonizing pre/post-2007 municipality codes.
	`BOL ‘i’`	Physical housing attributes (living area, rooms, building age).
	`EJVK ‘i’`	Official property tax valuations and administrative assessments.
Transactions	`EJSA2023`	Administrative real estate transaction register, containing property sales records up to 2023.

Table 14: Filtering rules for the housing characteristics dataset

Code	Label	Exclusion criterion
0	keep	The observation meets all validity criteria and is retained in the sample.
1	wrong property type	The unit is not a regular owner-occupied dwelling. Administrative use-codes and property-type codes are used to actively flag and drop commercial buildings, agricultural properties, and other non-residential units.
2	not privately owned	The ownership code indicates that the property is held by companies, the state, municipalities, or housing associations, ensuring the sample captures exclusively properties owned by private households.
3	DST duplicates	The observation is explicitly flagged as an administrative duplicate directly by Statistics Denmark.
4	too big / incomplete	The living area is suspiciously large (exceeding 2,000 square meters), the number of rooms is reported as greater than 10, or information regarding both bathrooms and toilets is simultaneously missing.
5	too small / incomplete	The living area is suspiciously small (below 20 square meters), the reported number of rooms is 0, or the room count is missing entirely.
6	address duplicates	Multiple identical combinations of municipality and address codes point to the exact same record, indicating data entry overlaps.
7	merged addresses (reform)	Two previously distinct addresses became one following the 2007 Danish municipality reform.
8	split addresses (reform)	Several addresses point to a single property due to the aforementioned 2007 municipality reform, creating unresolvable spatial ambiguity.

Table 15: Filtering rules for the transaction dataset

Code	Label	Exclusion criterion
Panel A: Pre-merge conditions		
0	keep	The observation is a valid market transaction.
1	not household trade	The variables identifying the buyer and seller types indicate non-private entities. We restrict the sample exclusively to private household trades.
2	extreme price	The observation is flagged by an administrative dummy as having an outlier purchase price.
3	special circumstances	The transaction includes unusual, non-monetary obligations, meaning the recorded price is not representative of the true market value of the house.
8	foreclosure	The transfer code indicates a forced sale or auction.
9	family trade	The transfer code indicates an intra-family transfer, which typically occurs at artificially low prices for tax purposes.
Panel B: Post-merge conditions		
4	characteristics not found	The property could not be matched with physical characteristics despite the multi-year imputation strategy.
5	real price $<$ 100,000	The transaction is unrealistically cheap, likely indicating an unregistered special circumstance.
6	real price $>$ 20,000,000	The transaction is unrealistically expensive and structurally unrepresentative of the standard housing market.
7	size does not make sense	The recorded living area is zero for properties that are not classified as apartments.

The first three tables present the datasets from which the raw data were sourced, alongside the filtering procedures applied. The subsequent tables, in turn, detail the variables adopted directly from the codebase of Andersen et al. (2022), as well as those derived independently for the thesis.

Table 16: Variables utilized in the analysis: price indices and open-market loss dataset.

Variable	Description
Panel A — Price Index Variables	
`repeat`	Dummy for repeated sales
`repeat_sales`	Total number of repeated sales
`kom_2`	Property municipality code
`bopikom_2`	Owner residence municipality code
`pdate`	Exact date of purchase
`sold_price`	Nominal transaction price
`sold_price_R`	Real transaction price adjusted for inflation (2015=100)
`Lsold_price`	Natural logarithm of the nominal price
`Lsold_price_R`	Natural logarithm of the real price adjusted for inflation (2015=100)
`LLiving`	Log of total living area in square meters
`Rooms`	Total number of rooms
`Bathrooms`	Total number of bathrooms
`BuildingAge`	Age of the building at the time of transaction
Panel B — Open-Market Loss Dataset Variables	
`year`	Year of purchase date
`month`	Month of purchase date
`sold_apartment`	Dummy for apartment
`unoccupied`	Dummy for unoccupied
`ejer_owners`	Number of owners in the house
`ejer_Threeormore`	Dummy for houses owned by more than 2 individuals
`h_members`	Number of adults living in the house (restricted to 2)
`h_female`	Mean for sex
`h_agemax and h_age`	Max and mean age of the adults in the household
`educlen`	Mean years of education of the adults in the household
`edu_college`	Mean of dummy university degree
`educlenmax`	Max years of education of the adults in the household
`edu_college_max`	Dummy equal to 1 if at least one adult has a university degree
`educ1–educ5`	Education dummies (1 = compulsory, 2 = upper sec., 3 = short tertiary, 4 = bachelor, 5 = master/PhD.
`ind_income_total`	Total household nominal income
`ind_netwealth_total`	Total household net wealth (assets minus debts)
`ind_liquidity_total`	Total household liquid assets (cash plus stocks)
`ind_finwealth_1_total`	Financial wealth 1 (cash + stocks + bonds − bank debt)
`ind_finwealth_2_total`	Financial wealth 2 (finwealth 1 + mortgage credits − mortgage debt)
`ind_finass_total`	Total gross financial assets (cash + stocks + bonds)
`ind_mortgage_total`	Total household mortgage debt based on outstanding bonds
`ind_bankdebt_total`	Total household bank debt. This is the OBLGAELD variable referred to in Open Market transactions
`ind_cash/stocks/bonds_total`	Total household cash, stocks, and bonds (three separate variables)
`splithousehold`	Dummy equal to 1 if the household dissolves
`properties`	Total number of properties owned by the household

Table 17: Derived variables utilized for the estimation of the Price Indices (Repeated Sales & Hedonic).

Variable	Description
Panel A: Repeated Sales Filtering & GLS Weights	
area	16 areas created by merging kommunes
k_months	Time elapsed between the first and second sale of a property (in months). Observations with k_months ≤ 6 are dropped.
growth_level_price	Raw price growth between the two sales.
comm_growth	Mean price growth of the specific area during the same time interval.
diff_growth	Difference between the individual property growth and the area growth. Used to drop extreme outliers.
w_gls	Weights constructed for the Generalized Least Squares (GLS) estimation, inversely proportional to the square root of mean squared residuals by k_months deciles.
Panel B: Fixed Effects & Final Indices (Areas vs. National)	
fe_sales_pair	Property-specific fixed effect used to control for unobserved time-invariant characteristics in the repeated sales model.
fe_loc_time	Location-by-time fixed effect (area $\times$ Month) extracted to construct the local indices.
fe_time	Time fixed effect (Month) extracted to construct the aggregate National index.
FE_phi_t	Exponentiated time fixed effects representing the price level.
price_index	Final Area-level repeated sales price index (Base 2015 = 100).
index_denmark	Final National-level aggregate repeated sales price index (Base 2015 = 100).
annual_comm_growth	Annualized price growth rate at the area level.
annual_growth	Annualized price growth rate at the aggregate National level.
Panel C: Hedonic Model Covariates	
LLiving2/3	Squared and cubed log of total living area.
Rooms2/3	Squared and cubed number of rooms.
Bathrooms2/3	Squared and cubed number of bathrooms.
BuildingAge2/3	Squared and cubed building age.

Table 18: Derived variables for the open-market loss analysis.

Variable	Description
Panel A: The Failure Event	
`loss_sale`	A dummy variable defined as 1 if the transaction meets the strict criteria of a market-loss liquidation: real individual growth is lower than real area growth, a nominal loss is realized, and the purchase occurs within the specified bubble period.
Panel B: Financial Variables	
`LTV`	Continuous leverage ratio calculated logarithmically using `ind_mortgage_total`, `ind_bankdebt_total`, and the nominal purchase price.
`LTV_cat`	Categorical leverage trigger divided into 4 risk bins derived from `LTV`.
`liquidity_dec`	Decile rank (1-10) constructed using total household liquid assets (`ind_liquidity_total`).
`wealth_dec`	Decile rank (1-10) constructed using net financial wealth (`ind_finwealth_2_total`).
`income_dec`	Decile rank (1-10) constructed using total household income (`ind_income_total`).
Panel C: Fixed Effects and Geographic Controls	
`area`	16 geographical areas created by merging kommunes.
`kom_2`	Kommune.
`year_1`	Year of purchase.
Panel D: Temporal Variables and Survival Time	
`bubble_start`	Static temporal marker defining the empirical onset of the housing bubble phase (April 2005), utilized to filter and restrict the analytical sample to purchases made during the speculative run-up.
`bubble_end`	Static temporal marker defining the conclusion of the expansionary bubble phase, aligning exactly with the market peak (January 2007).
`peak`	Static temporal marker defining the absolute peak of the housing bubble (August 2007), consistently identified by both the constructed price index and official Statistics Denmark (DST) data.
`trough`	Static temporal marker defining the absolute bottom of the housing market crash (January 2012), verified by the constructed index and DST, utilized as the right-censoring threshold.
`k_months`	Holding period measured in months from the exact purchase date to either the date of sale or right-censored at the market trough.
`k_years`	Annualized holding period (`k_months`/12), utilized as the primary survival time variable for the baseline models.
`exit_crisis`	Temporal boundary defining the distance between the sale date (or trough) and the market peak.
`duration_crisis`	Synchronized survival time metric, calculated as `exit_crisis` / 12. It measures the exact time spent under stress strictly within the acute deflationary phase (Peak-to-Trough).

Table 19: Household financial variables from Statistics Denmark (IND register).

Variable	Description
Panel A: Underlying DST register variables (year-end stocks, DKK)	
`BANKAKT`	Balance held in financial institutions (bank/savings deposits) as of December 31.
`KURSAKT`	Market value of Danish stocks and investment-fund certificates held in custody (depot) as of December 31.
`OBLAKT`	Market value of bonds held in custody (depot) as of December 31.
`PANTAKT`	Outstanding mortgage-deed (*pantebrevsgæld*) held in custody as receivables as of December 31.
`PANTGAELD`	Outstanding mortgage-deed debt (*pantebrevsgæld*) as of December 31.
`BANKGAELD`	Debt owed to financial institutions (bank debt) as of December 31.
`OBGAELD`	Market value of bond-based mortgage debt (*realkredit*) as of December 31.
Panel B: Constructed household financial variables	
`ind_cash_total`	Total household cash holdings, equal to `BANKAKT`.
`ind_stocks_total`	Total household stock holdings, equal to `KURSAKT`.
`ind_bonds_total`	Total household bond holdings, equal to `OBLAKT`.
`ind_bankdebt_total`	Total household bank debt, equal to `BANKGAELD`. This is the `OBGAELD` variable referred to in the Open Market transactions analysis.
`ind_mortgage_total`	Total mortgage debt, equal to `OBLGAELD` (outstanding bond-based mortgage debt).
`ind_liquidity_total`	Total household liquid assets: `BANKAKT` + `KURSAKT`.
`ind_finass_total`	Total gross financial assets: `BANKAKT` + `KURSAKT` + `OBLAKT`.
`ind_finwealth_1_total`	Financial wealth net of bank debt: `BANKAKT` + `KURSAKT` + `OBLAKT` − `BANKGAELD`.
`ind_finwealth_2_total`	Financial wealth net of bank debt and mortgage liabilities: `ind_finwealth_1_total` + `OBGAELD` − `PANTGAELD`.

Denmark area description

Denmark — Areas Overview

Spatial delineations of the Danish municipalities (kommuner) into custom areas. Colors match the map in Figure 2. Bornholm is excluded from the analysis.

Area	Name	Region	Municipality Codes	Size	Description
001	**Central Copenhagen**	Capital	101, 147, 155, 185	272,315	The municipalities forming the historic and administrative core of the Danish capital on the island of Zealand, including Copenhagen and Frederiksberg.
002	**Western Copenhagen Suburbs (Vestegnen)**	Capital	151, 153, 161, 163, 165, 167, 169, 175, 183, 187, 240, 250	139,164	The suburban municipalities directly west and southwest of Copenhagen between the capital and Roskilde.
003	**Northern Copenhagen Suburbs**	Capital	157, 159, 173, 190, 201, 223, 230	131,372	The suburban municipalities north of Copenhagen along the Øresund coast and the inland areas between Copenhagen and North Zealand.
004	**North Zealand**	Capital	210, 217, 219, 260, 270	114,320	The northern part of the island of Zealand, including the municipalities around Hillerød, Helsingør, and the northern coastal areas.
005	**Roskilde-Køge Area**	Zealand	253, 259, 265, 269, 329, 350	102,558	The municipalities southwest of Copenhagen centered around the towns of Roskilde and Køge along the central part of eastern Zealand.
006	**West Zealand**	Zealand	306, 316, 326, 330, 340	122,606	The western part of the island of Zealand, including the municipalities around Holbæk, Kalundborg, and Slagelse along the western coast.
007	**South Zealand and Lolland-Falster**	Zealand	320, 336, 360, 370, 376, 390	125,530	The southern part of Zealand together with the islands of Lolland and Falster in southern Denmark.
008	**Funen (Fyn)**	South	410, 420, 430, 440, 450, 461, 479, 480, 482, 492	162,220	The island of Funen and its surrounding smaller islands, located between Jutland and Zealand and centered around the city of Odense.
009	**South-West Jutland**	South	510, 540, 550, 561, 563, 573, 580	130,025	The southwestern part of the Jutland peninsula along the North Sea coast and the Danish–German border, including municipalities around Esbjerg and Sønderborg.
010	**Triangle Region (Trekantområdet)**	South	530, 575, 607, 621, 630	102,828	The area around Fredericia, Kolding, and Vejle in southeast Jutland where Jutland, Funen, and the Little Belt meet.
011	**East Jutland**	Central	615, 727, 740, 741, 746, 766	104,614	The inland and coastal municipalities of central eastern Jutland south of Aarhus, including towns such as Horsens, Skanderborg, and Silkeborg.
012	**West Jutland**	Central	657, 661, 665, 671, 756, 760	108,518	The central and northwestern part of the Jutland peninsula, including municipalities around Herning, Holstebro, and Lemvig.
013	**Randers-Djursland Area**	Central	706, 707, 710, 730, 779, 791	133,864	The area around the city of Randers and the Djursland peninsula in eastern Jutland.
014	**Aarhus Area**	Central	751	112,276	The municipality of Aarhus, on the eastern coast of Jutland.
015	**Northern Jutland**	North	773, 787, 810, 813, 825, 849, 860	106,882	The northernmost mainland part of Denmark, including Vendsyssel and Thy, north of the Limfjord.
016	**Aalborg Area**	North	820, 840, 846, 851	125,646	The municipalities surrounding Aalborg around the Limfjord in northern Jutland.
999	***Bornholm (excluded)***	*Capital*	*400*	*23,236*	*The Baltic island of Bornholm — administratively part of the Capital Region but geographically separate. Excluded from the analysis.*

Map colors: ***Red*** ***Blue*** ***Green*** ***Yellow*** ***Purple*** | *Purple marks* ***Bornholm****, which is excluded from the analysis.*

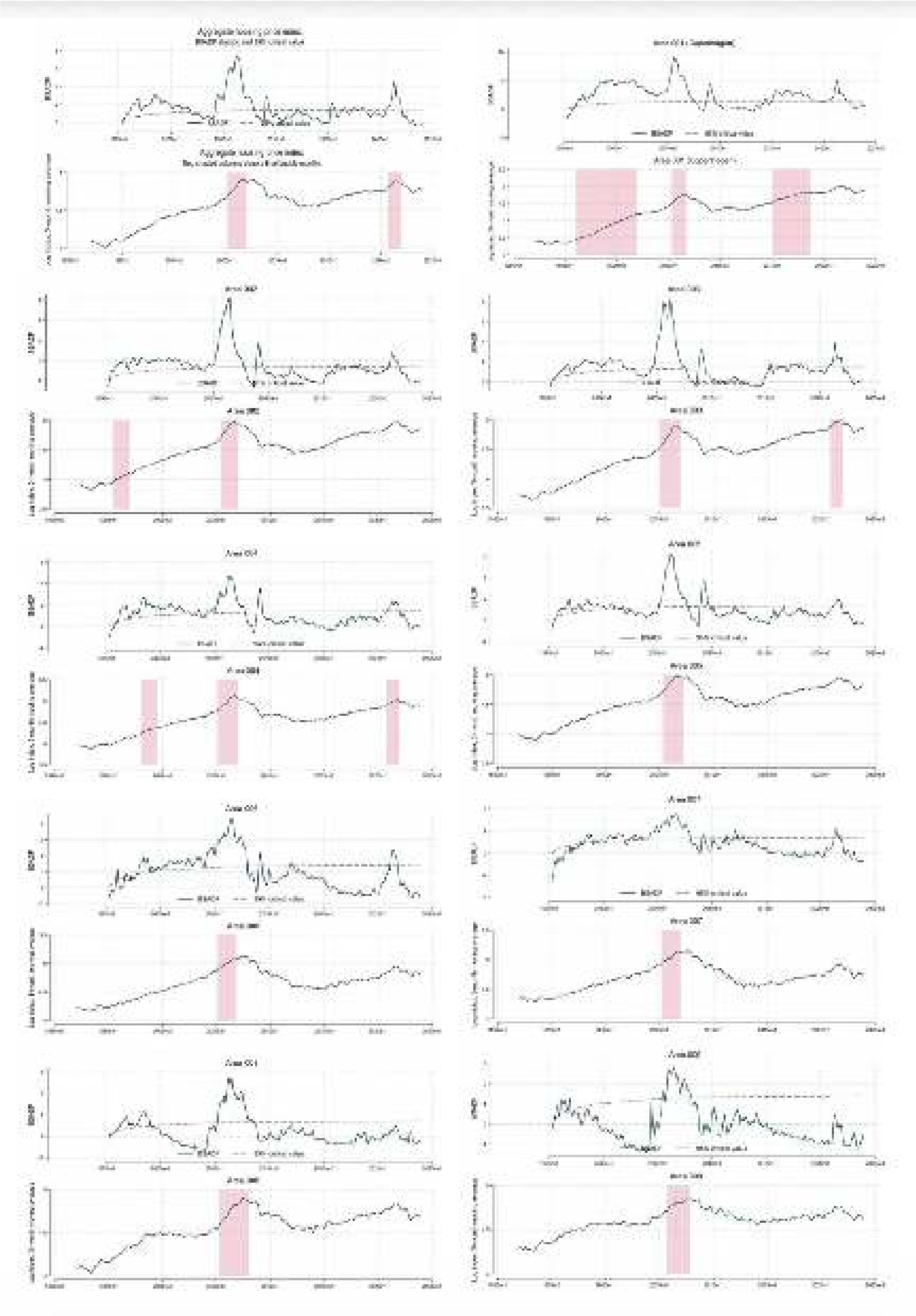
Bubble Detection

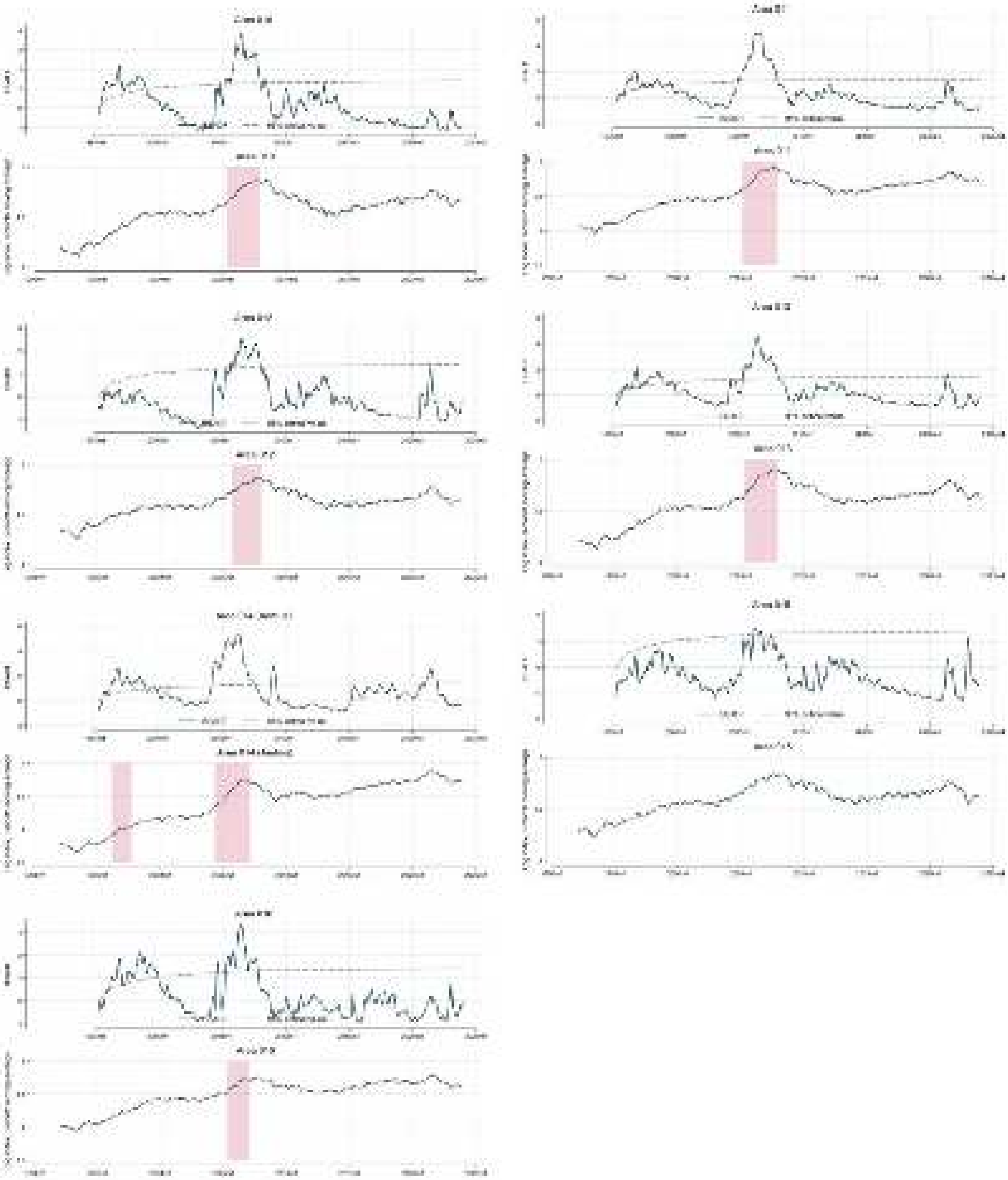
This appendix reports the summary table and graphical outputs for the bubble detection analysis in Section 5.2.1.

Table 20: GSADF statistics and final bubble episodes under the combined detection rule.

Series	GSADF	Bubble episode(s)
Aggregate index	7.364	2005m4–2007m1; 2020m9–2021m12
Area 001	9.132	1996m2–2001m12; 2005m5–2006m9; 2015m1–2018m9
Area 002	8.115	1995m7–1996m12; 2005m6–2007m1
Area 003	8.364	2005m2–2006m12; 2020m12–2021m12
Area 004	4.806	1998m2–1999m6; 2005m3–2007m1; 2020m11–2021m12
Area 005	6.299	2005m6–2007m3
Area 006	4.402	2005m3–2006m11
Area 007	3.618	2005m4–2006m12
Area 008	5.437	2005m5–2007m12
Area 009	2.850	2005m10–2007m10
Area 010	3.925	2005m4–2007m11
Area 011	5.025	2005m3–2007m11
Area 012	2.634	2005m10–2007m12
Area 013	4.644	2005m5–2007m11
Area 014	5.338	1996m3–1997m9; 2004m5–2007m2
Area 015	1.527	None
Area 016	3.355	2005m5–2007m1
95% critical value	**1.424**	

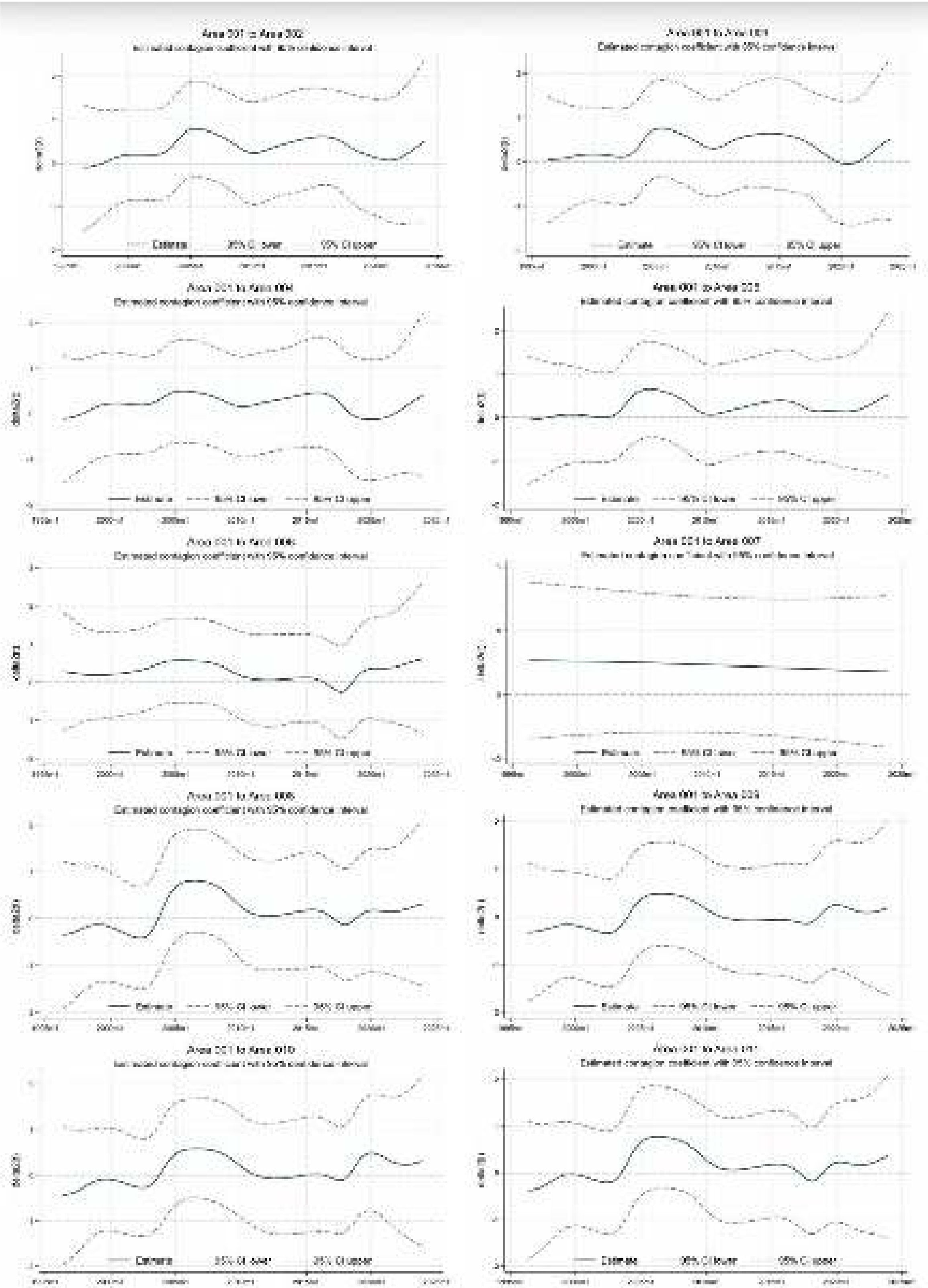
Combined detection graphs

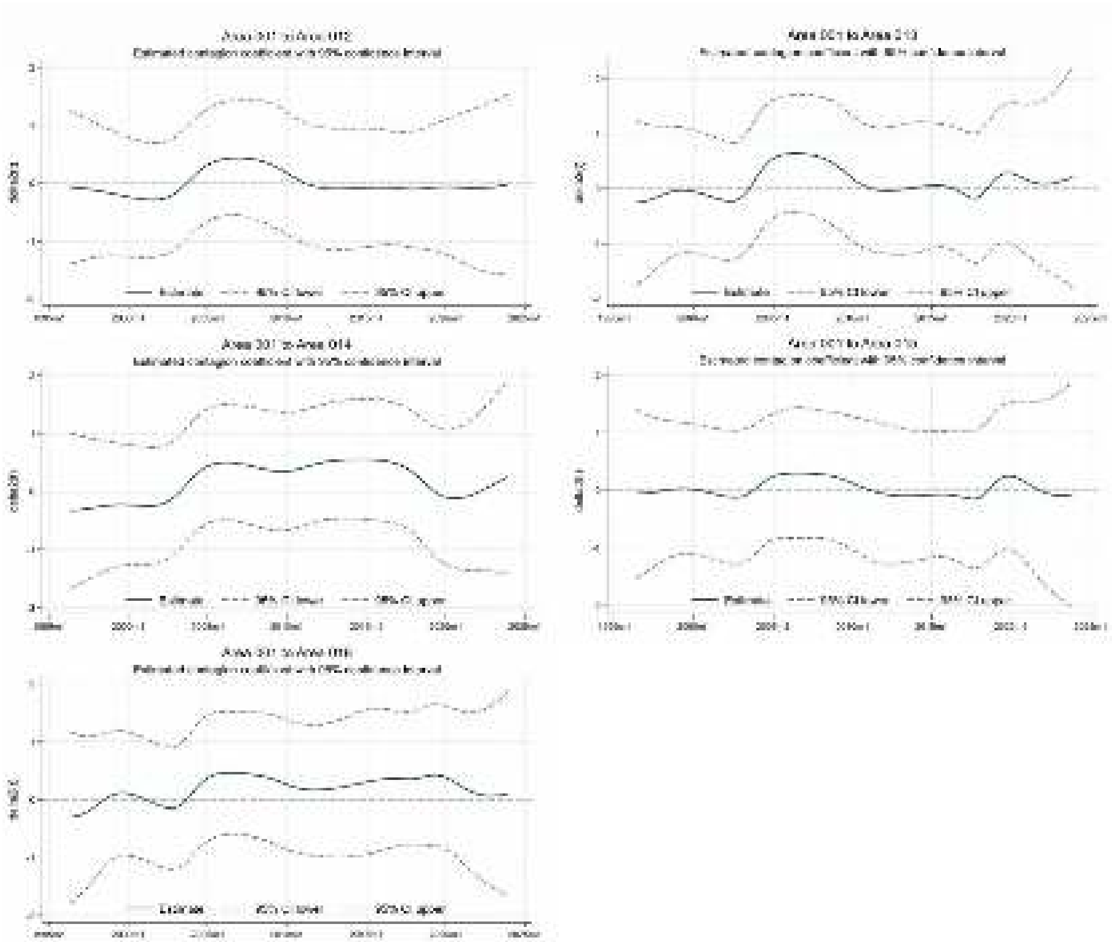




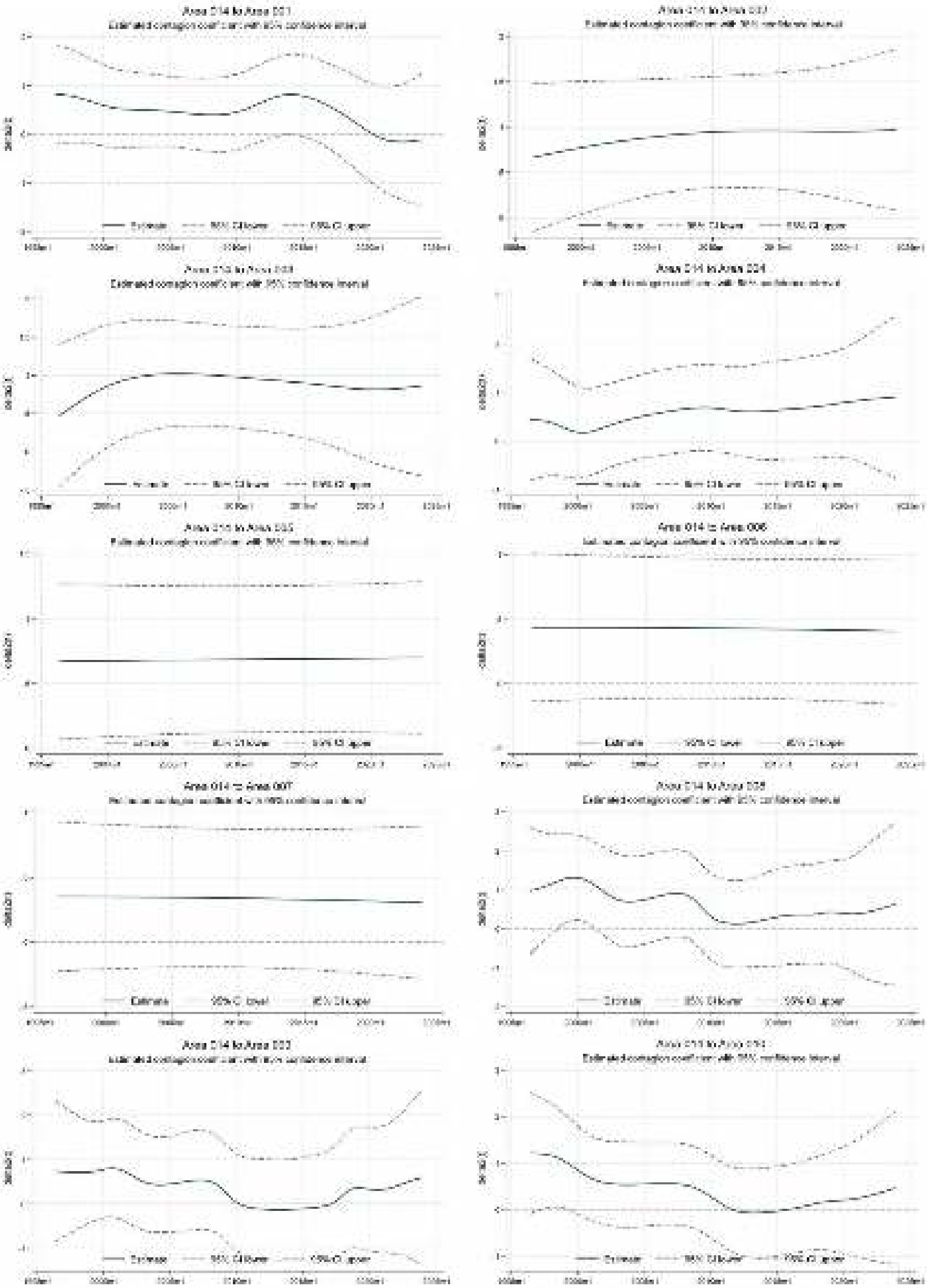
Bubble Contagion

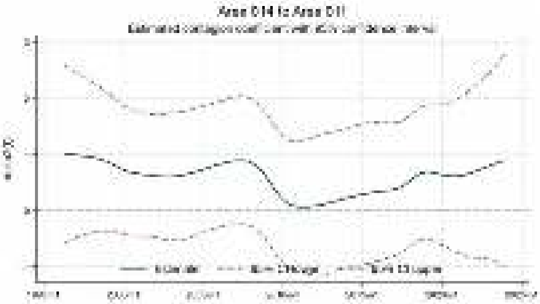
This appendix reports the graphical outputs for the contagion analysis in Section 5.2.2.

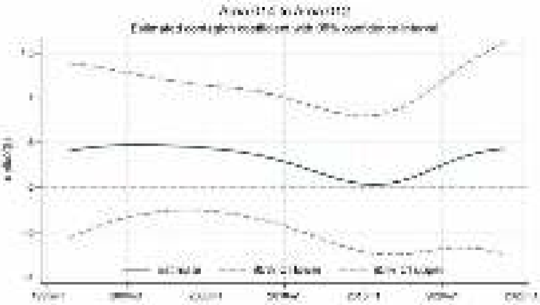


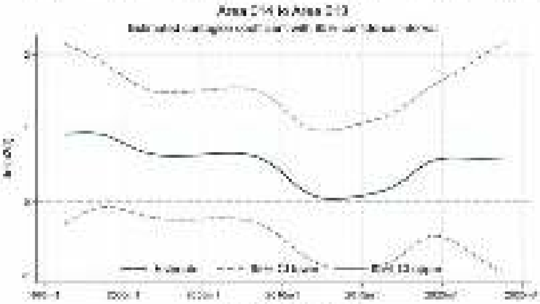


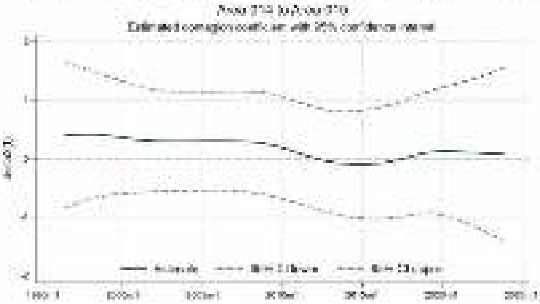
The following figures show the failure and hazard graphs estimated for the liquidity and LTV variables of models 5 (left) and 6 (right). Note that these graphs do not change across different model specifications.

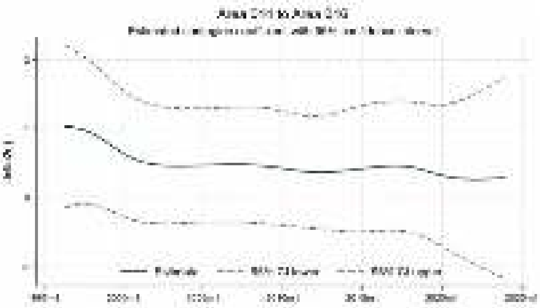

Area 014 to Area 001
Estimated contagion coefficient with 95% confidence interval
Area 014 to Area 004
Area 014 to Area 006
Area 014 to Area 007
Area 014 to Area 008
Estimate
95% CI lower
95% CI upper


Area 014 to Area 011



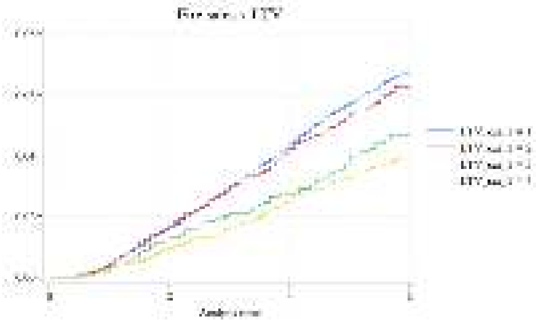

Area 014 to Area 013

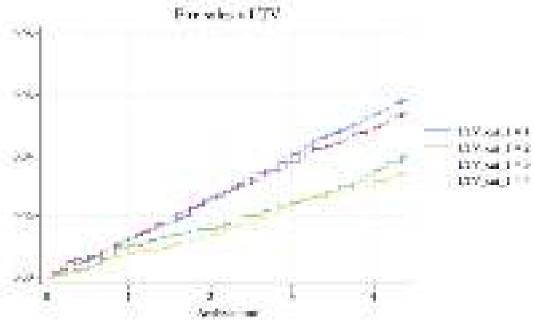

Estimated contagion coefficient with 95% confidence interval

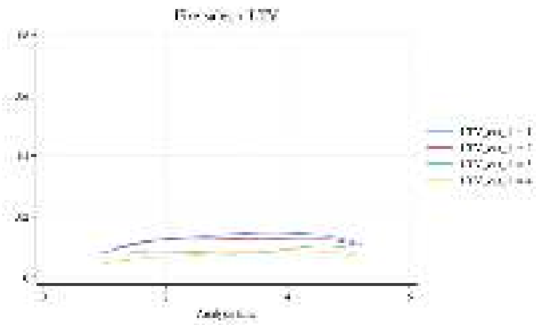


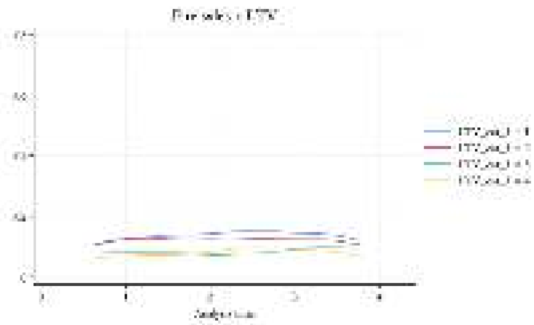
Open-Market Loss Transactions during the 2008 Crisis

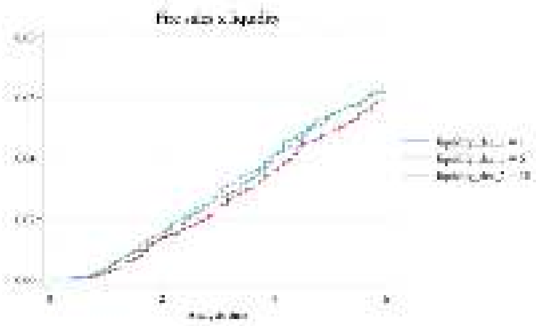
The following figures show the failure and hazard graphs estimated for the liquidity and LTV variables of models 5 (left) and 6 (right). Note that these graphs do not change across different model specifications.

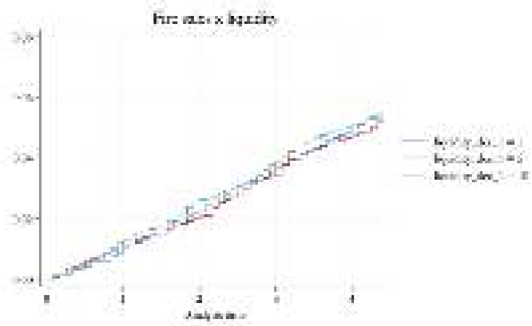


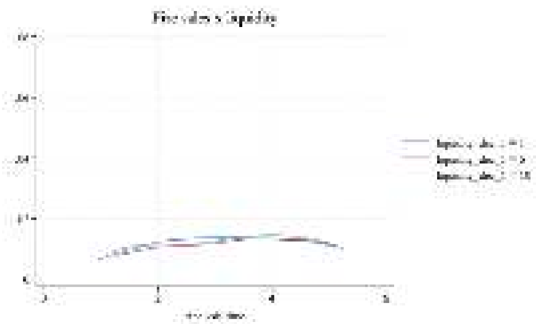


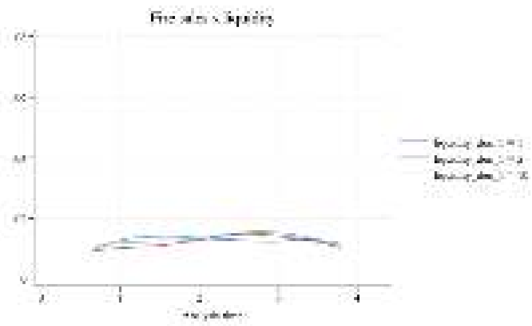




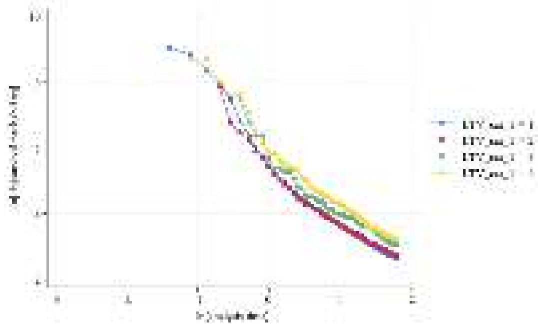


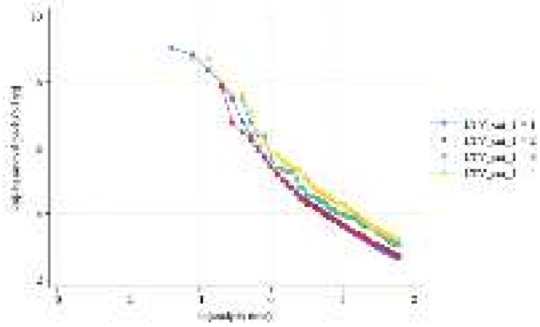


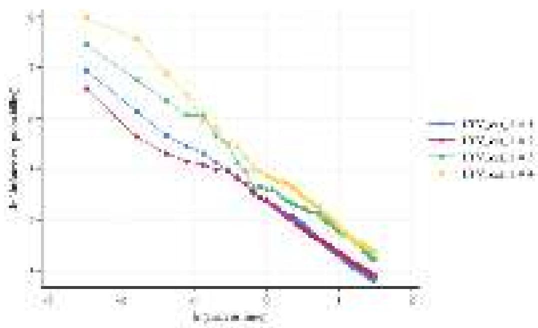


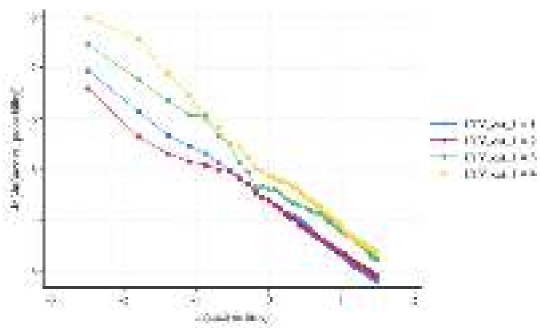


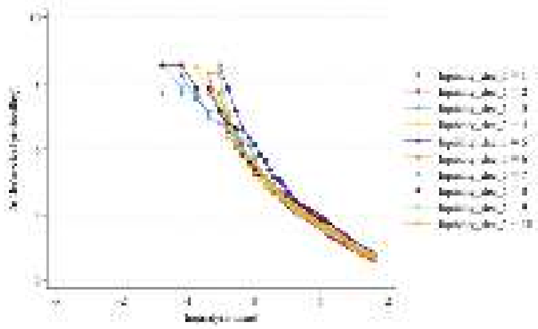
The following figures show the log-log survival plots estimated for all the financial variables. On the left you can see the plot of the baseline specification, on the right you can see the plot of the interaction term specification. For each variable, the first top two graphs show the plots of model 5, while the two below show the plots of model 6.

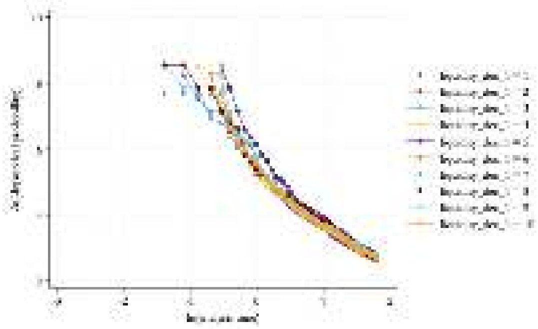


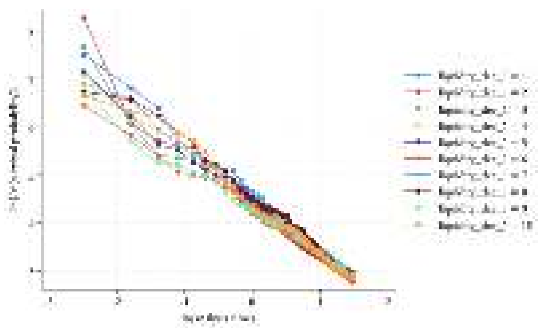


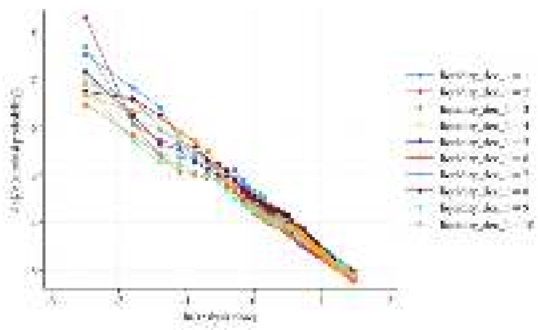


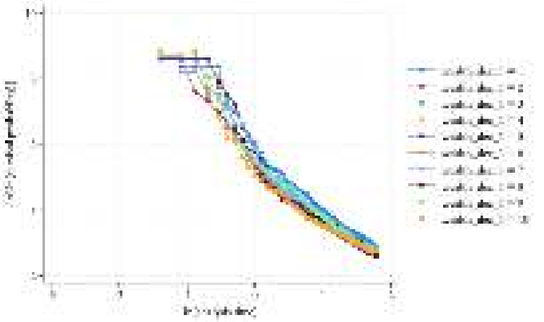


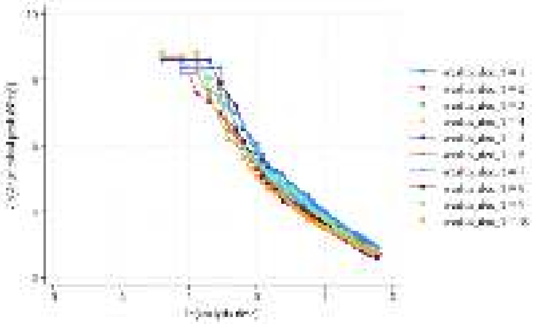


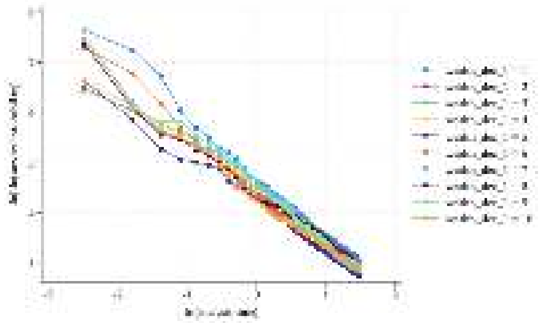


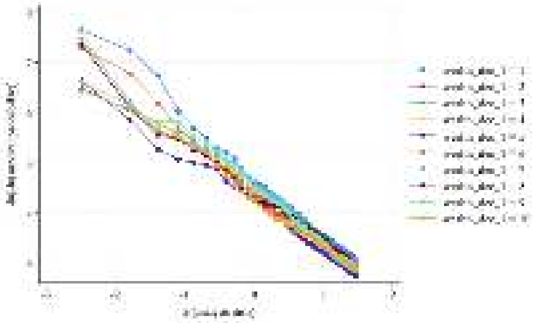


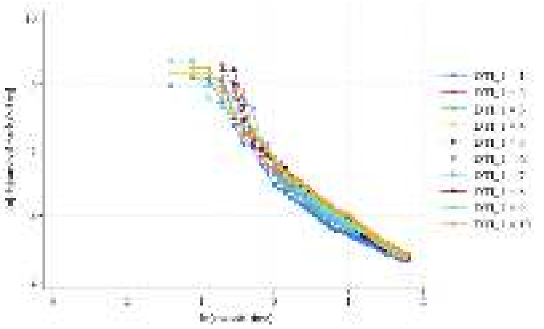


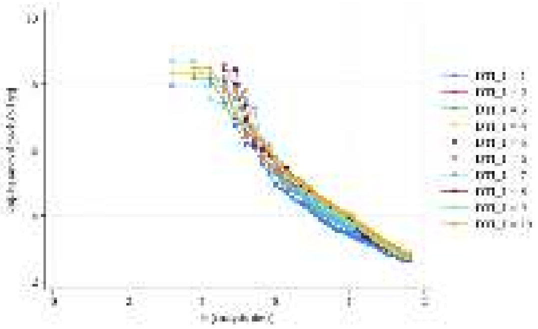


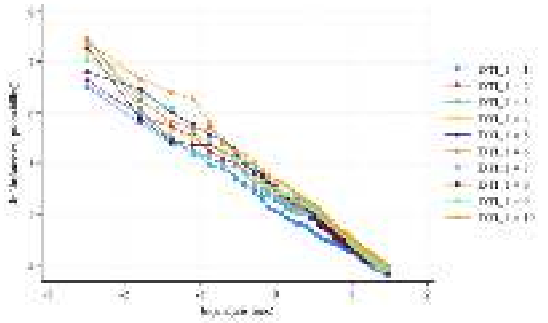


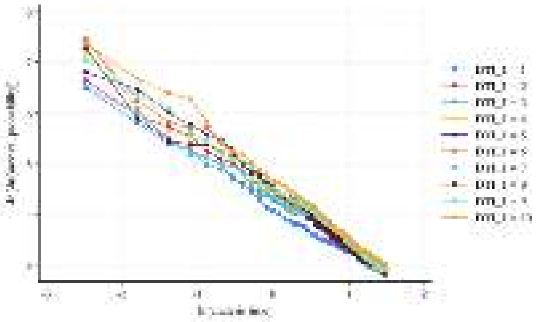












VIF statistic

Variable	VIF	1/VIF
liquidity ~1		
1	2.440	0.410
2	2.510	0.398
3	2.680	0.373
4	2.790	0.359
5	2.840	0.353
6	2.780	0.360
7	2.690	0.371
8	2.510	0.398
9	2.210	0.452
LTV cat 1		
2	1.100	0.909
3	1.110	0.900
4	1.440	0.696
wealth dec		
1	5	0.200
2	6.160	0.162
3	6.360	0.157
4	6.020	0.166
5	5.290	0.189
6	4.490	0.223
7	3.670	0.273
8	3.080	0.325
9	2.230	0.448
DTI 1		
1	3.630	0.276
2	2.690	0.371
3	2.400	0.416
4	2.380	0.419
5	2.410	0.414
6	2.440	0.411
7	2.400	0.417
8	2.320	0.432
9	2.100	0.476
unoccupiec	1.040	0.959
h agemax '	953.4	0.00105
h agemax '	4020	0.000249
h agemax '	1129	0.000886
educ1 1	2.580	0.387
educ2 1	3.280	0.305
educ3 1	1.550	0.647
educ4 1	2.500	0.399
properties '		
2	1.400	0.713
3	1.290	0.777
BuildingAg	40.18	0.0249
BuildingAg	171.8	0.00582
BuildingAg	62.14	0.0161
Bathrooms	138.8	0.00721
Bathrooms	452.9	0.00221
Bathrooms	110.4	0.00906
Rooms 1	136.6	0.00732
Rooms2	565.4	0.00177
Rooms3	172.6	0.00579
year 1		
2006	1.060	0.947
2007	1.050	0.955
Mean	VIF	158.1

Summary statistics discounted sales

LTV statistics model 1			LTV statistics model 3		LTV statistics model 5	
Summary	for	variables	for	variables	for	variables
Group	variable	LTV cat 1	variable	LTV cat 1	variable	LTV cat 1
LTV cat 1	Mean	N	Mean	N	Mean	N
1	0.0606	67059	0.0634	31354	0.0589	33775
2	0.0508	11101	0.0482	5524	0.0544	5198
3	0.0403	3445	0.0379	1530	0.0409	1761
4	0.0328	12279	0.0297	5622	0.0335	6027
Total	0.0551	93884	0.0563	44030	0.0544	46761

Liq statistics model 1			Liq statistics model 3		Liq statistics model 5	
Summary	for	variables	for	variables	for	variables
Group	variable	Liq dec 1	variable	iquidity dec 1	variable	iquidity dec 1
liquidity de	Mean	N	Mean	N	Mean	N
1	0.0532	7745	0.0525	3717	0.0545	3742
2	0.0553	8114	0.0572	3829	0.0545	4015
3	0.0535	9232	0.0576	4533	0.0503	4410
4	0.0545	10048	0.0513	4869	0.0569	4886
5	0.0549	10628	0.0604	5168	0.0502	5177
6	0.0599	10582	0.0595	5029	0.0602	5236
7	0.0553	10493	0.0596	4963	0.0523	5197
8	0.0560	10393	0.0573	4868	0.0557	5187
9	0.0531	9366	0.0536	4123	0.0541	4902
10	0.0535	7283	0.0505	2931	0.0551	4009
Total	0.0551	93884	0.0563	44030	0.0544	46761

DTI statistics model 1			DTI statistics model 3		DTI statistics model 5	
Summary	for	variables	for	variables	for	variables
Group	variable	DTI 1	variable	DTI 1	variable	DTI 1
DTI 1	Mean	N	Mean	N	Mean	N
1	0.0599	7217	0.0603	3633	0.0596	3421
2	0.0526	6821	0.0558	3492	0.0500	3162
3	0.0563	7671	0.0579	3854	0.0560	3608
4	0.0596	8730	0.0655	4380	0.0550	4126
5	0.0507	9803	0.0493	4933	0.0526	4618
6	0.0542	10768	0.0532	5202	0.0560	5265
7	0.0550	11275	0.0558	5269	0.0543	5671
8	0.0554	11562	0.0573	5198	0.0548	6002
9	0.0566	11003	0.0603	4609	0.0548	5934
10	0.0510	8918	0.0477	3418	0.0514	4887
Total	0.0551	93768	0.0563	43988	0.0544	46694

Wealth statistics model 1			Wealth statistics model 3		Wealth statistics model 5	
Summary	for	variables	for	variables	for	variables
Group	variable	wealth dec 1	variable	wealth dec 1	variable	wealth dec 1
wealth dec	Mean	N	Mean	N	Mean	N
1	0.0487	7603	0.0488	2764	0.0477	4273
2	0.0514	11480	0.0484	5004	0.0534	6007
3	0.0509	12826	0.0520	5807	0.0504	6645
4	0.0579	12529	0.0618	5984	0.0554	6152
5	0.0596	11098	0.0595	5411	0.0613	5397
6	0.0594	9343	0.0622	4565	0.0569	4520
7	0.0570	7695	0.0584	3889	0.0564	3598
8	0.0530	6792	0.0524	3419	0.0542	3213
9	0.0570	6650	0.0629	3370	0.0524	3129
10	0.0555	7868	0.0542	3817	0.0562	3827
Total	0.0551	93884	0.0563	44030	0.0544	46761

Model 1 regression 1 linktest

Failure	d	loss sale==1					
Analysis	time	t	k years				
Iteration	0	Log	likelihood	=	-58100		
Iteration	1	Log	likelihood	=	-57926		
Iteration	2	Log	likelihood	=	-57920		
Iteration	3	Log	likelihood	=	-57920		
Iteration	4	Log	likelihood	=	-57920		
Refining	estimates						
Iteration	0	Log	likelihood	=	-57920		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	93,768	Number of obs	=	93,768
No.	of	failures	=	5,163			
Time	at	risk	=	494,161.58			
LR	chi2(2)	=	359.8				
Log	likelihood	=	-57920	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	0.934	0.188	4.970	0	0.566	1.303	
hatsq	-0.155	0.170	-0.910	0.361	-0.488	0.178	

Model 1 regression 2 linktest

Failure	d	loss sale==1					
Analysis	time	t	k years				
Iteration	0	Log	likelihood	=	-58100		
Iteration	1	Log	likelihood	=	-57903		
Iteration	2	Log	likelihood	=	-57895		
Iteration	3	Log	likelihood	=	-57895		
Iteration	4	Log	likelihood	=	-57895		
Refining	estimates						
Iteration	0	Log	likelihood	=	-57895		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	93,768	umber of ob	=	93,768
No.	of	failures	=	5,163			
Time	at	risk	=	494,161.58			
LR	chi2(2)	=	410.4				
Log	likelihood	=	-57895	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	1.034	0.123	8.430	0	0.794	1.274	
hatsq	-0.0575	0.130	-0.440	0.658	-0.312	0.197	

Model 2 regression 1 linktest

Failure	d	loss sale==1					
Analysis	time	t	uration cris				
Iteration	0	Log	likelihood	=	-48754		
Iteration	1	Log	likelihood	=	-48571		
Iteration	2	Log	likelihood	=	-48562		
Iteration	3	Log	likelihood	=	-48562		
Iteration	4	Log	likelihood	=	-48562		
Refining	estimates						
Iteration	0	Log	likelihood	=	-48562		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	88,495	Number of obs	=	88,495
No.	of	failures	=	4,319			
Time	at	risk	=	352,480.82			
LR	chi2(2)	=	383.3				
Log	likelihood	=	-48562	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	1.128	0.0620	18.20	0	1.006	1.249	
hatsq	-0.101	0.137	-0.740	0.460	-0.370	0.168	

Model 2 regression 2 linktest

Failure	d	loss sale==1					
Analysis	time	t	uration cris				
Iteration	0	Log	likelihood	=	-48754		
Iteration	1	Log	likelihood	=	-48571		
Iteration	2	Log	likelihood	=	-48562		
Iteration	3	Log	likelihood	=	-48562		
Iteration	4	Log	likelihood	=	-48562		
Refining	estimates						
Iteration	0	Log	likelihood	=	-48562		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	88,495	umber of ob	=	88,495
No.	of	failures	=	4,319			
Time	at	risk	=	352,480.82			
LR	chi2(2)	=	383.3				
Log	likelihood	=	-48562	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	1.128	0.0620	18.20	0	1.006	1.249	
hatsq	-0.101	0.137	-0.740	0.460	-0.370	0.168	

Model 3 regression 1 linktest

Failure	d	loss sale==1					
Analysis	time	t	k years				
Iteration	0	Log	likelihood	=	-26031		
Iteration	1	Log	likelihood	=	-25892		
Iteration	2	Log	likelihood	=	-25886		
Iteration	3	Log	likelihood	=	-25886		
Iteration	4	Log	likelihood	=	-25886		
Refining	estimates						
Iteration	0	Log	likelihood	=	-25886		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	43,988	Number of obs	=	43,988
No.	of	failures	=	2,477			
Time	at	risk	=	243,738.75			
LR	chi2(2)	=	289.5				
Log	likelihood	=	-25886	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	1.262	0.0778	16.23	0	1.110	1.415	
hatsq	0.421	0.161	2.620	0.00900	0.106	0.737	

Model 3 regression 2 linktest

Failure	d	loss sale==1					
Analysis	time	t	k years				
Iteration	0	Log	likelihood	=	-26031		
Iteration	1	Log	likelihood	=	-25873		
Iteration	2	Log	likelihood	=	-25869		
Iteration	3	Log	likelihood	=	-25869		
Iteration	4	Log	likelihood	=	-25869		
Refining	estimates						
Iteration	0	Log	likelihood	=	-25869		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	43,988	umber of ob	=	43,988
No.	of	failures	=	2,477			
Time	at	risk	=	243,738.75			
LR	chi2(2)	=	323.9				
Log	likelihood	=	-25869	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	1.174	0.0678	17.32	0	1.041	1.307	
hatsq	0.246	0.104	2.360	0.0180	0.0416	0.451	

Model 4 regression 1 linktest

Failure	d	loss sale==1					
Analysis	time	t	uration cris				
Iteration	0	Log	likelihood	=	-19823		
Iteration	1	Log	likelihood	=	-19669		
Iteration	2	Log	likelihood	=	-19664		
Iteration	3	Log	likelihood	=	-19664		
Iteration	4	Log	likelihood	=	-19664		
Refining	estimates						
Iteration	0	Log	likelihood	=	-19664		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	39,987	Number of obs	=	39,987
No.	of	failures	=	1,891			
Time	at	risk	=	157,321.41			
LR	chi2(2)	=	318.0				
Log	likelihood	=	-19664	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	0.703	0.207	3.400	0.00100	0.298	1.108	
hatsq	0.315	0.118	2.680	0.00700	0.0850	0.546	

Model 4 regression 2 linktest

Failure	d	loss sale==1					
Analysis	time	t	uration cris				
Iteration	0	Log	likelihood	=	-19823		
Iteration	1	Log	likelihood	=	-19669		
Iteration	2	Log	likelihood	=	-19664		
Iteration	3	Log	likelihood	=	-19664		
Iteration	4	Log	likelihood	=	-19664		
Refining	estimates						
Iteration	0	Log	likelihood	=	-19664		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	39,987	umber of ob	=	39,987
No.	of	failures	=	1,891			
Time	at	risk	=	157,321.41			
LR	chi2(2)	=	318.0				
Log	likelihood	=	-19664	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	0.703	0.207	3.400	0.00100	0.298	1.108	
hatsq	0.315	0.118	2.680	0.00700	0.0850	0.546	

Model 5 regression 1 linktest

Failure	d	loss sale==1					
Analysis	time	t	k years				
Iteration	0	Log	likelihood	=	-26969		
Iteration	1	Log	likelihood	=	-26893		
Iteration	2	Log	likelihood	=	-26889		
Iteration	3	Log	likelihood	=	-26888		
Iteration	4	Log	likelihood	=	-26888		
Iteration	5	Log	likelihood	=	-26888		
Iteration	6	Log	likelihood	=	-26888		
Refining	estimates						
Iteration	0	Log	likelihood	=	-26888		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	46,694	Number of obs	=	46,694
No.	of	failures	=	2,542			
Time	at	risk	=	235,873.58			
LR	chi2(2)	=	161.2				
Log	likelihood	=	-26888	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	0.845	0.214	3.950	0	0.426	1.265	
hatsq	-0.154	0.186	-0.830	0.409	-0.519	0.211	

Model 5 regression 2 linktest

Failure	d	loss sale==1					
Analysis	time	t	k years				
Iteration	0	Log	likelihood	=	-26969		
Iteration	1	Log	likelihood	=	-26870		
Iteration	2	Log	likelihood	=	-26863		
Iteration	3	Log	likelihood	=	-26862		
Iteration	4	Log	likelihood	=	-26862		
Iteration	5	Log	likelihood	=	-26862		
Refining	estimates						
Iteration	0	Log	likelihood	=	-26862		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	46,694	umber of ob	=	46,694
No.	of	failures	=	2,542			
Time	at	risk	=	235,873.58			
LR	chi2(2)	=	213.0				
Log	likelihood	=	-26862	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	0.952	0.135	7.080	0	0.689	1.216	
hatsq	-0.0581	0.117	-0.500	0.620	-0.288	0.172	

Model 6 regression 1 linktest

Failure	d	loss sale==1					
Analysis	time	t	uration cris				
Iteration	0	Log	likelihood	=	-24278		
Iteration	1	Log	likelihood	=	-24208		
Iteration	2	Log	likelihood	=	-24203		
Iteration	3	Log	likelihood	=	-24201		
Iteration	4	Log	likelihood	=	-24200		
Iteration	5	Log	likelihood	=	-24200		
Iteration	6	Log	likelihood	=	-24200		
Refining	estimates						
Iteration	0	Log	likelihood	=	-24200		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	45,422	Number of obs	=	45,422
No.	of	failures	=	2,284			
Time	at	risk	=	182,410.32			
LR	chi2(2)	=	154.2				
Log	likelihood	=	-24200	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	0.910	0.158	5.740	0	0.599	1.220	
hatsq	-0.162	0.181	-0.900	0.371	-0.518	0.193	

Model 6 regression 2 linktest

Failure	d	loss sale==1					
Analysis	time	t	uration cris				
Iteration	0	Log	likelihood	=	-24278		
Iteration	1	Log	likelihood	=	-24189		
Iteration	2	Log	likelihood	=	-24181		
Iteration	3	Log	likelihood	=	-24179		
Iteration	4	Log	likelihood	=	-24179		
Iteration	5	Log	likelihood	=	-24179		
Iteration	6	Log	likelihood	=	-24179		
Refining	estimates						
Iteration	0	Log	likelihood	=	-24179		
Cox	regression	with	Breslow	method	for	ties	
No.	of	subjects	=	45,422	umber of ob	=	45,422
No.	of	failures	=	2,284			
Time	at	risk	=	182,410.32			
LR	chi2(2)	=	198.0				
Log	likelihood	=	-24179	Prob>chi2	=	0	
t	Coefficient	Std.	err.	z	P>|z|	95%	
hat	0.967	0.108	8.990	0	0.756	1.178	
hatsq	-0.0909	0.121	-0.750	0.453	-0.328	0.147	

Ph test model 1 regression 1

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	-0.00425	0.160	1	0.694
2.liquidit~1	-0.0258	12.39	1	0.000400
3.liquidit~1	0.00625	0.250	1	0.619
4.liquidit~1	0.0288	6.560	1	0.0104
5.liquidit~1	0.0306	6.040	1	0.0140
6.liquidit~1	0.000510	0	1	0.957
7.liquidit~1	0.0227	4.060	1	0.0439
8.liquidit~1	0.0377	12.29	1	0.000500
9.liquidit~1	0.0317	8.070	1	0.00450
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	-0.00354	0.160	1	0.691
3.LTV cat 1	0.0303	10.42	1	0.00120
4.LTV cat 1	0.0392	23.44	1	0
1.wealth d~	0.0256	10.33	1	0.00130
2.wealth d~	0.00902	1.030	1	0.310
3.wealth d~	0.0240	11.14	1	0.000800
4.wealth d~	0.0259	17.43	1	0
5.wealth d~	0.0321	34.08	1	0
6.wealth d~	0.0342	30.18	1	0
7.wealth d~	0.0342	33.15	1	0
8.wealth d~	0.00959	0.950	1	0.329
9.wealth d~	0.0297	9.070	1	0.00260
10b.wealth	.	.	1	.
1.DTI 1	0.0265	25.07	1	0
2.DTI 1	0.0256	25.28	1	0
3.DTI 1	0.00234	0.0900	1	0.761
4.DTI 1	0.0209	9.360	1	0.00220
5.DTI 1	0.0316	13.14	1	0.000300
6.DTI 1	0.0270	20.09	1	0
7.DTI 1	-0.00881	0.530	1	0.469
8.DTI 1	0.0246	10.53	1	0.00120
9.DTI 1	0.00594	0.590	1	0.444
10b.DTI 1	.	.	1	.
unoccupie	0.0322	16.93	1	0
h agemax 1	0.0423	32.66	1	0
h agemax 1	-0.0416	33.66	1	0
h agemax 1	0.0408	33.01	1	0
educ1 1	-0.0331	28.80	1	0
educ2 1	-0.0236	20.69	1	0
educ3 1	-0.0341	44.03	1	0
educ4 1	-0.0212	16.14	1	0.000100
o.educ5 1	.	.	1	.
1b.propert	.	.	1	.
2.properti~	-0.00722	0.670	1	0.412
3.properti~	0.0110	1.720	1	0.190
BuildingAg	0.0147	7.700	1	0.00550
BuildingAg	-0.0196	15.25	1	0.000100
BuildingAg	0.0208	17.35	1	0
Bathrooms	0.0296	16.01	1	0.000100
Bathrooms	-0.0247	10.63	1	0.00110
Bathrooms	0.0173	5.210	1	0.0225
Rooms 1	0.0354	49.54	1	0
Rooms2	-0.0382	61.17	1	0
Rooms3	0.0383	60.87	1	0
2005b.year	.	.	1	.
2006.year 1	-0.0307	44.33	1	0
2007.year 1	-0.0240	17.65	1	0
Global	test	226.1	51	0
Note	Robust	ıceb covar	matrix	used.

Ph test model 1 regression 2

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	-0.00672	0.540	1	0.464
2.liquidit~1	-0.0156	6.040	1	0.0140
3.liquidit~1	0.0139	1.910	1	0.167
4.liquidit~1	0.0183	3.310	1	0.0689
5.liquidit~1	0.0159	3.070	1	0.0796
6.liquidit~1	0.000560	0.0100	1	0.936
7.liquidit~1	0.0120	1.710	1	0.192
8.liquidit~1	0.0172	5.380	1	0.0203
9.liquidit~1	0.0181	3.820	1	0.0507
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	-0.00178	0.0500	1	0.815
3.LTV cat 1	-0.00280	0.0800	1	0.772
4.LTV cat 1	0.0167	4.400	1	0.0360
1o.liquidi~	.	.	1	.
1.liquidit~1	0.00947	0.840	1	0.360
1.liquidit~1	-0.00324	0.120	1	0.725
1.liquidit~1	0.0155	2.850	1	0.0916
2o.liquidi~	.	.	1	.
2.liquidit~1	0.00244	0.110	1	0.742
2.liquidit~1	-0.0150	1.880	1	0.171
2.liquidit~1	-0.00849	0.970	1	0.325
3o.liquidi~	.	.	1	.
3.liquidit~1	-0.0247	5.330	1	0.0210
3.liquidit~1	0.00156	0.0200	1	0.879
3.liquidit~1	-0.00415	0.240	1	0.622
4o.liquidi~	.	.	1	.
4.liquidit~1	0.00359	0.140	1	0.711
4.liquidit~1	-0.00225	0.0500	1	0.816
4.liquidit~1	-0.0180	3.360	1	0.0668
5o.liquidi~	.	.	1	.
5.liquidit~1	-0.00898	0.940	1	0.332
5.liquidit~1	0.0244	4.490	1	0.0341
5.liquidit~1	0.0135	2.940	1	0.0864
6o.liquidi~	.	.	1	.
6.liquidit~1	-0.000810	0.0200	1	0.899
6.liquidit~1	-0.00361	0.180	1	0.671
6.liquidit~1	-0.0147	2.760	1	0.0965
7o.liquidi~	.	.	1	.
7.liquidit~1	0.00560	0.410	1	0.521
7.liquidit~1	0.0208	4.170	1	0.0412
7.liquidit~1	0.00494	0.430	1	0.511
8o.liquidi~	.	.	1	.
8.liquidit~1	0.00108	0.0200	1	0.885
8.liquidit~1	0.0161	3.380	1	0.0661
8.liquidit~1	0.00226	0.100	1	0.749
9o.liquidi~	.	.	1	.
9.liquidit~1	0.00745	1.410	1	0.235
9.liquidit~1	0.0164	5.860	1	0.0155
9.liquidit~1	-0.0132	2.830	1	0.0926
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
1.wealth d~	0.0209	8.700	1	0.00320
2.wealth d~	0.00359	0.190	1	0.665
3.wealth d~	0.0148	4.770	1	0.0289
4.wealth d~	0.0188	10.52	1	0.00120
5.wealth d~	0.0238	21.15	1	0
6.wealth d~	0.0278	22.84	1	0

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
7.wealth d~	0.0249	20	1	0
8.wealth d~	0.00523	0.340	1	0.560
9.wealth d~	0.0248	7.940	1	0.00480
10b.wealth	.	.	1	.
1.DTI 1	0.0174	11.87	1	0.000600
2.DTI 1	0.0173	12.44	1	0.000400
3.DTI 1	-0.00198	0.0700	1	0.785
4.DTI 1	0.0115	3.140	1	0.0762
5.DTI 1	0.0212	6.490	1	0.0109
6.DTI 1	0.0183	10.13	1	0.00150
7.DTI 1	-0.00753	0.460	1	0.498
8.DTI 1	0.0166	5.540	1	0.0186
9.DTI 1	-0.00346	0.240	1	0.623
10b.DTI 1	.	.	1	.
unoccupie	0.0252	12.33	1	0.000400
h agemax 1	0.0338	25.97	1	0
h agemax 1	-0.0326	25.84	1	0
h agemax 1	0.0314	24.60	1	0
educ1 1	-0.0226	14.52	1	0.000100
educ2 1	-0.0146	8.660	1	0.00330
educ3 1	-0.0199	16.97	1	0
educ4 1	-0.0140	7.670	1	0.00560
o.educ5 1	.	.	1	.
1b.propert	.	.	1	.
2.properti~	-0.00422	0.280	1	0.596
3.properti~	0.00251	0.110	1	0.743
BuildingAg	0.00428	0.730	1	0.394
BuildingAg	-0.00848	3.180	1	0.0744
BuildingAg	0.00925	3.840	1	0.0502
Bathrooms	0.0259	13.21	1	0.000300
Bathrooms	-0.0216	8.820	1	0.00300
Bathrooms	0.0153	4.400	1	0.0360
Rooms 1	0.0263	29.98	1	0
Rooms2	-0.0291	38.52	1	0
Rooms3	0.0289	37.43	1	0
2005b.year	.	.	1	.
2006.year 1	-0.0182	17.33	1	0
2007.year 1	-0.0119	4.690	1	0.0303
Global	test	216.8	78	0
Note	Robust	ıceb covar	matrix	used.

Ph test model 2 regression 1

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	-0.0121	1.220	1	0.270
2.liquidit~1	-0.00536	0.490	1	0.485
3.liquidit~1	-0.00598	0.220	1	0.636
4.liquidit~1	-0.00742	0.370	1	0.544
5.liquidit~1	0.00840	0.480	1	0.489
6.liquidit~1	0.0116	2.050	1	0.152
7.liquidit~1	0.00213	0.0300	1	0.873
8.liquidit~1	0.0277	6.620	1	0.0101
9.liquidit~1	-0.00734	0.380	1	0.537
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	0.00584	0.300	1	0.585
3.LTV cat 1	0.0255	5.430	1	0.0198
4.LTV cat 1	0.0174	4.130	1	0.0422
1.wealth d~	-0.0155	3.430	1	0.0642
2.wealth d~	-0.0108	1.410	1	0.234
3.wealth d~	-0.00858	1.300	1	0.254
4.wealth d~	-0.00752	1.370	1	0.242
5.wealth d~	-0.00462	0.660	1	0.416
6.wealth d~	-0.00481	0.550	1	0.458
7.wealth d~	0.000360	0	1	0.956
8.wealth d~	-0.0135	1.470	1	0.226
9.wealth d~	0.00597	0.470	1	0.494
10b.wealth	.	.	1	.
1.DTI 1	-0.0182	10.08	1	0.00150
2.DTI 1	-0.0160	9.320	1	0.00230
3.DTI 1	-0.0224	10.06	1	0.00150
4.DTI 1	-0.0244	14.42	1	0.000100
5.DTI 1	-0.0166	3.350	1	0.0673
6.DTI 1	-0.0126	5	1	0.0253
7.DTI 1	-0.0214	3.840	1	0.0500
8.DTI 1	-0.0137	2.920	1	0.0873
9.DTI 1	-0.0174	6.210	1	0.0127
10b.DTI 1	.	.	1	.
unoccupie	-0.00645	0.410	1	0.524
h agemax 1	-0.00623	0.860	1	0.355
h agemax 1	0.00638	0.910	1	0.340
h agemax 1	-0.00673	0.990	1	0.319
educ1 1	0.00113	0.0200	1	0.879
educ2 1	0.0136	5.280	1	0.0216
educ3 1	0.00774	2.170	1	0.140
educ4 1	0.0101	3.220	1	0.0725
o.educ5 1	.	.	1	.
1b.propert	.	.	1	.
2.properti~	-0.0116	1.540	1	0.215
3.properti~	-0.0112	1.800	1	0.179
BuildingAg	-0.0109	4.920	1	0.0265
BuildingAg	0.0103	4.770	1	0.0290
BuildingAg	-0.0102	4.770	1	0.0289
Bathrooms	-0.00198	0.0500	1	0.816
Bathrooms	0.00372	0.190	1	0.664
Bathrooms	-0.00672	0.630	1	0.428
Rooms 1	0.00540	1.060	1	0.303
Rooms2	-0.00485	0.900	1	0.343
Rooms3	0.00386	0.560	1	0.452
2005b.year	.	.	1	.
2006.year 1	0.00830	3.780	1	0.0517
2007.year 1	0.0175	12.32	1	0.000400
Global	test	83.43	51	0.00280
Note	Robust	ıceb covar	matrix	used.

Ph test model 2 regression 2

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	-0.00760	0.750	1	0.387
2.liquidit~1	0.00471	0.550	1	0.460
3.liquidit~1	0.00982	0.880	1	0.348
4.liquidit~1	0.00221	0.0500	1	0.817
5.liquidit~1	0.0122	2.440	1	0.119
6.liquidit~1	0.0120	4.090	1	0.0431
7.liquidit~1	0.000510	0	1	0.952
8.liquidit~1	0.0205	8.510	1	0.00350
9.liquidit~1	0.00722	0.700	1	0.402
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	0.0183	5.570	1	0.0183
3.LTV cat 1	0.00427	0.150	1	0.699
4.LTV cat 1	0.0259	9.820	1	0.00170
1o.liquidi~	.	.	1	.
1.liquidit~1	-0.00454	0.170	1	0.676
1.liquidit~1	-0.00849	0.790	1	0.375
1.liquidit~1	0.00842	0.920	1	0.336
2o.liquidi~	.	.	1	.
2.liquidit~1	-0.0148	3.830	1	0.0505
2.liquidit~1	0.0133	1.110	1	0.291
2.liquidit~1	-0.0268	8.100	1	0.00440
3o.liquidi~	.	.	1	.
3.liquidit~1	-0.0107	1.030	1	0.309
3.liquidit~1	-0.00899	0.730	1	0.393
3.liquidit~1	-0.0117	1.990	1	0.158
4o.liquidi~	.	.	1	.
4.liquidit~1	-0.0282	7.200	1	0.00730
4.liquidit~1	-0.00616	0.340	1	0.563
4.liquidit~1	-0.0450	20.06	1	0
5o.liquidi~	.	.	1	.
5.liquidit~1	-0.0236	6.210	1	0.0127
5.liquidit~1	0.0123	0.990	1	0.320
5.liquidit~1	-0.0175	5.610	1	0.0179
6o.liquidi~	.	.	1	.
6.liquidit~1	-0.0257	18.21	1	0
6.liquidit~1	-0.00205	0.0400	1	0.850
6.liquidit~1	-0.0220	6.030	1	0.0141
7o.liquidi~	.	.	1	.
7.liquidit~1	-0.000810	0.0100	1	0.924
7.liquidit~1	0.0289	7.370	1	0.00660
7.liquidit~1	-0.00531	0.690	1	0.406
8o.liquidi~	.	.	1	.
8.liquidit~1	-0.00899	1.290	1	0.256
8.liquidit~1	0.00656	0.500	1	0.481
8.liquidit~1	-0.0164	5.690	1	0.0171
9o.liquidi~	.	.	1	.
9.liquidit~1	-0.0253	13.25	1	0.000300
9.liquidit~1	-0.00751	1.350	1	0.245
9.liquidit~1	-0.00686	0.560	1	0.453
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
1.wealth d~	-0.0160	4.650	1	0.0311
2.wealth d~	-0.0177	4.440	1	0.0351
3.wealth d~	-0.0177	6.410	1	0.0114
4.wealth d~	-0.0117	3.870	1	0.0493
5.wealth d~	-0.0106	4.020	1	0.0448
6.wealth d~	-0.00926	2.390	1	0.122

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
7.wealth d~	-0.00666	1.270	1	0.260
8.wealth d~	-0.0247	6.600	1	0.0102
9.wealth d~	-0.00390	0.240	1	0.621
10b.wealth	.	.	1	.
1.DTI 1	-0.0191	12.05	1	0.000500
2.DTI 1	-0.0169	11.20	1	0.000800
3.DTI 1	-0.0208	9.660	1	0.00190
4.DTI 1	-0.0238	14.72	1	0.000100
5.DTI 1	-0.0200	5.280	1	0.0215
6.DTI 1	-0.0140	6.740	1	0.00940
7.DTI 1	-0.0187	3.320	1	0.0684
8.DTI 1	-0.0150	4	1	0.0455
9.DTI 1	-0.0198	8.730	1	0.00310
10b.DTI 1	.	.	1	.
unoccupie	-0.00709	0.630	1	0.428
h agemax 1	-0.00529	0.740	1	0.390
h agemax 1	0.00528	0.750	1	0.386
h agemax 1	-0.00554	0.820	1	0.366
educ1 1	0.00603	0.710	1	0.399
educ2 1	0.0169	8.860	1	0.00290
educ3 1	0.0141	8.080	1	0.00450
educ4 1	0.0120	4.970	1	0.0258
o.educ5 1	.	.	1	.
1b.propert	.	.	1	.
2.properti~	-0.0109	1.550	1	0.213
3.properti~	-0.0189	5.990	1	0.0144
BuildingAg	-0.0150	10.31	1	0.00130
BuildingAg	0.0150	11.23	1	0.000800
BuildingAg	-0.0154	11.84	1	0.000600
Bathrooms	-0.00201	0.0600	1	0.808
Bathrooms	0.00316	0.140	1	0.704
Bathrooms	-0.00479	0.340	1	0.561
Rooms 1	0.000750	0.0200	1	0.879
Rooms2	-0.000820	0.0300	1	0.866
Rooms3	-0.000190	0	1	0.969
2005b.year	.	.	1	.
2006.year 1	0.0128	9.810	1	0.00170
2007.year 1	0.0197	16.82	1	0
Global	test	111.8	78	0.00720
Note	Robust	ıceb covar	matrix	used.

Ph test model 3 regression 1

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	0.0201	1.770	1	0.184
2.liquidit~1	0.00956	0.510	1	0.475
3.liquidit~1	0.0550	26.46	1	0
4.liquidit~1	0.0564	22.33	1	0
5.liquidit~1	0.0486	7.940	1	0.00480
6.liquidit~1	0.00641	0.240	1	0.627
7.liquidit~1	0.0618	29.70	1	0
8.liquidit~1	0.0667	35.90	1	0
9.liquidit~1	0.0339	4.710	1	0.0300
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	-0.00663	0.330	1	0.568
3.LTV cat 1	-0.0172	1.970	1	0.161
4.LTV cat 1	0.0490	15.40	1	0.000100
1.wealth d~	0.0328	4.020	1	0.0450
2.wealth d~	0.0215	2.910	1	0.0881
3.wealth d~	0.0385	8.910	1	0.00280
4.wealth d~	0.0423	14.41	1	0.000100
5.wealth d~	0.0506	29.65	1	0
6.wealth d~	0.0398	10	1	0.00160
7.wealth d~	0.0406	12.17	1	0.000500
8.wealth d~	-0.0190	1.670	1	0.196
9.wealth d~	0.0389	7.420	1	0.00650
10b.wealth	.	.	1	.
1.DTI 1	0.0422	17.66	1	0
2.DTI 1	0.0583	33.97	1	0
3.DTI 1	0.00418	0.0800	1	0.772
4.DTI 1	-0.0249	2.770	1	0.0958
5.DTI 1	0.0172	1.350	1	0.246
6.DTI 1	0.0272	5.320	1	0.0211
7.DTI 1	-0.0440	13.67	1	0.000200
8.DTI 1	0.000310	0	1	0.980
9.DTI 1	-0.000940	0.0100	1	0.937
10b.DTI 1	.	.	1	.
unoccupie	-0.0197	1.840	1	0.175
h agemax 1	0.0450	10.46	1	0.00120
h agemax 1	-0.0480	12.05	1	0.000500
h agemax 1	0.0481	11.66	1	0.000600
educ1 1	-0.0466	23.01	1	0
educ2 1	-0.0456	24.35	1	0
educ3 1	-0.0578	43.09	1	0
educ4 1	-0.0418	17.75	1	0
o.educ5 1	.	.	1	.
1b.propert~	.	.	1	.
2.properti~	-0.0384	18.35	1	0
3.properti~	0.0497	14.42	1	0.000100
BuildingAg~	0.0212	4.680	1	0.0305
BuildingAg	-0.0310	10.09	1	0.00150
BuildingAg	0.0336	10.93	1	0.000900
Bathrooms	0.0606	26.58	1	0
Bathrooms	-0.0537	19.90	1	0
Bathrooms	0.0428	12.36	1	0.000400
Rooms 1	0.0492	37.93	1	0
Rooms2	-0.0539	52.02	1	0
Rooms3	0.0556	56.15	1	0
2005o.year	.	.	1	.
Global	test	179.2	49	0
Note	Robust	ıceb covar	matrix	used.

Ph test model 3 regression 2

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	0.00716	0.390	1	0.535
2.liquidit~1	0.00224	0.0400	1	0.835
3.liquidit~1	0.0469	23	1	0
4.liquidit~1	0.0367	13.31	1	0.000300
5.liquidit~1	0.0165	1.440	1	0.231
6.liquidit~1	0.00196	0.0300	1	0.858
7.liquidit~1	0.0407	15.74	1	0.000100
8.liquidit~1	0.0240	7.010	1	0.00810
9.liquidit~1	0.0128	1.160	1	0.281
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	-0.0199	3.020	1	0.0821
3.LTV cat 1	-0.00578	0.200	1	0.655
4.LTV cat 1	0.00421	0.160	1	0.689
1o.liquidi~	.	.	1	.
1.liquidit~1	0.0221	2.310	1	0.129
1.liquidit~1	-0.0253	2.860	1	0.0906
1.liquidit~1	0.0160	2.570	1	0.109
2o.liquidi~	.	.	1	.
2.liquidit~1	0.0322	6.480	1	0.0109
2.liquidit~1	-0.0356	6.230	1	0.0126
2.liquidit~1	-0.0101	0.890	1	0.345
3o.liquidi~	.	.	1	.
3.liquidit~1	-0.0200	1.790	1	0.181
3.liquidit~1	0.00375	0.0500	1	0.818
3.liquidit~1	-0.00164	0.0200	1	0.881
4o.liquidi~	.	.	1	.
4.liquidit~1	0.0197	2.800	1	0.0945
4.liquidit~1	-0.00357	0.0600	1	0.809
4.liquidit~1	-0.00874	0.620	1	0.430
5o.liquidi~	.	.	1	.
5.liquidit~1	0.0297	5.290	1	0.0215
5.liquidit~1	0.00482	0.0900	1	0.762
5.liquidit~1	0.0492	14.91	1	0.000100
6o.liquidi~	.	.	1	.
6.liquidit~1	0.0143	1.450	1	0.229
6.liquidit~1	-0.00735	0.310	1	0.579
6.liquidit~1	-0.00466	0.140	1	0.709
7o.liquidi~	.	.	1	.
7.liquidit~1	0.0360	7.960	1	0.00480
7.liquidit~1	0.00390	0.0800	1	0.778
7.liquidit~1	-0.00822	0.430	1	0.512
8o.liquidi~	.	.	1	.
8.liquidit~1	0.00969	0.870	1	0.351
8.liquidit~1	-0.00353	0.0800	1	0.775
8.liquidit~1	0.0231	5.850	1	0.0156
9o.liquidi~	.	.	1	.
9.liquidit~1	0.0190	2.720	1	0.0990
9.liquidit~1	0.0409	21.20	1	0
9.liquidit~1	-0.00662	0.440	1	0.508
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
1.wealth d~	0.0218	2.280	1	0.131
2.wealth d~	-0.00302	0.0700	1	0.793
3.wealth d~	0.0132	1.230	1	0.268
4.wealth d~	0.0174	2.880	1	0.0894

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
5.wealth d~	0.0296	11.92	1	0.000600
6.wealth d~	0.0195	2.780	1	0.0957
7.wealth d~	0.0146	1.960	1	0.161
8.wealth d~	-0.0325	5.390	1	0.0202
9.wealth d~	0.0176	1.830	1	0.176
10b.wealth	.	.	1	.
1.DTI 1	0.0257	8.010	1	0.00470
2.DTI 1	0.0449	25.31	1	0
3.DTI 1	-0.0135	1.210	1	0.272
4.DTI 1	-0.0277	4.610	1	0.0318
5.DTI 1	-0.000590	0	1	0.962
6.DTI 1	0.0160	2.420	1	0.120
7.DTI 1	-0.0462	17.57	1	0
8.DTI 1	-0.0114	1.100	1	0.293
9.DTI 1	-0.0111	1.110	1	0.292
10b.DTI 1	.	.	1	.
unoccupie	-0.0169	1.700	1	0.193
h agemax 1	0.0466	14.86	1	0.000100
h agemax 1	-0.0484	16.02	1	0.000100
h agemax 1	0.0478	15.06	1	0.000100
educ1 1	-0.0362	15.86	1	0.000100
educ2 1	-0.0319	14.68	1	0.000100
educ3 1	-0.0370	23.21	1	0
educ4 1	-0.0272	9.420	1	0.00210
o.educ5 1	.	.	1	.
1b.propert~	.	.	1	.
2.properti~	-0.0300	13.37	1	0.000300
3.properti~	0.0553	24.45	1	0
BuildingAg~	0.0256	7.950	1	0.00480
BuildingAg	-0.0319	12.53	1	0.000400
BuildingAg	0.0324	12.07	1	0.000500
Bathrooms	0.0590	27.39	1	0
Bathrooms	-0.0506	19.79	1	0
Bathrooms	0.0393	12.17	1	0.000500
Rooms 1	0.0469	39.46	1	0
Rooms2	-0.0528	56.43	1	0
Rooms3	0.0555	62.57	1	0
2005o.year	.	.	1	.
Global	test	207.5	76	0
Note	Robust	ıceb covar	matrix	used.

Ph test model 4 regression 1

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	-0.00128	0.0100	1	0.938
2.liquidit~1	0.0126	0.630	1	0.427
3.liquidit~1	0.0251	4.750	1	0.0294
4.liquidit~1	0.00161	0.0100	1	0.915
5.liquidit~1	0.0243	1.660	1	0.197
6.liquidit~1	0.0262	3.380	1	0.0661
7.liquidit~1	0.0238	2.860	1	0.0906
8.liquidit~1	0.0560	15.10	1	0.000100
9.liquidit~1	-0.0236	1.520	1	0.217
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	0.0153	1.640	1	0.201
3.LTV cat 1	0.0241	3.950	1	0.0469
4.LTV cat 1	0.0444	11.30	1	0.000800
1.wealth d~	-0.00522	0.0800	1	0.779
2.wealth d~	0.0266	3.930	1	0.0473
3.wealth d~	0.0300	3.910	1	0.0479
4.wealth d~	0.0335	7.690	1	0.00550
5.wealth d~	0.0290	8.350	1	0.00390
6.wealth d~	0.0325	4.610	1	0.0319
7.wealth d~	0.0303	5.420	1	0.0199
8.wealth d~	0.0146	0.620	1	0.430
9.wealth d~	0.0454	11.51	1	0.000700
10b.wealth	.	.	1	.
1.DTI 1	0.00675	0.370	1	0.542
2.DTI 1	0.0175	2.670	1	0.102
3.DTI 1	0.00565	0.140	1	0.707
4.DTI 1	-0.0595	13.25	1	0.000300
5.DTI 1	-0.0190	1.400	1	0.236
6.DTI 1	0.00357	0.0700	1	0.788
7.DTI 1	-0.0346	4.320	1	0.0377
8.DTI 1	-0.00148	0.0100	1	0.907
9.DTI 1	0.0192	2.110	1	0.146
10b.DTI 1	.	.	1	.
unoccupie(	0.00906	0.340	1	0.561
h agemax 1	-0.00182	0.0200	1	0.887
h agemax 1	-0.000280	0	1	0.983
h agemax 1	0.000800	0	1	0.953
educ1 1	-0.0393	13	1	0.000300
educ2 1	-0.0373	13.31	1	0.000300
educ3 1	-0.0294	10.69	1	0.00110
educ4 1	-0.0437	16.68	1	0
o.educ5 1	.	.	1	.
1b.propert~	.	.	1	.
2.properti~	0.0132	1.280	1	0.259
3.properti~	-0.0127	0.710	1	0.401
BuildingAg~	0.00522	0.370	1	0.545
BuildingAge	-0.0134	2.410	1	0.121
BuildingAge	0.0192	4.640	1	0.0313
Bathrooms	0.000110	0	1	0.994
Bathrooms	0.0117	0.610	1	0.434
Bathrooms	-0.0248	3.110	1	0.0780
Rooms 1	0.0198	5.440	1	0.0197
Rooms2	-0.0138	2.860	1	0.0908
Rooms3	0.00895	1.190	1	0.275
2005o.year	.	.	1	.
Global	test	83.81	49	0.00140
Note	Robust	ıceb covar	matrix	used.

Ph test model 4 regression 2

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	-0.00180	0.0200	1	0.886
2.liquidit~1	0.0125	1.180	1	0.278
3.liquidit~1	0.0186	2.470	1	0.116
4.liquidit~1	-0.000170	0	1	0.989
5.liquidit~1	0.00543	0.130	1	0.716
6.liquidit~1	0.0129	1.400	1	0.236
7.liquidit~1	0.0133	1.080	1	0.300
8.liquidit~1	0.0151	1.650	1	0.199
9.liquidit~1	-0.00556	0.150	1	0.696
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	0.00294	0.0400	1	0.847
3.LTV cat 1	0.0260	3.940	1	0.0472
4.LTV cat 1	0.000520	0	1	0.965
1o.liquidi~	.	.	1	.
1.liquidit~1	-0.00186	0.0100	1	0.913
1.liquidit~1	-0.0339	3.090	1	0.0789
1.liquidit~1	0.0137	1.430	1	0.231
2o.liquidi~	.	.	1	.
2.liquidit~1	-0.00154	0.0100	1	0.916
2.liquidit~1	-0.0471	7.070	1	0.00780
2.liquidit~1	-0.00933	0.700	1	0.402
3o.liquidi~	.	.	1	.
3.liquidit~1	0.00860	0.290	1	0.592
3.liquidit~1	-0.00203	0.0100	1	0.914
3.liquidit~1	0.00368	0.0900	1	0.759
4o.liquidi~	.	.	1	.
4.liquidit~1	0.00577	0.130	1	0.714
4.liquidit~1	-0.0217	1.530	1	0.216
4.liquidit~1	7.00e-05	0	1	0.995
5o.liquidi~	.	.	1	.
5.liquidit~1	0.00516	0.100	1	0.748
5.liquidit~1	-0.0148	0.900	1	0.343
5.liquidit~1	0.0222	3.170	1	0.0749
6o.liquidi~	.	.	1	.
6.liquidit~1	0.000690	0	1	0.959
6.liquidit~1	-0.0352	4.910	1	0.0267
6.liquidit~1	-0.00573	0.230	1	0.632
7o.liquidi~	.	.	1	.
7.liquidit~1	0.0134	0.950	1	0.331
7.liquidit~1	-0.0323	3.950	1	0.0468
7.liquidit~1	0.00478	0.210	1	0.649
8o.liquidi~	.	.	1	.
8.liquidit~1	-0.0131	0.930	1	0.334
8.liquidit~1	-0.00477	0.120	1	0.724
8.liquidit~1	0.0286	6.180	1	0.0129
9o.liquidi~	.	.	1	.
9.liquidit~1	-0.00544	0.200	1	0.651
9.liquidit~1	0.00141	0.0200	1	0.884
9.liquidit~1	0.0198	2.370	1	0.123
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
1.wealth d~	-0.0253	2.730	1	0.0987
2.wealth d~	-0.00775	0.410	1	0.522
3.wealth d~	-0.00189	0.0200	1	0.892
4.wealth d~	0.000110	0	1	0.992

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
5.wealth d~	0.00613	0.450	1	0.501
6.wealth d~	-0.00141	0.0100	1	0.918
7.wealth d~	-0.00732	0.400	1	0.528
8.wealth d~	-0.0231	1.680	1	0.195
9.wealth d~	0.00579	0.230	1	0.631
10b.wealth	.	.	1	.
1.DTI 1	-0.0154	2.380	1	0.123
2.DTI 1	0.00369	0.150	1	0.703
3.DTI 1	-0.0233	3.520	1	0.0606
4.DTI 1	-0.0534	13.91	1	0.000200
5.DTI 1	-0.0388	8.200	1	0.00420
6.DTI 1	-0.0118	1.060	1	0.304
7.DTI 1	-0.0460	9.140	1	0.00250
8.DTI 1	-0.0173	2.320	1	0.128
9.DTI 1	0.00126	0.0100	1	0.916
10b.DTI 1	.	.	1	.
unoccupie(	0.0180	1.790	1	0.181
h agemax 1	0.0142	1.570	1	0.211
h agemax 1	-0.0155	1.820	1	0.178
h agemax 1	0.0164	1.870	1	0.172
educ1 1	-0.0265	6.730	1	0.00950
educ2 1	-0.0179	3.820	1	0.0505
educ3 1	-0.0140	2.990	1	0.0839
educ4 1	-0.0256	7.030	1	0.00800
o.educ5 1	.	.	1	.
1b.propert~	.	.	1	.
2.properti~	0.0210	4.170	1	0.0411
3.properti~	-0.00402	0.100	1	0.757
BuildingAg~	0.00940	1.390	1	0.238
BuildingAge	-0.0146	3.360	1	0.0666
BuildingAge	0.0183	4.980	1	0.0257
Bathrooms	0.00660	0.200	1	0.651
Bathrooms	0.000460	0	1	0.974
Bathrooms	-0.00875	0.450	1	0.504
Rooms 1	0.0149	3.670	1	0.0555
Rooms2	-0.0109	2.090	1	0.148
Rooms3	0.00750	0.980	1	0.322
2005o.year	.	.	1	.
Global	test	91.87	76	0.104
Note	Robust	ıceb covar	matrix	used.

Ph test model 5 regression 1

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	-0.0316	4.740	1	0.0295
2.liquidit~1	-0.0317	7.890	1	0.00500
3.liquidit~1	-0.0205	3.030	1	0.0817
4.liquidit~1	-0.0292	3.050	1	0.0809
5.liquidit~1	0.00622	0.180	1	0.672
6.liquidit~1	-0.0175	1.170	1	0.280
7.liquidit~1	-0.00398	0.0900	1	0.767
8.liquidit~1	0.0126	0.750	1	0.386
9.liquidit~1	-0.0172	0.970	1	0.324
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	-0.00531	0.110	1	0.743
3.LTV cat 1	0.0146	1.380	1	0.240
4.LTV cat 1	0.0314	5.860	1	0.0155
1.wealth d~	-0.00453	0.120	1	0.734
2.wealth d~	-0.00535	0.130	1	0.723
3.wealth d~	-0.00311	0.0500	1	0.826
4.wealth d~	-0.00421	0.0900	1	0.758
5.wealth d~	0.00383	0.0800	1	0.774
6.wealth d~	-0.00266	0.0400	1	0.846
7.wealth d~	0.0150	1.380	1	0.241
8.wealth d~	0.000870	0	1	0.956
9.wealth d~	-0.00259	0.0300	1	0.859
10b.wealth	.	.	1	.
1.DTI 1	-0.0175	2.070	1	0.150
2.DTI 1	-0.0290	7.140	1	0.00750
3.DTI 1	-0.0346	8.370	1	0.00380
4.DTI 1	-0.0198	3.030	1	0.0816
5.DTI 1	-0.0112	0.640	1	0.425
6.DTI 1	-0.0155	2.210	1	0.137
7.DTI 1	-0.0315	5.690	1	0.0171
8.DTI 1	-0.0151	1.740	1	0.188
9.DTI 1	-0.00965	0.480	1	0.488
10b.DTI 1	.	.	1	.
unoccupie	0.00219	0.0300	1	0.860
h agemax 1	0.0369	6.550	1	0.0105
h agemax 1	-0.0334	5.900	1	0.0151
h agemax 1	0.0304	5.200	1	0.0226
educ1 1	-0.00237	0.0300	1	0.860
educ2 1	0.0211	3.690	1	0.0548
educ3 1	-0.00962	0.520	1	0.471
educ4 1	0.0131	1.590	1	0.208
o.educ5 1	.	.	1	.
1b.propert	.	.	1	.
2.properti~	-0.0184	3.260	1	0.0711
3.properti~	-0.0336	6.740	1	0.00940
BuildingAg	-0.0103	1.030	1	0.310
BuildingAg	0.0116	1.480	1	0.224
BuildingAg	-0.0125	1.780	1	0.182
Bathrooms	0.0260	3.420	1	0.0642
Bathrooms	-0.0257	3.360	1	0.0666
Bathrooms	0.0249	3.190	1	0.0742
Rooms 1	0.0179	2.900	1	0.0883
Rooms2	-0.0212	3.950	1	0.0468
Rooms3	0.0216	3.790	1	0.0517
2006o.year	.	.	1	.
Global	test	54.27	49	0.281
Note	Robust	ıceb covar	matrix	used.

Ph test model 5 regression 2

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	-0.00180	0.0200	1	0.891
2.liquidit~1	-0.00340	0.0900	1	0.758
3.liquidit~1	0.0181	2.750	1	0.0975
4.liquidit~1	0.0139	0.780	1	0.376
5.liquidit~1	0.0325	6.750	1	0.00940
6.liquidit~1	0.0124	0.650	1	0.421
7.liquidit~1	0.0172	2.390	1	0.122
8.liquidit~1	0.0295	7.670	1	0.00560
9.liquidit~1	0.0173	1.150	1	0.284
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	0.0374	6.640	1	0.00990
3.LTV cat 1	0.00969	0.460	1	0.496
4.LTV cat 1	0.0481	14.63	1	0.000100
1o.liquidi~	.	.	1	.
1.liquidit~1	-0.0242	2.720	1	0.0993
1.liquidit~1	0.0106	1	1	0.317
1.liquidit~1	-0.0223	3.020	1	0.0820
2o.liquidi~	.	.	1	.
2.liquidit~1	-0.0299	5.200	1	0.0226
2.liquidit~1	0.0363	5.340	1	0.0208
2.liquidit~1	-0.0193	1.730	1	0.188
3o.liquidi~	.	.	1	.
3.liquidit~1	-0.0365	6.790	1	0.00920
3.liquidit~1	-0.0162	1.580	1	0.208
3.liquidit~1	-0.0158	1.250	1	0.264
4o.liquidi~	.	.	1	.
4.liquidit~1	-0.0172	1.460	1	0.227
4.liquidit~1	-0.0441	11.27	1	0.000800
4.liquidit~1	-0.0523	20.16	1	0
5o.liquidi~	.	.	1	.
5.liquidit~1	-0.0270	3.480	1	0.0622
5.liquidit~1	0.0385	7.500	1	0.00620
5.liquidit~1	-0.0487	15.16	1	0.000100
6o.liquidi~	.	.	1	.
6.liquidit~1	-0.0336	6.880	1	0.00870
6.liquidit~1	0.00361	0.100	1	0.748
6.liquidit~1	-0.0243	3.690	1	0.0548
7o.liquidi~	.	.	1	.
7.liquidit~1	-0.0456	11.22	1	0.000800
7.liquidit~1	0.0317	7.530	1	0.00610
7.liquidit~1	-0.00702	0.430	1	0.513
8o.liquidi~	.	.	1	.
8.liquidit~1	-0.00719	0.360	1	0.548
8.liquidit~1	-0.0209	3.150	1	0.0757
8.liquidit~1	-0.0354	10.30	1	0.00130
9o.liquidi~	.	.	1	.
9.liquidit~1	-0.0300	4.600	1	0.0320
9.liquidit~1	-0.0274	2.690	1	0.101
9.liquidit~1	-0.0136	1.090	1	0.296
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
1.wealth d~	-0.0222	3.900	1	0.0483
2.wealth d~	-0.0208	2.400	1	0.121
3.wealth d~	-0.0269	4.730	1	0.0297
4.wealth d~	-0.0209	2.970	1	0.0850

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
5.wealth d~	-0.0215	3.160	1	0.0753
6.wealth d~	-0.0185	2.270	1	0.132
7.wealth d~	0.00263	0.0600	1	0.814
8.wealth d~	-0.0156	1.380	1	0.241
9.wealth d~	-0.000380	0	1	0.977
10b.wealth	.	.	1	.
1.DTI 1	-0.0264	5.580	1	0.0182
2.DTI 1	-0.0457	21.03	1	0
3.DTI 1	-0.0329	8.950	1	0.00280
4.DTI 1	-0.0317	9.020	1	0.00270
5.DTI 1	-0.0288	4.840	1	0.0279
6.DTI 1	-0.0230	5.500	1	0.0190
7.DTI 1	-0.0307	6.550	1	0.0105
8.DTI 1	-0.0219	4.510	1	0.0337
9.DTI 1	-0.0226	3.640	1	0.0564
10b.DTI 1	.	.	1	.
unoccupie	-0.0135	1.430	1	0.232
h agemax 1	0.0409	9.960	1	0.00160
h agemax 1	-0.0376	9.100	1	0.00260
h agemax 1	0.0342	7.910	1	0.00490
educ1 1	-0.000830	0	1	0.947
educ2 1	0.0244	5.630	1	0.0177
educ3 1	0.000860	0	1	0.944
educ4 1	0.0173	3.170	1	0.0749
o.educ5 1	.	.	1	.
1b.propert	.	.	1	.
2.properti~	-0.0326	12.02	1	0.000500
3.properti~	-0.0309	9.010	1	0.00270
BuildingAg	-0.0275	8.940	1	0.00280
BuildingAg	0.0281	10.65	1	0.00110
BuildingAg	-0.0283	11.25	1	0.000800
Bathrooms	0.0126	0.950	1	0.331
Bathrooms	-0.0140	1.170	1	0.280
Bathrooms	0.0157	1.440	1	0.229
Rooms 1	0.00248	0.0700	1	0.798
Rooms2	-0.00391	0.160	1	0.693
Rooms3	0.00249	0.0600	1	0.808
2006o.year	.	.	1	.
Global	test	92.75	76	0.0929
Note	Robust	ıceb covar	matrix	used.

Ph test model 6 regression 1

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	-0.0231	2.390	1	0.122
2.liquidit~1	-0.0201	2.750	1	0.0973
3.liquidit~1	-0.0176	1.950	1	0.163
4.liquidit~1	-0.0282	2.910	1	0.0881
5.liquidit~1	-0.00992	0.430	1	0.513
6.liquidit~1	-0.0138	0.780	1	0.376
7.liquidit~1	0.0161	1.400	1	0.237
8.liquidit~1	0.0213	2.300	1	0.130
9.liquidit~1	-0.0111	0.370	1	0.543
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	-0.00710	0.210	1	0.649
3.LTV cat 1	0.0202	2.540	1	0.111
4.LTV cat 1	0.0244	3.310	1	0.0688
1.wealth d~	-0.00635	0.200	1	0.657
2.wealth d~	-0.00236	0.0200	1	0.883
3.wealth d~	-0.00790	0.260	1	0.607
4.wealth d~	-0.0173	1.290	1	0.255
5.wealth d~	-0.0117	0.590	1	0.443
6.wealth d~	-0.0106	0.490	1	0.483
7.wealth d~	0.00956	0.430	1	0.511
8.wealth d~	0.00162	0.0100	1	0.922
9.wealth d~	-0.00557	0.140	1	0.710
10b.wealth	.	.	1	.
1.DTI 1	-0.0267	3.960	1	0.0466
2.DTI 1	-0.0370	8.440	1	0.00370
3.DTI 1	-0.0368	9.540	1	0.00200
4.DTI 1	-0.0272	5.310	1	0.0213
5.DTI 1	-0.0115	0.570	1	0.449
6.DTI 1	-0.0142	1.670	1	0.196
7.DTI 1	-0.0274	4.370	1	0.0366
8.DTI 1	-0.0307	6.730	1	0.00950
9.DTI 1	-0.00898	0.360	1	0.549
10b.DTI 1	.	.	1	.
unoccupie	-0.0196	1.760	1	0.184
h agemax 1	0.0368	6.200	1	0.0128
h agemax 1	-0.0345	6.040	1	0.0140
h agemax 1	0.0328	5.840	1	0.0156
educ1 1	0.0157	1.220	1	0.270
educ2 1	0.0316	7.400	1	0.00650
educ3 1	-0.00290	0.0400	1	0.850
educ4 1	0.0206	3.780	1	0.0520
o.educ5 1	.	.	1	.
1b.propert	.	.	1	.
2.properti~	-0.0270	6.850	1	0.00890
3.properti~	-0.0193	2.070	1	0.150
BuildingAg	-0.00834	0.630	1	0.428
BuildingAg	0.0141	2.040	1	0.153
BuildingAg	-0.0178	3.330	1	0.0682
Bathrooms	0.0302	4.880	1	0.0272
Bathrooms	-0.0301	4.830	1	0.0280
Bathrooms	0.0291	4.470	1	0.0344
Rooms 1	0.0198	2.980	1	0.0843
Rooms2	-0.0242	4.260	1	0.0391
Rooms3	0.0259	4.410	1	0.0357
2006o.year	.	.	1	.
Global	test	50.92	49	0.398
Note	Robust	ıceb covar	matrix	used.

Ph test model 6 regression 2

Test	of	proportional-hazards		
Time	function	Analysis	time	
rho	chi2	df	Prob>chi2	
1.liquidit~1	-0.00694	0.250	1	0.621
2.liquidit~1	-0.000320	0	1	0.979
3.liquidit~1	0.00882	0.600	1	0.437
4.liquidit~1	0.0104	0.470	1	0.495
5.liquidit~1	0.0129	0.960	1	0.327
6.liquidit~1	0.00563	0.150	1	0.695
7.liquidit~1	0.0209	3.500	1	0.0615
8.liquidit~1	0.0291	7.370	1	0.00660
9.liquidit~1	0.0127	0.580	1	0.448
10b.liquid~	.	.	1	.
1b.LTV cat	.	.	1	.
2.LTV cat 1	0.0237	2.720	1	0.0994
3.LTV cat 1	0.00329	0.0500	1	0.820
4.LTV cat 1	0.0488	13.59	1	0.000200
1o.liquidi~	.	.	1	.
1.liquidit~1	-0.0235	2.590	1	0.108
1.liquidit~1	0.00578	0.300	1	0.584
1.liquidit~1	-0.00667	0.240	1	0.626
2o.liquidi~	.	.	1	.
2.liquidit~1	-0.0262	3.650	1	0.0561
2.liquidit~1	0.0364	5.140	1	0.0234
2.liquidit~1	-0.0425	8.420	1	0.00370
3o.liquidi~	.	.	1	.
3.liquidit~1	-0.0217	2.370	1	0.123
3.liquidit~1	-0.0112	0.690	1	0.405
3.liquidit~1	-0.0163	1.250	1	0.263
4o.liquidi~	.	.	1	.
4.liquidit~1	-0.0289	3.590	1	0.0581
4.liquidit~1	-0.0517	17.95	1	0
4.liquidit~1	-0.0623	29.58	1	0
5o.liquidi~	.	.	1	.
5.liquidit~1	-0.0315	4.100	1	0.0428
5.liquidit~1	0.0341	5.400	1	0.0201
5.liquidit~1	-0.0353	7.130	1	0.00760
6o.liquidi~	.	.	1	.
6.liquidit~1	-0.0355	7.340	1	0.00670
6.liquidit~1	0.00126	0.0100	1	0.917
6.liquidit~1	-0.0263	3.860	1	0.0494
7o.liquidi~	.	.	1	.
7.liquidit~1	-0.0207	2.180	1	0.140
7.liquidit~1	0.0359	8.220	1	0.00410
7.liquidit~1	-0.00885	0.650	1	0.422
8o.liquidi~	.	.	1	.
8.liquidit~1	0.00598	0.230	1	0.631
8.liquidit~1	-0.000130	0	1	0.992
8.liquidit~1	-0.0336	8.950	1	0.00280
9o.liquidi~	.	.	1	.
9.liquidit~1	-0.0286	3.540	1	0.0600
9.liquidit~1	-0.0182	1.150	1	0.283
9.liquidit~1	-0.0223	2.700	1	0.100
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
10b.liquid~	.	.	1	.
1.wealth d~	-0.0154	1.640	1	0.201
2.wealth d~	-0.0128	0.770	1	0.380
3.wealth d~	-0.0241	3.170	1	0.0748
4.wealth d~	-0.0237	3.140	1	0.0766

Test	of	proportional-hazards		
5.wealth d~	-0.0256	3.420	1	0.0643
6.wealth d~	-0.0230	2.970	1	0.0850
7.wealth d~	0.00140	0.0100	1	0.912
8.wealth d~	-0.0237	2.780	1	0.0952
9.wealth d~	-0.00600	0.190	1	0.663
10b.wealth	.	.	1	.
1.DTI 1	-0.0297	6.090	1	0.0136
2.DTI 1	-0.0434	14.28	1	0.000200
3.DTI 1	-0.0299	7.180	1	0.00740
4.DTI 1	-0.0325	8.790	1	0.00300
5.DTI 1	-0.0202	1.930	1	0.164
6.DTI 1	-0.0200	3.780	1	0.0518
7.DTI 1	-0.0265	5.070	1	0.0244
8.DTI 1	-0.0308	8.330	1	0.00390
9.DTI 1	-0.0146	1.330	1	0.249
10b.DTI 1	.	.	1	.
unoccupie	-0.0221	2.820	1	0.0929
h agemax 1	0.0373	7.770	1	0.00530
h agemax 1	-0.0352	7.590	1	0.00590
h agemax 1	0.0330	7.020	1	0.00800
educ1 1	0.0228	2.970	1	0.0851
educ2 1	0.0342	9.900	1	0.00160
educ3 1	0.0104	0.550	1	0.458
educ4 1	0.0242	6.010	1	0.0142
o.educ5 1	.	.	1	.
1b.propert	.	.	1	.
2.properti~	-0.0360	13.86	1	0.000200
3.properti~	-0.0265	6	1	0.0143
BuildingAg	-0.0156	2.690	1	0.101
BuildingAg	0.0226	6.320	1	0.0120
BuildingAg	-0.0271	9.320	1	0.00230
Bathrooms	0.0157	1.560	1	0.212
Bathrooms	-0.0163	1.680	1	0.196
Bathrooms	0.0166	1.710	1	0.191
Rooms 1	0.00563	0.290	1	0.593
Rooms2	-0.00807	0.560	1	0.455
Rooms3	0.00672	0.350	1	0.553
2006o.year	.	.	1	.
Global	test	92.44	76	0.0967
Note	Robust	ıceb covar	matrix	used.

REGRESSION 1 (BASELINE SPECIFICATION)

	MODEL 1			MODEL 2		
	(1)	(2)	(3)	(1)	(2)	(3)
	_t			_t		
VARIABLES	coefEform	seEform	pval	coefEform	seEform	pval
_t						
1.liquidity_dec_1	0.836**	(0.0626)	0.0168	0.849**	(0.0682)	0.0420
2.liquidity_dec_1	0.872	(0.0757)	0.113	0.872	(0.0852)	0.161
3.liquidity_dec_1	0.849**	(0.0598)	0.0199	0.873*	(0.0688)	0.0856
4.liquidity_dec_1	0.889*	(0.0613)	0.0888	0.903	(0.0673)	0.171
5.liquidity_dec_1	0.896*	(0.0593)	0.0983	0.923	(0.0638)	0.248
6.liquidity_dec_1	0.988	(0.0684)	0.863	1.034	(0.0813)	0.675
7.liquidity_dec_1	0.921	(0.0600)	0.209	0.942	(0.0625)	0.370
8.liquidity_dec_1	0.959	(0.0671)	0.551	0.969	(0.0760)	0.687
9.liquidity_dec_1	0.938	(0.0566)	0.292	0.963	(0.0669)	0.590
2.LTV_cat_1	0.860***	(0.0461)	0.00494	0.844***	(0.0483)	0.00303
3.LTV_cat_1	0.640***	(0.0536)	1.00e-07	0.645***	(0.0562)	4.89e-07
4.LTV_cat_1	0.526***	(0.0369)	0	0.543***	(0.0398)	0
1.wealth_dec_1	1.336***	(0.137)	0.00459	1.257*	(0.149)	0.0537
2.wealth_dec_1	1.220*	(0.125)	0.0525	1.125	(0.131)	0.310
3.wealth_dec_1	1.143	(0.122)	0.211	1.091	(0.125)	0.449
4.wealth_dec_1	1.266**	(0.148)	0.0435	1.197	(0.155)	0.166
5.wealth_dec_1	1.248*	(0.147)	0.0593	1.221	(0.157)	0.121
6.wealth_dec_1	1.190*	(0.122)	0.0883	1.146	(0.125)	0.211
7.wealth_dec_1	1.142	(0.122)	0.214	1.084	(0.123)	0.480
8.wealth_dec_1	1.082	(0.0944)	0.367	1.018	(0.0906)	0.841
9.wealth_dec_1	1.119	(0.0823)	0.125	1.134	(0.0973)	0.143
1.DTI_1	1.242*	(0.138)	0.0515	1.133	(0.144)	0.325
2.DTI_1	1.029	(0.120)	0.804	0.999	(0.121)	0.991
3.DTI_1	1.071	(0.0911)	0.422	1.017	(0.100)	0.864
4.DTI_1	1.126	(0.0858)	0.120	1.078	(0.0913)	0.373
5.DTI_1	0.963	(0.0693)	0.600	0.971	(0.0717)	0.690
6.DTI_1	1.033	(0.0875)	0.703	1.025	(0.0942)	0.787
7.DTI_1	1.052	(0.0658)	0.420	1.014	(0.0694)	0.845
8.DTI_1	1.061	(0.0763)	0.412	1.057	(0.0836)	0.486
9.DTI_1	1.094	(0.0716)	0.169	1.078	(0.0776)	0.299
unoccupied_1	0.898	(0.0592)	0.103	0.911	(0.0629)	0.177
h_agemax_1	0.942**	(0.0254)	0.0267	0.960	(0.0283)	0.162
h_agemax_1_sq	1.001*	(0.000538)	0.0770	1.001	(0.000591)	0.309
h_agemax_1_cb	1.000	(3.34e-06)	0.149	1.000	(3.71e-06)	0.452
educ1_1	1.150**	(0.0727)	0.0266	1.165**	(0.0798)	0.0253
educ2_1	1.136**	(0.0649)	0.0257	1.156**	(0.0727)	0.0216
educ3_1	1.158*	(0.0958)	0.0761	1.172*	(0.108)	0.0848
educ4_1	1.113*	(0.0714)	0.0967	1.149**	(0.0798)	0.0456
o.educ5_1	-		-	-		-
2.properties_1	1.115***	(0.0470)	0.00984	1.143***	(0.0543)	0.00478
3.properties_1	1.074	(0.0968)	0.430	1.100	(0.113)	0.355
BuildingAge_1	1.006*	(0.00361)	0.0884	1.004	(0.00442)	0.405
BuildingAge2	1.000	(4.08e-05)	0.220	1.000	(5.00e-05)	0.572
BuildingAge3	1.000	(1.20e-07)	0.298	1.000	(1.49e-07)	0.644
Bathrooms_1	2.479**	(0.970)	0.0203	4.327***	(2.073)	0.00223
Bathrooms2	0.599**	(0.130)	0.0183	0.450***	(0.116)	0.00189
Bathrooms3	1.079**	(0.0407)	0.0444	1.126***	(0.0479)	0.00536
Rooms_1	0.975	(0.146)	0.866	0.969	(0.162)	0.850
Rooms2	0.991	(0.0355)	0.809	0.983	(0.0384)	0.663
Rooms3	1.001	(0.00241)	0.613	1.002	(0.00259)	0.398
2006.year_1	1.055	(0.0611)	0.357	1.048	(0.0793)	0.531
2007.year_1	1.004	(0.0995)	0.966	0.973	(0.113)	0.816
Observations	93,768			88,495		

*** p<0.01, ** p<0.05, * p<0.1

REGRESSION 2 (INTERACTION TERM SPECIFICATION)

	MODEL 1			MODEL 2		
	(1)	(2)	(3)	(1)	(2)	(3)
	_t			_t		
VARIABLES	coefEform	seEform	pval	coefEform	seEform	pval
_t						
1.liquidity_dec_1	1.037	(0.0904)	0.681	1.067	(0.104)	0.505
2.liquidity_dec_1	1.085	(0.109)	0.418	1.114	(0.134)	0.368
3.liquidity_dec_1	1.038	(0.0878)	0.663	1.099	(0.0996)	0.300
4.liquidity_dec_1	1.132*	(0.0816)	0.0846	1.149*	(0.0897)	0.0761
5.liquidity_dec_1	1.113	(0.0853)	0.161	1.177*	(0.0993)	0.0541
6.liquidity_dec_1	1.218**	(0.0985)	0.0146	1.312***	(0.130)	0.00625
7.liquidity_dec_1	1.126	(0.0883)	0.131	1.178*	(0.104)	0.0649
8.liquidity_dec_1	1.138	(0.109)	0.174	1.184	(0.131)	0.126
9.liquidity_dec_1	1.118	(0.0836)	0.136	1.171*	(0.101)	0.0653
2.LTV_cat_1	1.532**	(0.254)	0.0103	1.593**	(0.289)	0.0102
3.LTV_cat_1	0.924	(0.173)	0.671	0.868	(0.189)	0.514
4.LTV_cat_1	0.985	(0.154)	0.921	1.085	(0.179)	0.622
1o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
1.liquidity_dec_1#2.LTV_cat_1	0.548***	(0.111)	0.00307	0.550***	(0.126)	0.00922
1.liquidity_dec_1#3.LTV_cat_1	0.691	(0.266)	0.337	0.777	(0.333)	0.556
1.liquidity_dec_1#4.LTV_cat_1	0.446***	(0.118)	0.00227	0.392***	(0.115)	0.00146
2o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
2.liquidity_dec_1#2.LTV_cat_1	0.624**	(0.117)	0.0116	0.561**	(0.131)	0.0130
2.liquidity_dec_1#3.LTV_cat_1	0.357**	(0.162)	0.0236	0.376**	(0.184)	0.0462
2.liquidity_dec_1#4.LTV_cat_1	0.319***	(0.0959)	0.000145	0.332***	(0.106)	0.000524
3o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
3.liquidity_dec_1#2.LTV_cat_1	0.632**	(0.123)	0.0187	0.572**	(0.134)	0.0174
3.liquidity_dec_1#3.LTV_cat_1	0.406**	(0.176)	0.0373	0.425*	(0.200)	0.0696
3.liquidity_dec_1#4.LTV_cat_1	0.459***	(0.110)	0.00112	0.430***	(0.109)	0.000834
4o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
4.liquidity_dec_1#2.LTV_cat_1	0.427***	(0.0826)	1.08e-05	0.484***	(0.101)	0.000518
4.liquidity_dec_1#3.LTV_cat_1	0.492**	(0.159)	0.0278	0.514*	(0.179)	0.0560
4.liquidity_dec_1#4.LTV_cat_1	0.510***	(0.113)	0.00235	0.488***	(0.112)	0.00173
5o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
5.liquidity_dec_1#2.LTV_cat_1	0.494***	(0.0970)	0.000329	0.425***	(0.0948)	0.000126
5.liquidity_dec_1#3.LTV_cat_1	0.829	(0.211)	0.461	0.912	(0.261)	0.747
5.liquidity_dec_1#4.LTV_cat_1	0.470***	(0.0945)	0.000174	0.458***	(0.102)	0.000483
6o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
6.liquidity_dec_1#2.LTV_cat_1	0.502***	(0.110)	0.00166	0.451***	(0.112)	0.00132
6.liquidity_dec_1#3.LTV_cat_1	0.731	(0.237)	0.333	0.836	(0.274)	0.585
6.liquidity_dec_1#4.LTV_cat_1	0.521***	(0.106)	0.00139	0.453***	(0.102)	0.000412
7o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
7.liquidity_dec_1#2.LTV_cat_1	0.580**	(0.127)	0.0127	0.550**	(0.142)	0.0202
7.liquidity_dec_1#3.LTV_cat_1	0.569*	(0.181)	0.0761	0.721	(0.250)	0.344
7.liquidity_dec_1#4.LTV_cat_1	0.513***	(0.0914)	0.000181	0.444***	(0.0956)	0.000163
8o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
8.liquidity_dec_1#2.LTV_cat_1	0.513***	(0.124)	0.00562	0.459***	(0.124)	0.00398
8.liquidity_dec_1#3.LTV_cat_1	1.175	(0.279)	0.496	1.145	(0.313)	0.620
8.liquidity_dec_1#4.LTV_cat_1	0.552**	(0.137)	0.0169	0.508**	(0.137)	0.0119

9o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
9.liquidity_dec_1#2.LTV_cat_1	0.597**	(0.152)	0.0430	0.558**	(0.149)	0.0288
9.liquidity_dec_1#3.LTV_cat_1	0.579*	(0.166)	0.0572	0.604	(0.195)	0.118
9.liquidity_dec_1#4.LTV_cat_1	0.549***	(0.117)	0.00507	0.522***	(0.116)	0.00340
10b.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
10b.liquidity_dec_1#2o.LTV_cat_1	1	(0)		1	(0)	
10b.liquidity_dec_1#3o.LTV_cat_1	1	(0)		1	(0)	
10b.liquidity_dec_1#4o.LTV_cat_1	1	(0)		1	(0)	
1.wealth_dec_1	1.212*	(0.122)	0.0575	1.131	(0.136)	0.305
2.wealth_dec_1	1.112	(0.113)	0.294	1.017	(0.116)	0.883
3.wealth_dec_1	1.042	(0.112)	0.700	0.986	(0.114)	0.900
4.wealth_dec_1	1.154	(0.133)	0.215	1.081	(0.138)	0.543
5.wealth_dec_1	1.137	(0.132)	0.269	1.102	(0.140)	0.446
6.wealth_dec_1	1.085	(0.109)	0.418	1.036	(0.113)	0.748
7.wealth_dec_1	1.043	(0.110)	0.690	0.982	(0.111)	0.869
8.wealth_dec_1	0.992	(0.0872)	0.931	0.926	(0.0833)	0.392
9.wealth_dec_1	1.040	(0.0755)	0.592	1.047	(0.0884)	0.588
1.DTI_1	1.235*	(0.135)	0.0539	1.122	(0.141)	0.361
2.DTI_1	1.035	(0.119)	0.764	1.002	(0.121)	0.988
3.DTI_1	1.081	(0.0913)	0.359	1.025	(0.101)	0.800
4.DTI_1	1.136*	(0.0857)	0.0897	1.087	(0.0906)	0.315
5.DTI_1	0.971	(0.0694)	0.682	0.978	(0.0716)	0.761
6.DTI_1	1.040	(0.0870)	0.640	1.031	(0.0934)	0.732
7.DTI_1	1.060	(0.0663)	0.352	1.020	(0.0698)	0.768
8.DTI_1	1.067	(0.0767)	0.370	1.063	(0.0841)	0.440
9.DTI_1	1.098	(0.0726)	0.158	1.081	(0.0781)	0.284
unoccupied_1	0.898	(0.0591)	0.103	0.911	(0.0628)	0.177
h_agemax_1	0.943**	(0.0256)	0.0303	0.961	(0.0285)	0.175
h_agemax_1_sq	1.001*	(0.000543)	0.0847	1.001	(0.000597)	0.326
h_agemax_1_cb	1.000	(3.37e-06)	0.159	1.000	(3.75e-06)	0.466
educ1_1	1.148**	(0.0724)	0.0282	1.164**	(0.0795)	0.0258
educ2_1	1.138**	(0.0652)	0.0246	1.158**	(0.0730)	0.0199
educ3_1	1.160*	(0.0975)	0.0768	1.175*	(0.109)	0.0834
educ4_1	1.114*	(0.0717)	0.0928	1.151**	(0.0800)	0.0422
o.educ5_1	-		-	-		-
2.properties_1	1.126***	(0.0480)	0.00521	1.157***	(0.0555)	0.00240
3.properties_1	1.067	(0.0967)	0.474	1.092	(0.113)	0.397
BuildingAge_1	1.006*	(0.00362)	0.0870	1.004	(0.00443)	0.396
BuildingAge2	1.000	(4.09e-05)	0.217	1.000	(5.02e-05)	0.563
BuildingAge3	1.000	(1.21e-07)	0.293	1.000	(1.49e-07)	0.633
Bathrooms_1	2.471**	(0.957)	0.0195	4.277***	(2.018)	0.00207
Bathrooms2	0.601**	(0.129)	0.0176	0.453***	(0.114)	0.00173
Bathrooms3	1.078**	(0.0402)	0.0429	1.125***	(0.0469)	0.00487
Rooms_1	0.981	(0.147)	0.900	0.975	(0.163)	0.878
Rooms2	0.991	(0.0354)	0.793	0.982	(0.0383)	0.650
Rooms3	1.001	(0.00240)	0.601	1.002	(0.00258)	0.389
2006.year_1	1.054	(0.0608)	0.358	1.048	(0.0790)	0.530
2007.year_1	1.003	(0.0976)	0.979	0.972	(0.111)	0.806
Observations	93,768			88,495		

Robust seeform in parentheses

*** p<0.01, ** p<0.05, * p<0.1

REGRESSION 1 (BASELINE SPECIFICATION)

	MODEL 3			MODEL 4		
	(1)	(2)	(3)	(1)	(2)	(3)
	_t			_t		
VARIABLES	coefEform	seEform	pval	coefEform	seEform	pval
_t						
1.liquidity_dec_1	0.873	(0.100)	0.236	0.936	(0.129)	0.631
2.liquidity_dec_1	0.955	(0.117)	0.710	1.000	(0.138)	0.999
3.liquidity_dec_1	0.966	(0.123)	0.787	1.052	(0.161)	0.740
4.liquidity_dec_1	0.906	(0.0994)	0.369	0.966	(0.116)	0.775
5.liquidity_dec_1	1.053	(0.0952)	0.567	1.120	(0.127)	0.317
6.liquidity_dec_1	1.047	(0.113)	0.670	1.165	(0.144)	0.217
7.liquidity_dec_1	1.049	(0.112)	0.653	1.150	(0.125)	0.197
8.liquidity_dec_1	1.043	(0.116)	0.703	1.104	(0.139)	0.433
9.liquidity_dec_1	1.013	(0.0915)	0.884	1.091	(0.119)	0.422
2.LTV_cat_1	0.849*	(0.0716)	0.0520	0.814**	(0.0815)	0.0396
3.LTV_cat_1	0.623***	(0.0829)	0.000378	0.664**	(0.107)	0.0106
4.LTV_cat_1	0.503***	(0.0473)	0	0.533***	(0.0588)	1.17e-08
1.wealth_dec_1	1.420***	(0.183)	0.00633	1.335*	(0.220)	0.0798
2.wealth_dec_1	1.168	(0.163)	0.267	1.055	(0.188)	0.763
3.wealth_dec_1	1.181	(0.150)	0.191	1.113	(0.176)	0.498
4.wealth_dec_1	1.362**	(0.202)	0.0373	1.248	(0.223)	0.216
5.wealth_dec_1	1.235	(0.178)	0.142	1.164	(0.213)	0.406
6.wealth_dec_1	1.213	(0.159)	0.141	1.132	(0.175)	0.422
7.wealth_dec_1	1.152	(0.155)	0.293	1.021	(0.168)	0.898
8.wealth_dec_1	1.034	(0.146)	0.814	0.931	(0.141)	0.637
9.wealth_dec_1	1.221**	(0.115)	0.0343	1.230*	(0.150)	0.0896
1.DTI_1	1.352*	(0.215)	0.0574	1.142	(0.218)	0.486
2.DTI_1	1.170	(0.156)	0.239	1.140	(0.176)	0.397
3.DTI_1	1.212*	(0.131)	0.0765	1.141	(0.142)	0.289
4.DTI_1	1.366***	(0.131)	0.00117	1.295**	(0.144)	0.0199
5.DTI_1	1.031	(0.0939)	0.734	1.070	(0.100)	0.472
6.DTI_1	1.109	(0.134)	0.390	1.123	(0.148)	0.380
7.DTI_1	1.172	(0.121)	0.122	1.105	(0.112)	0.322
8.DTI_1	1.192**	(0.106)	0.0467	1.160	(0.123)	0.162
9.DTI_1	1.260**	(0.144)	0.0430	1.237*	(0.151)	0.0809
unoccupied_1	0.858	(0.0954)	0.168	0.781*	(0.107)	0.0715
h_agemax_1	0.970	(0.0337)	0.379	1.009	(0.0396)	0.824
h_agemax_1_sq	1.000	(0.000695)	0.541	1.000	(0.000781)	0.630
h_agemax_1_cb	1.000	(4.31e-06)	0.700	1.000	(4.81e-06)	0.475
educ1_1	1.111	(0.119)	0.328	1.160	(0.148)	0.244
educ2_1	1.171	(0.113)	0.103	1.183	(0.142)	0.162
educ3_1	1.228	(0.165)	0.127	1.257	(0.217)	0.185
educ4_1	1.118	(0.104)	0.231	1.176	(0.131)	0.145
o.educ5_1	-		-	-		-
2.properties_1	1.043	(0.0820)	0.596	1.051	(0.0878)	0.548
3.properties_1	1.092	(0.139)	0.488	1.219	(0.186)	0.193
BuildingAge_1	1.009**	(0.00418)	0.0286	1.005	(0.00579)	0.385
BuildingAge2	1.000	(4.61e-05)	0.147	1.000	(6.24e-05)	0.626
BuildingAge3	1.000	(1.36e-07)	0.224	1.000	(1.81e-07)	0.687
Bathrooms_1	1.369	(0.623)	0.491	2.721*	(1.415)	0.0543
Bathrooms2	0.850	(0.221)	0.533	0.598*	(0.174)	0.0778
Bathrooms3	1.014	(0.0479)	0.764	1.069	(0.0557)	0.204
Rooms_1	0.955	(0.177)	0.804	1.045	(0.225)	0.838
Rooms2	0.991	(0.0464)	0.843	0.951	(0.0496)	0.338
Rooms3	1.001	(0.00331)	0.717	1.005	(0.00357)	0.191
2005o.year_1	-		-	-		-
Observations	43,988			39,987		

Robust seeform in parentheses

*** p<0.01, ** p<0.05, * p<0.1

REGRESSION 2 (INTERACTION TERM SPECIFICATION)

	MODEL 3			MODEL 4		
	(1)	(2)	(3)	(1)	(2)	(3)
	_t			_t		
VARIABLES	coefEform	seEform	pval	coefEform	seEform	pval
_t						
1.liquidity_dec_1	1.048	(0.142)	0.731	1.077	(0.184)	0.666
2.liquidity_dec_1	1.128	(0.172)	0.428	1.206	(0.217)	0.298
3.liquidity_dec_1	1.117	(0.167)	0.459	1.233	(0.208)	0.214
4.liquidity_dec_1	1.089	(0.138)	0.498	1.123	(0.153)	0.394
5.liquidity_dec_1	1.181	(0.126)	0.121	1.261	(0.178)	0.101
6.liquidity_dec_1	1.166	(0.157)	0.254	1.320*	(0.218)	0.0928
7.liquidity_dec_1	1.186	(0.148)	0.172	1.312**	(0.178)	0.0457
8.liquidity_dec_1	1.179	(0.177)	0.274	1.262	(0.218)	0.177
9.liquidity_dec_1	1.194*	(0.127)	0.0958	1.294**	(0.168)	0.0477
2.LTV_cat_1	1.727*	(0.496)	0.0567	1.756*	(0.538)	0.0662
3.LTV_cat_1	0.728	(0.248)	0.351	0.489	(0.283)	0.216
4.LTV_cat_1	0.790	(0.231)	0.421	0.909	(0.293)	0.767
1o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
1.liquidity_dec_1#2.LTV_cat_1	0.391***	(0.123)	0.00286	0.460**	(0.169)	0.0348
1.liquidity_dec_1#3.LTV_cat_1	1.683	(0.805)	0.276	2.970*	(1.705)	0.0579
1.liquidity_dec_1#4.LTV_cat_1	0.264**	(0.165)	0.0334	0.330*	(0.214)	0.0878
2o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
2.liquidity_dec_1#2.LTV_cat_1	0.503**	(0.160)	0.0311	0.409**	(0.166)	0.0278
2.liquidity_dec_1#3.LTV_cat_1	0.382	(0.311)	0.237	0.387	(0.449)	0.414
2.liquidity_dec_1#4.LTV_cat_1	0.395*	(0.190)	0.0535	0.402*	(0.222)	0.0994
3o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
3.liquidity_dec_1#2.LTV_cat_1	0.500**	(0.151)	0.0215	0.411**	(0.157)	0.0201
3.liquidity_dec_1#3.LTV_cat_1	0.171*	(0.183)	0.0977	0.331	(0.381)	0.337
3.liquidity_dec_1#4.LTV_cat_1	0.712	(0.271)	0.371	0.684	(0.285)	0.361
4o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
4.liquidity_dec_1#2.LTV_cat_1	0.352***	(0.135)	0.00630	0.434**	(0.166)	0.0291
4.liquidity_dec_1#3.LTV_cat_1	0.893	(0.409)	0.805	1.547	(0.982)	0.492
4.liquidity_dec_1#4.LTV_cat_1	0.540	(0.223)	0.136	0.523	(0.235)	0.148
5o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
5.liquidity_dec_1#2.LTV_cat_1	0.553**	(0.160)	0.0409	0.457**	(0.160)	0.0250
5.liquidity_dec_1#3.LTV_cat_1	1.307	(0.506)	0.489	2.065	(1.328)	0.260
5.liquidity_dec_1#4.LTV_cat_1	0.609	(0.201)	0.133	0.673	(0.261)	0.307
6o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
6.liquidity_dec_1#2.LTV_cat_1	0.566	(0.212)	0.128	0.502	(0.215)	0.108
6.liquidity_dec_1#3.LTV_cat_1	0.653	(0.370)	0.452	1.209	(0.896)	0.797
6.liquidity_dec_1#4.LTV_cat_1	0.801	(0.255)	0.485	0.654	(0.254)	0.274
7o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
7.liquidity_dec_1#2.LTV_cat_1	0.552*	(0.192)	0.0874	0.539	(0.208)	0.109
7.liquidity_dec_1#3.LTV_cat_1	0.896	(0.443)	0.823	1.724	(1.149)	0.414
7.liquidity_dec_1#4.LTV_cat_1	0.636	(0.199)	0.148	0.462**	(0.173)	0.0389
8o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
8.liquidity_dec_1#2.LTV_cat_1	0.436**	(0.182)	0.0470	0.414**	(0.182)	0.0448
8.liquidity_dec_1#3.LTV_cat_1	1.300	(0.603)	0.571	1.765	(1.097)	0.361
8.liquidity_dec_1#4.LTV_cat_1	0.688	(0.281)	0.360	0.592	(0.255)	0.223

9o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
9.liquidity_dec_1#2.LTV_cat_1	0.356**	(0.149)	0.0134	0.340**	(0.162)	0.0234
9.liquidity_dec_1#3.LTV_cat_1	0.418	(0.291)	0.210	0.785	(0.664)	0.775
9.liquidity_dec_1#4.LTV_cat_1	0.608	(0.237)	0.202	0.531	(0.217)	0.121
10b.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
10b.liquidity_dec_1#2o.LTV_cat_1	1	(0)		1	(0)	
10b.liquidity_dec_1#3o.LTV_cat_1	1	(0)		1	(0)	
10b.liquidity_dec_1#4o.LTV_cat_1	1	(0)		1	(0)	
1.wealth_dec_1	1.359**	(0.180)	0.0207	1.280	(0.222)	0.154
2.wealth_dec_1	1.129	(0.157)	0.382	1.025	(0.183)	0.889
3.wealth_dec_1	1.142	(0.145)	0.294	1.079	(0.170)	0.632
4.wealth_dec_1	1.317*	(0.192)	0.0594	1.208	(0.213)	0.283
5.wealth_dec_1	1.195	(0.171)	0.213	1.129	(0.206)	0.504
6.wealth_dec_1	1.172	(0.154)	0.227	1.099	(0.169)	0.542
7.wealth_dec_1	1.114	(0.147)	0.411	0.990	(0.159)	0.953
8.wealth_dec_1	1.002	(0.140)	0.989	0.904	(0.135)	0.498
9.wealth_dec_1	1.188*	(0.109)	0.0609	1.198	(0.141)	0.124
1.DTI_1	1.363*	(0.217)	0.0517	1.146	(0.219)	0.477
2.DTI_1	1.177	(0.158)	0.224	1.140	(0.177)	0.397
3.DTI_1	1.215*	(0.133)	0.0746	1.138	(0.144)	0.304
4.DTI_1	1.372***	(0.131)	0.000970	1.294**	(0.143)	0.0197
5.DTI_1	1.036	(0.0941)	0.699	1.069	(0.100)	0.475
6.DTI_1	1.112	(0.135)	0.381	1.122	(0.149)	0.385
7.DTI_1	1.174	(0.119)	0.114	1.103	(0.111)	0.326
8.DTI_1	1.191**	(0.105)	0.0484	1.159	(0.122)	0.163
9.DTI_1	1.254**	(0.144)	0.0478	1.230*	(0.150)	0.0896
unoccupied_1	0.858	(0.0946)	0.165	0.778*	(0.107)	0.0680
h_agemax_1	0.971	(0.0341)	0.408	1.011	(0.0401)	0.784
h_agemax_1_sq	1.000	(0.000702)	0.577	1.000	(0.000790)	0.594
h_agemax_1_cb	1.000	(4.35e-06)	0.738	1.000	(4.87e-06)	0.447
educ1_1	1.107	(0.119)	0.342	1.157	(0.147)	0.251
educ2_1	1.169	(0.114)	0.108	1.181	(0.143)	0.168
educ3_1	1.230	(0.167)	0.128	1.261	(0.220)	0.184
educ4_1	1.117	(0.105)	0.239	1.176	(0.132)	0.148
o.educ5_1	-		-	-		-
2.properties_1	1.050	(0.0836)	0.539	1.057	(0.0904)	0.514
3.properties_1	1.088	(0.137)	0.500	1.214	(0.184)	0.201
BuildingAge_1	1.009**	(0.00422)	0.0299	1.005	(0.00582)	0.378
BuildingAge2	1.000	(4.65e-05)	0.153	1.000	(6.28e-05)	0.620
BuildingAge3	1.000	(1.37e-07)	0.230	1.000	(1.82e-07)	0.680
Bathrooms_1	1.355	(0.610)	0.500	2.641*	(1.358)	0.0590
Bathrooms2	0.857	(0.219)	0.545	0.609*	(0.174)	0.0831
Bathrooms3	1.013	(0.0468)	0.780	1.065	(0.0542)	0.214
Rooms_1	0.959	(0.176)	0.820	1.050	(0.225)	0.821
Rooms2	0.991	(0.0462)	0.842	0.951	(0.0496)	0.333
Rooms3	1.001	(0.00331)	0.725	1.005	(0.00358)	0.190
2005o.year_1	-		-	-		-
Observations	43,988			39,987		

Robust seeform in parentheses

*** p<0.01, ** p<0.05, * p<0.1

REGRESSION 1 (BASELINE SPECIFICATION)

	MODEL 5			MODEL 6		
	(1)	(2)	(3)	(1)	(2)	(3)
	_t			_t		
VARIABLES	coefEform	seEform	pval	coefEform	seEform	pval
_t						
1.liquidity_dec_1	0.800**	(0.0795)	0.0248	0.782**	(0.0853)	0.0244
2.liquidity_dec_1	0.799*	(0.0931)	0.0542	0.778**	(0.0966)	0.0433
3.liquidity_dec_1	0.745***	(0.0784)	0.00507	0.745**	(0.0892)	0.0140
4.liquidity_dec_1	0.852*	(0.0699)	0.0510	0.832**	(0.0769)	0.0470
5.liquidity_dec_1	0.763***	(0.0745)	0.00565	0.792**	(0.0770)	0.0166
6.liquidity_dec_1	0.928	(0.0788)	0.376	0.934	(0.0849)	0.450
7.liquidity_dec_1	0.826**	(0.0711)	0.0267	0.816**	(0.0762)	0.0298
8.liquidity_dec_1	0.904	(0.0753)	0.225	0.896	(0.0837)	0.241
9.liquidity_dec_1	0.906	(0.0700)	0.202	0.913	(0.0764)	0.279
2.LTV_cat_1	0.904	(0.0594)	0.126	0.911	(0.0649)	0.193
3.LTV_cat_1	0.637***	(0.0853)	0.000753	0.619***	(0.0929)	0.00140
4.LTV_cat_1	0.518***	(0.0486)	0	0.523***	(0.0513)	0
1.wealth_dec_1	1.384**	(0.218)	0.0393	1.349*	(0.222)	0.0687
2.wealth_dec_1	1.339**	(0.199)	0.0500	1.256	(0.187)	0.126
3.wealth_dec_1	1.172	(0.182)	0.307	1.140	(0.174)	0.389
4.wealth_dec_1	1.257	(0.195)	0.141	1.236	(0.196)	0.181
5.wealth_dec_1	1.346**	(0.200)	0.0454	1.361**	(0.201)	0.0370
6.wealth_dec_1	1.201	(0.174)	0.206	1.188	(0.172)	0.234
7.wealth_dec_1	1.175	(0.179)	0.290	1.191	(0.181)	0.252
8.wealth_dec_1	1.168	(0.154)	0.238	1.133	(0.155)	0.361
9.wealth_dec_1	1.055	(0.125)	0.648	1.100	(0.137)	0.442
1.DTI_1	1.259*	(0.164)	0.0775	1.218	(0.174)	0.168
2.DTI_1	0.983	(0.146)	0.907	0.947	(0.136)	0.704
3.DTI_1	1.037	(0.115)	0.746	0.989	(0.117)	0.926
4.DTI_1	0.996	(0.113)	0.969	0.974	(0.113)	0.818
5.DTI_1	0.962	(0.0968)	0.701	0.935	(0.0939)	0.504
6.DTI_1	1.037	(0.117)	0.747	1.000	(0.118)	1.000
7.DTI_1	1.002	(0.0949)	0.979	0.972	(0.0986)	0.780
8.DTI_1	1.016	(0.105)	0.881	1.024	(0.115)	0.835
9.DTI_1	1.027	(0.0853)	0.750	1.014	(0.0880)	0.871
unoccupied_1	0.917	(0.0815)	0.328	0.960	(0.0836)	0.639
h_agemax_1	0.920**	(0.0336)	0.0220	0.926*	(0.0368)	0.0528
h_agemax_1_sq	1.001*	(0.000756)	0.0652	1.001	(0.000825)	0.111
h_agemax_1_cb	1.000	(4.88e-06)	0.124	1.000	(5.37e-06)	0.168
educ1_1	1.211**	(0.0953)	0.0151	1.185**	(0.0942)	0.0330
educ2_1	1.092	(0.0825)	0.244	1.114	(0.0864)	0.163
educ3_1	1.110	(0.110)	0.291	1.118	(0.109)	0.253
educ4_1	1.108	(0.0819)	0.167	1.125	(0.0858)	0.124
o.educ5_1	-		-	-		-
2.properties_1	1.145**	(0.0769)	0.0442	1.172**	(0.0835)	0.0264
3.properties_1	1.013	(0.127)	0.916	0.955	(0.130)	0.735
BuildingAge_1	1.003	(0.00434)	0.561	1.001	(0.00466)	0.772
BuildingAge2	1.000	(4.96e-05)	0.661	1.000	(5.48e-05)	0.859
BuildingAge3	1.000	(1.51e-07)	0.736	1.000	(1.69e-07)	0.903
Bathrooms_1	7.435***	(4.710)	0.00154	10.04***	(7.585)	0.00226
Bathrooms2	0.332***	(0.107)	0.000628	0.284***	(0.108)	0.000925
Bathrooms3	1.184***	(0.0572)	0.000478	1.212***	(0.0669)	0.000506
Rooms_1	0.979	(0.180)	0.910	0.921	(0.170)	0.657
Rooms2	0.994	(0.0415)	0.883	1.005	(0.0421)	0.905
Rooms3	1.001	(0.00279)	0.691	1.001	(0.00280)	0.857
2006o.year_1	-		-	-		-
Observations	46,694			45,422		

Robust seeform in parentheses

*** p<0.01, ** p<0.05, * p<0.1

REGRESSION 2 (INTERACTION TERM SPECIFICATION)

	MODEL 5			MODEL 6		
	(1)	(2)	(3)	(1)	(2)	(3)
	_t			_t		
VARIABLES	coefEform	seEform	pval	coefEform	seEform	pval
_t						
1.liquidity_dec_1	0.967	(0.127)	0.800	0.990	(0.135)	0.939
2.liquidity_dec_1	0.995	(0.130)	0.971	0.982	(0.142)	0.901
3.liquidity_dec_1	0.917	(0.114)	0.482	0.939	(0.127)	0.641
4.liquidity_dec_1	1.083	(0.101)	0.390	1.065	(0.112)	0.551
5.liquidity_dec_1	0.999	(0.115)	0.991	1.055	(0.122)	0.645
6.liquidity_dec_1	1.200*	(0.116)	0.0591	1.227**	(0.124)	0.0427
7.liquidity_dec_1	1.034	(0.113)	0.759	1.036	(0.124)	0.769
8.liquidity_dec_1	1.092	(0.125)	0.441	1.118	(0.141)	0.379
9.liquidity_dec_1	1.049	(0.103)	0.629	1.083	(0.111)	0.440
2.LTV_cat_1	1.440*	(0.303)	0.0832	1.541**	(0.337)	0.0480
3.LTV_cat_1	1.125	(0.294)	0.653	1.145	(0.312)	0.620
4.LTV_cat_1	0.934	(0.193)	0.739	0.985	(0.207)	0.945
1o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
1.liquidity_dec_1#2.LTV_cat_1	0.722	(0.199)	0.237	0.631	(0.187)	0.120
1.liquidity_dec_1#3.LTV_cat_1	0.114**	(0.118)	0.0369	0.123**	(0.129)	0.0454
1.liquidity_dec_1#4.LTV_cat_1	0.772	(0.287)	0.487	0.604	(0.240)	0.205
2o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
2.liquidity_dec_1#2.LTV_cat_1	0.767	(0.208)	0.329	0.722	(0.214)	0.271
2.liquidity_dec_1#3.LTV_cat_1	0.359*	(0.197)	0.0621	0.406	(0.224)	0.102
2.liquidity_dec_1#4.LTV_cat_1	0.252***	(0.117)	0.00288	0.277***	(0.133)	0.00742
3o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
3.liquidity_dec_1#2.LTV_cat_1	0.802	(0.244)	0.468	0.757	(0.250)	0.399
3.liquidity_dec_1#3.LTV_cat_1	0.487	(0.268)	0.192	0.402	(0.239)	0.126
3.liquidity_dec_1#4.LTV_cat_1	0.288***	(0.117)	0.00223	0.263***	(0.114)	0.00216
4o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
4.liquidity_dec_1#2.LTV_cat_1	0.510**	(0.144)	0.0173	0.549**	(0.162)	0.0416
4.liquidity_dec_1#3.LTV_cat_1	0.320**	(0.161)	0.0239	0.270**	(0.149)	0.0174
4.liquidity_dec_1#4.LTV_cat_1	0.613	(0.209)	0.151	0.585	(0.223)	0.159
5o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
5.liquidity_dec_1#2.LTV_cat_1	0.409***	(0.132)	0.00576	0.413***	(0.135)	0.00697
5.liquidity_dec_1#3.LTV_cat_1	0.498	(0.220)	0.114	0.523	(0.231)	0.142
5.liquidity_dec_1#4.LTV_cat_1	0.469***	(0.126)	0.00488	0.405***	(0.108)	0.000730
6o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
6.liquidity_dec_1#2.LTV_cat_1	0.438***	(0.126)	0.00401	0.424***	(0.130)	0.00526
6.liquidity_dec_1#3.LTV_cat_1	0.653	(0.290)	0.338	0.641	(0.277)	0.303
6.liquidity_dec_1#4.LTV_cat_1	0.420***	(0.120)	0.00243	0.401***	(0.117)	0.00175
7o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
7.liquidity_dec_1#2.LTV_cat_1	0.597*	(0.182)	0.0911	0.558*	(0.186)	0.0803
7.liquidity_dec_1#3.LTV_cat_1	0.341**	(0.177)	0.0386	0.379*	(0.195)	0.0591
7.liquidity_dec_1#4.LTV_cat_1	0.544**	(0.131)	0.0115	0.539**	(0.136)	0.0147
8o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
8.liquidity_dec_1#2.LTV_cat_1	0.563*	(0.178)	0.0690	0.473**	(0.168)	0.0353
8.liquidity_dec_1#3.LTV_cat_1	1.009	(0.344)	0.979	0.885	(0.314)	0.730
8.liquidity_dec_1#4.LTV_cat_1	0.502**	(0.142)	0.0145	0.484**	(0.147)	0.0169

9o.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
9.liquidity_dec_1#2.LTV_cat_1	0.752	(0.227)	0.346	0.643	(0.193)	0.140
9.liquidity_dec_1#3.LTV_cat_1	0.604	(0.219)	0.164	0.522*	(0.195)	0.0814
9.liquidity_dec_1#4.LTV_cat_1	0.679	(0.188)	0.162	0.688	(0.194)	0.186
10b.liquidity_dec_1#1b.LTV_cat_1	1	(0)		1	(0)	
10b.liquidity_dec_1#2o.LTV_cat_1	1	(0)		1	(0)	
10b.liquidity_dec_1#3o.LTV_cat_1	1	(0)		1	(0)	
10b.liquidity_dec_1#4o.LTV_cat_1	1	(0)		1	(0)	
1.wealth_dec_1	1.214	(0.190)	0.215	1.179	(0.194)	0.319
2.wealth_dec_1	1.181	(0.175)	0.262	1.102	(0.164)	0.513
3.wealth_dec_1	1.030	(0.161)	0.852	0.995	(0.154)	0.975
4.wealth_dec_1	1.103	(0.173)	0.530	1.079	(0.172)	0.634
5.wealth_dec_1	1.183	(0.178)	0.265	1.188	(0.178)	0.248
6.wealth_dec_1	1.056	(0.152)	0.703	1.039	(0.151)	0.794
7.wealth_dec_1	1.038	(0.155)	0.804	1.046	(0.157)	0.765
8.wealth_dec_1	1.038	(0.142)	0.784	1.002	(0.142)	0.991
9.wealth_dec_1	0.958	(0.115)	0.719	0.995	(0.126)	0.968
1.DTI_1	1.227	(0.160)	0.115	1.186	(0.170)	0.236
2.DTI_1	0.978	(0.146)	0.882	0.943	(0.137)	0.686
3.DTI_1	1.042	(0.115)	0.712	0.995	(0.118)	0.969
4.DTI_1	1.000	(0.112)	0.998	0.979	(0.113)	0.854
5.DTI_1	0.966	(0.0961)	0.729	0.940	(0.0938)	0.534
6.DTI_1	1.040	(0.115)	0.724	1.004	(0.117)	0.975
7.DTI_1	1.007	(0.0951)	0.943	0.976	(0.0998)	0.814
8.DTI_1	1.017	(0.105)	0.868	1.026	(0.116)	0.818
9.DTI_1	1.029	(0.0859)	0.735	1.016	(0.0892)	0.853
unoccupied_1	0.914	(0.0810)	0.310	0.957	(0.0831)	0.616
h_agemax_1	0.921**	(0.0335)	0.0234	0.926*	(0.0365)	0.0522
h_agemax_1_sq	1.001*	(0.000754)	0.0688	1.001	(0.000821)	0.110
h_agemax_1_cb	1.000	(4.88e-06)	0.128	1.000	(5.35e-06)	0.166
educ1_1	1.211**	(0.0958)	0.0157	1.185**	(0.0947)	0.0338
educ2_1	1.098	(0.0833)	0.216	1.121	(0.0873)	0.143
educ3_1	1.116	(0.111)	0.270	1.123	(0.110)	0.237
educ4_1	1.113	(0.0826)	0.151	1.130	(0.0865)	0.111
o.educ5_1	-		-	-		-
2.properties_1	1.166**	(0.0782)	0.0219	1.195**	(0.0855)	0.0126
3.properties_1	1.006	(0.128)	0.965	0.945	(0.131)	0.684
BuildingAge_1	1.003	(0.00433)	0.532	1.002	(0.00466)	0.745
BuildingAge2	1.000	(4.95e-05)	0.628	1.000	(5.48e-05)	0.828
BuildingAge3	1.000	(1.50e-07)	0.698	1.000	(1.69e-07)	0.869
Bathrooms_1	7.503***	(4.761)	0.00150	10.08***	(7.601)	0.00220
Bathrooms2	0.331***	(0.107)	0.000615	0.284***	(0.108)	0.000890
Bathrooms3	1.185***	(0.0573)	0.000459	1.212***	(0.0667)	0.000470
Rooms_1	0.982	(0.181)	0.923	0.926	(0.172)	0.680
Rooms2	0.994	(0.0417)	0.888	1.005	(0.0421)	0.910
Rooms3	1.001	(0.00280)	0.700	1.001	(0.00280)	0.857
2006o.year_1	-		-	-		-
Observations	46,694			45,422		

Robust seeform in parentheses

*** p<0.01, ** p<0.05, * p<0.1